

Federated Adaptive Causal Estimation with High Dimensional Covariates (FACE-HD)

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Abstract

High-dimensional observational data, such as electronic health records (EHRs), present distinctive challenges for causal inference and comparative effectiveness research. When patient-level data are distributed across multiple clinical sites and covariate distributions differ across sites, naive pooling or isolated site-specific analyses can yield unstable or biased estimates of target-population treatment effects. We develop a federated adaptive causal estimation framework for high-dimensional covariates (FACE-HD), extending Federated Adaptive Causal Estimation (FACE) to settings where confounding adjustment must be learned from many candidate variables. The contribution is threefold. First, FACE-HD uses a merged propensity score–density ratio calibration model, replacing separate treatment and transport models by a single high-dimensional tilting parameter for each source-treatment pair. Second, it combines two-step penalized nuisance estimation, calibration, sample splitting, and cross-fitting to construct doubly robust estimators of the target average treatment effect (TATE). Third, it provides a communication-efficient federated algorithm and a variance-aware adaptive aggregation rule that borrows information from transportable sources while downweighting non-transportable ones. Under sparsity, overlap, and regularity conditions, FACE-HD yields root- N consistent and asymptotically normal estimators when either the outcome regression model or the merged propensity score–density ratio model is correctly specified, and it attains semiparametric efficiency when both are correct. Simulation studies evaluate the efficiency gains and robustness of FACE-HD relative to distributed and target-only alternatives.

Keywords: causal inference, federated learning, high-dimensional inference, transfer learning, doubly robust estimation

1 Introduction

The increasing availability of real-world data, including electronic health records (EHRs), administrative claims, and patient registries, offers an unprecedented opportunity to es-

timate causal effects in large and diverse populations (Hernán & Robins 2016, Sherman et al. 2016). In practice, however, these data rarely arrive as one clean pooled dataset. Patient-level records are often held locally by clinical sites because of privacy, governance, and infrastructure constraints, while the number of candidate confounders can be large enough that standard low-dimensional causal methods become unstable (Belloni et al. 2014, Chernozhukov et al. 2018, Tan 2020). Multi-site observational studies add a further complication: covariate distributions may shift across institutions, so a source site that is useful for precision can also be harmful if its population is not transportable to the target site (Cole & Stuart 2010, Pearl & Bareinboim 2014, Dahabreh et al. 2019).

In this paper, we extend FACE to high-dimensional settings. The proposed FACE-HD framework begins from the target-site estimand, the TATE, and treats source sites as auxiliary information that must earn its way into the final estimator. For each target-source pair, FACE-HD uses a merged propensity score–density ratio calibration model, so transport and treatment weighting are handled by one exponential-tilting nuisance parameter rather than by two separately estimated high-dimensional models. It then fits the outcome regression and the merged propensity score–density ratio model through a two-step penalized calibration procedure with sample splitting and cross-fitting. This construction makes the pairwise source-assisted estimator doubly robust: first-order nuisance bias is removed when either the outcome regression or the weighting model is correctly specified, in the spirit of classical doubly robust and semiparametric estimating equations (Robins et al. 1994, Bang & Robins 2005, Tsiatis 2006).

The final step is adaptive aggregation. Source-assisted estimators share the same target sample, so their errors are correlated through target-side outcome-regression components. FACE-HD estimates these covariance terms explicitly and chooses aggregation weights by minimizing an estimated variance with an adaptive penalty based on disagreement between the target-only and source-assisted estimators. The result is a federated procedure that sends summary statistics and fitted nuisance parameters rather than patient-level records, borrows strength from transportable sources, and shrinks toward the target-only estimator when source information appears unreliable.

The remainder of the paper follows this story. [Section 2](#) discusses related work in high-dimensional causal inference, transportability, and federated multi-site estimation. [Section 3](#) introduces the multi-site notation, target estimand, calibration weights, pairwise doubly

robust estimators, cross-calibrated nuisance losses, adaptive aggregation objective, federated communication protocol, and tuning rules. [Section 4](#) gives the regularity conditions and asymptotic guarantees behind the method, [Section 5](#) describes the simulation study, and [Section 6](#) closes with implications and limitations. Proofs and additional technical details are provided in the Supplemental Material.

2 Related Work

FACE-HD builds on the high-dimensional causal inference literature, where regularized nuisance estimation must be combined with bias correction to support valid inference. Double-selection, debiased, and double machine learning methods show how outcome and propensity-score nuisance fits can be used without letting first-stage regularization bias dominate the target effect estimator ([Belloni et al. 2014](#), [Chernozhukov et al. 2018](#)). For augmented inverse probability weighting and doubly robust estimation, the same theme appears through orthogonal estimating equations: if the first-order sensitivity to the nuisance functions is removed, the final estimator can remain valid when one nuisance model is correctly specified and the product of nuisance errors is small ([Robins et al. 1994](#), [Bang & Robins 2005](#), [Tsiatis 2006](#), [Tan 2020](#), [Smucler et al. 2019](#)). FACE-HD follows this orthogonalization logic, but the nuisance functions are fitted under federated information constraints and are calibrated to a target-site distribution rather than to a single pooled population.

A second line of work studies generalizability, transportability, and multi-site causal inference. When source and target populations differ, source data can improve precision only if the relevant outcome and treatment-response structure transports after adjustment for observed covariates ([Cole & Stuart 2010](#), [Pearl & Bareinboim 2014](#), [Dahabreh et al. 2019](#)). This creates the negative-transfer problem: borrowing from a non-transportable source may reduce variance while introducing bias. The original FACE framework addresses this problem by adaptively aggregating target-only and source-assisted estimators for the TATE ([Han et al. 2025](#)). FACE-HD keeps the FACE idea of adaptive borrowing, but extends it to high-dimensional confounding adjustment and explicitly accounts for the covariance induced by shared target data across source-assisted estimators.

Finally, the communication pattern is related to federated learning and privacy-preserving

distributed analysis, where sites exchange model updates or low-dimensional summaries instead of individual-level records (McMahan et al. 2017, Kairouz et al. 2021). Recent work has begun to study federated or distributed ATE estimation directly, including differentially private multi-site aggregation of site-level treatment-effect estimates (Koga et al. 2024), one-shot and multi-shot federated plug-in estimators beyond meta-analysis (Khellaf et al. 2024), and federated propensity-score aggregation for IPW and AIPW estimation (Khellaf et al. 2025). Closely related high-dimensional distributed causal methods calibrate propensity score and outcome models under covariate shift (Wang et al. 2024, Tong et al. 2025), while multiply robust federated estimators use transfer-learning ideas to combine information across sites without pooling records (Han et al. 2023). FACE-HD differs by merging the treatment and density-ratio components into a single exponential-tilting calibration model for each source-treatment pair, then using a variance- and discrepancy-aware aggregation rule to decide how much each source should contribute to the target estimand.

3 Method

We address a causal inference problem involving a single target site (t) and multiple source sites ($\mathcal{S} = \{s_1, s_2, \dots, s_K\}$). Our primary objective is to estimate the target average treatment effect (TATE) by leveraging data from both the target site and auxiliary information from the source sites, following the target-population perspective used in generalizability and FACE-style multi-site estimation (Cole & Stuart 2010, Dahabreh et al. 2019, Han et al. 2025). Throughout, $\mathbb{E}_r[\cdot]$ denotes expectation conditional on site membership $R = r$. We define the target-site potential outcome means $\mu^1 := \mathbb{E}_t[Y(1)]$ and $\mu^0 := \mathbb{E}_t[Y(0)]$, and the TATE

$$\tau := \mu^1 - \mu^0.$$

Throughout the remainder of the Method section, unless explicitly stated otherwise, we develop the construction for the treated potential-outcome mean μ^1 . Thus the nuisance losses, influence-function components, aggregation summaries, and algorithmic steps below are written for μ^1 ; the corresponding objects for μ^0 are obtained by symmetry, replacing treatment level 1 by 0 and using the corresponding nuisance parameters. The TATE estimator is then obtained by differencing:

$$\hat{\tau} := \hat{\mu}^1 - \hat{\mu}^0.$$

Each observation contains site membership $R \in \{t, s_1, \dots, s_K\}$, raw covariates $\mathbf{X}$, treatment $A \in \{0, 1\}$, and outcome Y . We write $\mathcal{D}_t$ and $\mathcal{D}_{s_j}$ for the target and source samples, with sizes N_t and N_{s_j} , and $N_{\text{all}} := N_t + \sum_{j=1}^K N_{s_j}$. The empirical average over site r is denoted by $\tilde{E}_r(\cdot)$. To allow flexible high-dimensional modeling, the nuisance models use a fixed working feature map $\phi(\mathbf{X})$; one may also read $\phi(\mathbf{X})$ simply as the transformed covariate vector used by the analysis.

For each source site $s_j \in \mathcal{S}$ and treatment level $a \in \{0, 1\}$, we construct *calibration weights* that reweight the $(R = s_j, A = a)$ group to match the target covariate distribution. We use an *exponential tilting* (log-linear) *working model* for the tilting function, in line with calibrated weighting and density-ratio ideas for high-dimensional treatment-effect estimation (Tan 2020, Tong et al. 2025, Wang et al. 2024):

$$r_{s_j,a}(\mathbf{X}; \boldsymbol{\gamma}_{s_j,a}) := \exp\{\phi(\mathbf{X})^T \boldsymbol{\gamma}_{s_j,a}\}. \quad (1)$$

This form is convenient because it yields convex calibration losses and strictly positive weights. We do *not* interpret Eq. (1) as a literal probabilistic model for $P(R, A \mid \mathbf{X})$; instead, the key requirement for our theory is that the implied weights satisfy the covariate-balance identity defined below. Crucially, for the pairwise DR estimator $\hat{\mu}_{t,s_j}^1$, only the single parameter vector $\boldsymbol{\gamma}_{s_j,1}$ (for the treated group at source s_j) is required.

Given Eq. (1), we define the (unnormalized) calibration weight for source-site treated observations as

$$w(\mathbf{X}; \boldsymbol{\gamma}_{s_j,1}) := I(A = 1) \exp\{-\phi(\mathbf{X})^T \boldsymbol{\gamma}_{s_j,1}\} = \frac{I(A = 1)}{e^{\phi(\mathbf{X})^T \boldsymbol{\gamma}_{s_j,1}}}. \quad (2)$$

Because $\phi(\mathbf{X})$ includes an intercept, any multiplicative normalizing constant (equivalently, an additive constant in the log-tilt) is absorbed into the intercept component of $\boldsymbol{\gamma}_{s_j,1}$. Consequently, when $\boldsymbol{\gamma}_{s_j,1}^*$ is the population solution of the calibration/balancing equation, the operational weight satisfies the *balancing property*: for every integrable function h ,

$$\mathbb{E}_{s_j}[w(\mathbf{X}; \boldsymbol{\gamma}_{s_j,1}^*) h(\mathbf{X})] = \mathbb{E}_t[h(\mathbf{X})], \quad (3)$$

which is the key identity underlying the pairwise doubly robust estimator introduced below. The identity for *all* integrable h corresponds to the ideal population density-ratio relationship. In the high-dimensional analysis, balance is enforced over a rich but finite-dimensional function class, typically the linear span of the chosen features $h(\mathbf{X}) \in \text{span}\{\phi_1(\mathbf{X}), \dots, \phi_p(\mathbf{X})\}$

(including an intercept). The corresponding “weight-model correctness” condition in the Supplemental Material should be interpreted relative to this function class.

Remark 1 (Calibration weights vs. full probabilistic model). The exponential-tilting form Eq. (1) can be motivated by a multinomial logistic model for $(R, A \mid \mathbf{X})$ (with $R = t$ as a reference category), in which case the implied weights approximate inverse odds of site/treatment membership. In this paper, however, we treat Eq. (1) as a *calibration working model*: $\gamma_{s_j, a}$ is defined by the (regularized) calibration loss, and the substantive requirement is the resulting covariate-balance moment condition Eq. (3).

We use generalized linear models for the potential outcomes. Let $\psi(\cdot; \cdot)$ be a GLM response function with a canonical link, mapping features to the conditional mean of the outcome.

$$m^a(\mathbf{X}; \boldsymbol{\alpha}^a) = \psi(\phi(\mathbf{X}); \boldsymbol{\alpha}^a), \quad a \in \{0, 1\}$$

The doubly robust estimators combine outcome modeling and inverse probability weighting, following the classical AIPW/semiparametric estimating-equation construction (Robins et al. 1994, Bang & Robins 2005, Tsiatis 2006). The estimator for μ^1 assisted by source site s_j is:

$$\hat{\mu}_{t, s_j}^1 = \tilde{E}_t(\psi(\phi(\mathbf{X}); \hat{\boldsymbol{\alpha}}_{t, s_j}^1)) + \tilde{E}_{s_j} \left(\frac{I(A=1)}{e^{\phi(\mathbf{X})^T \hat{\gamma}_{s_j, 1}}} (Y - \psi(\phi(\mathbf{X}); \hat{\boldsymbol{\alpha}}_{t, s_j}^1)) \right) \quad (4)$$

The estimator for the control group, $\hat{\mu}_{t, s_j}^0$, is defined symmetrically by using the nuisance parameters $(\boldsymbol{\alpha}^0, \gamma_{s_j, 0})$ and replacing $I(A=1)$ by $I(A=0)$ in the weighted residual term:

$$\hat{\mu}_{t, s_j}^0 = \tilde{E}_t(\psi(\phi(\mathbf{X}); \hat{\boldsymbol{\alpha}}_{t, s_j}^0)) + \tilde{E}_{s_j} \left(\frac{I(A=0)}{e^{\phi(\mathbf{X})^T \hat{\gamma}_{s_j, 0}}} (Y - \psi(\phi(\mathbf{X}); \hat{\boldsymbol{\alpha}}_{t, s_j}^0)) \right).$$

Finally, the (pairwise) TATE estimator is $\hat{\tau}_{t, s_j} := \hat{\mu}_{t, s_j}^1 - \hat{\mu}_{t, s_j}^0$, and the target-only TATE estimator is $\hat{\tau}_{ot} := \hat{\mu}_{ot}^1 - \hat{\mu}_{ot}^0$.

The target-only estimator $\hat{\mu}_{ot}^1$ is a standard AIPW estimator on the target data alone and requires the *target-site propensity score*,

$$\pi_{ot}(\mathbf{X}) := P(A=1 \mid R=t, \mathbf{X}), \quad (5)$$

which models treatment assignment *within* the target population. This is conceptually and operationally distinct from the cross-site calibration/tilting parameters $\gamma_{s_j, a}$ defined above: π_{ot} is estimated locally at the target site (e.g., via ℓ_1 -penalized logistic regression on $(A, \mathbf{X})$ pairs from $\mathcal{D}_t$), and does not enter the source-assisted estimator $\hat{\mu}_{t, s_j}^1$ or its loss functions.

3.1 Cross-Calibrated Nuisance Estimation

In this framework, the outcome model parameters α_{t,s_j}^a and the site/treatment model parameters $\gamma_{s_j,a}$ are interdependent. This coupling arises because the optimal choice for α depends on weights derived from γ , and vice-versa. Solving this coupled system jointly is computationally challenging in a federated setting where data is decentralized.

To derive loss functions for a federated, iterative estimation process, we follow the principle of constructing estimators that solve efficient score equations, as identified by influence-function and orthogonal estimating-equation analysis (Tsiatis 2006, Chernozhukov et al. 2018, Tan 2020, Smucler et al. 2019).

Let $\bar{\alpha}_{t,s_j}^1$ and $\bar{\gamma}_{s_j,1}$ denote the population targets (probability limits / pseudo-true parameters) associated with the nuisance estimation step; under correct specification of the corresponding nuisance model, these coincide with the true parameters. Let $\bar{\mu}_{t,s_j}^1$ be the source-assisted functional obtained by evaluating the pairwise estimator at these population targets. The asymptotic error of $\hat{\mu}_{t,s_j}^1$ can be decomposed into terms related to the estimation errors of the parameters α and γ :

$$\hat{\mu}_{t,s_j}^1 - \bar{\mu}_{t,s_j}^1 = (\hat{\alpha}_{t,s_j}^1 - \bar{\alpha}_{t,s_j}^1)^T \mathbf{U}_\alpha + (\hat{\gamma}_{s_j,1} - \bar{\gamma}_{s_j,1})^T \mathbf{U}_\gamma + \text{h.o.t.} \quad (6)$$

The vectors $\mathbf{U}_\alpha$ and $\mathbf{U}_\gamma$ are the score vectors:

$$\mathbf{U}_\alpha = \tilde{E}_t \left[\nabla_\alpha \psi(\phi(\mathbf{X}); \bar{\alpha}_{t,s_j}^1) \right] - \tilde{E}_{s_j} \left[\frac{I(A=1)}{e^{\phi(\mathbf{X})^T \bar{\gamma}_{s_j,1}}} \nabla_\alpha \psi(\phi(\mathbf{X}); \bar{\alpha}_{t,s_j}^1) \right] \quad (7)$$

$$\mathbf{U}_\gamma = \tilde{E}_{s_j} \left[-\frac{I(A=1)}{e^{\phi(\mathbf{X})^T \bar{\gamma}_{s_j,1}}} \phi(\mathbf{X}) (Y - \psi(\phi(\mathbf{X}); \bar{\alpha}_{t,s_j}^1)) \right] \quad (8)$$

For the remainder in Eq. (6) to be second-order in the nuisance errors (Neyman orthogonality / debiasing), the estimators must be constructed so that $\mathbf{U}_\alpha = \mathbf{0}$ at $\gamma = \hat{\gamma}$ and $\mathbf{U}_\gamma = \mathbf{0}$ at $\alpha = \hat{\alpha}$. The cross-calibration strategy achieves this by constructing the loss for γ so that its first-order condition is the score equation $\mathbf{U}_\alpha = \mathbf{0}$, and symmetrically for α .

Let $\hat{\alpha}_{\text{init}}$ and $\hat{\gamma}_{\text{init}}$ denote initial plug-in estimates of the outcome and site/treatment parameters, respectively. To ensure numerical stability and satisfy the positivity regularity conditions used in the theory, define the truncation function

$$\mathcal{T}(x) := \text{sign}(x) \min\{|x|, M_\tau\},$$

where M_τ may depend on sample size and is chosen sufficiently large so truncation is asymptotically inactive at the population limit. The subscript recalls the TATE parameter τ defined above, and helps distinguish this numerical threshold from the treatment level. For exponential-family GLMs with canonical link, write the negative log-likelihood as $-Y\theta + \Psi(\theta)$, where $\theta = \phi(\mathbf{X})^T \boldsymbol{\alpha}$ is the linear predictor, $\psi(\theta) = \Psi'(\theta)$ is the conditional mean, and $\psi'(\theta)$ denotes the derivative with respect to θ . Let $\lambda_\gamma \geq 0$ and $\lambda_\alpha \geq 0$ denote the ℓ_1 regularization parameters for the site/treatment and outcome calibrated losses, respectively.

The calibrated exponential-tilting loss for $\boldsymbol{\gamma}_{s_j,1}$ is

$$\begin{aligned} \ell_{\gamma,\text{cal}}(\boldsymbol{\gamma}_{s_j,1}; \hat{\boldsymbol{\alpha}}_{\text{init}}) &= \tilde{E}_t \left[\nabla_{\boldsymbol{\alpha}} \psi(\phi(\mathbf{X}); \hat{\boldsymbol{\alpha}}_{\text{init}})^T \boldsymbol{\gamma}_{s_j,1} \right] \\ &\quad + \tilde{E}_{s_j} \left[I(A=1) e^{-\phi(\mathbf{X})^T \boldsymbol{\gamma}_{s_j,1}} \psi'(\mathcal{T}(\phi(\mathbf{X})^T \hat{\boldsymbol{\alpha}}_{\text{init}})) \right] \\ &\quad + \lambda_\gamma \|\boldsymbol{\gamma}_{s_j,1}\|_1. \end{aligned} \tag{9}$$

Symmetrically, the calibrated weighted GLM loss for $\boldsymbol{\alpha}_{t,s_j}^1$ is

$$\begin{aligned} \ell_{\alpha,\text{cal}}(\boldsymbol{\alpha}_{t,s_j}^1; \hat{\boldsymbol{\gamma}}_{\text{init}}) &= \tilde{E}_{s_j} \left[\frac{I(A=1)}{e^{\mathcal{T}(\phi(\mathbf{X})^T \hat{\boldsymbol{\gamma}}_{\text{init}})}} \left\{ -Y \cdot (\phi(\mathbf{X})^T \boldsymbol{\alpha}_{t,s_j}^1) + \Psi(\phi(\mathbf{X})^T \boldsymbol{\alpha}_{t,s_j}^1) \right\} \right] \\ &\quad + \lambda_\alpha \|\boldsymbol{\alpha}_{t,s_j}^1\|_1. \end{aligned} \tag{10}$$

To verify the cross-calibration, take the gradient of Eq. (9) with respect to $\boldsymbol{\gamma}$ (setting aside the ℓ_1 subdifferential):

$$\nabla_{\boldsymbol{\gamma}} \ell = \tilde{E}_t [\nabla_{\boldsymbol{\alpha}} \psi(\phi(\mathbf{X}); \bar{\boldsymbol{\alpha}})] - \tilde{E}_{s_j} \left[\frac{I(A=1)}{e^{\phi(\mathbf{X})^T \boldsymbol{\gamma}}} \psi'(\mathcal{T}(\phi(\mathbf{X})^T \bar{\boldsymbol{\alpha}})) \phi(\mathbf{X}) \right].$$

For a canonical-link GLM, $\nabla_{\boldsymbol{\alpha}} \psi(\phi(\mathbf{X}); \boldsymbol{\alpha}) = \psi'(\phi(\mathbf{X})^T \boldsymbol{\alpha}) \phi(\mathbf{X})$, so under asymptotically inactive truncation the two terms become

$$\nabla_{\boldsymbol{\gamma}} \ell = \tilde{E}_t [\psi'(\phi^T \bar{\boldsymbol{\alpha}}) \phi(\mathbf{X})] - \tilde{E}_{s_j} [w(\mathbf{X}; \boldsymbol{\gamma}) \psi'(\phi^T \bar{\boldsymbol{\alpha}}) \phi(\mathbf{X})],$$

which matches the score vector $\mathbf{U}_\alpha$ in Eq. (7) up to an asymptotically negligible truncation term. *Hence the population first-order condition of the calibrated $\boldsymbol{\gamma}$ -loss enforces the same first-order orthogonality equation $\mathbf{U}_\alpha = \mathbf{0}$.* Moreover, the calibrated loss Eq. (9) is strictly convex in $\boldsymbol{\gamma}$ (because $t \mapsto e^{-t}$ is convex), guaranteeing a unique minimizer under standard regularity.

The outcome model is fitted by minimizing Eq. (10), given a plug-in site/treatment model $\hat{\gamma}$. The corresponding gradient is

$$\nabla_{\alpha} \ell = \tilde{E}_{s_j} \left[\frac{I(A=1)}{e^{\mathcal{T}(\phi(\mathbf{X})^T \hat{\gamma})}} \left(-Y + \psi(\phi(\mathbf{X})^T \alpha) \right) \phi(\mathbf{X}) \right],$$

and under asymptotically inactive truncation this matches $-\mathbf{U}_{\gamma}(\alpha, \hat{\gamma})$ at first order, so the first-order condition $\nabla_{\alpha} \ell = \mathbf{0}$ enforces $\mathbf{U}_{\gamma} = \mathbf{0}$.

Together, the first-order terms in the remainder Eq. (6) vanish at the population level, leaving a second-order error $O_p(\|\hat{\alpha} - \alpha^*\| \cdot \|\hat{\gamma} - \gamma^*\|)$ that is negligible under standard high-dimensional rates (Bühlmann & Van De Geer 2011, Negahban et al. 2012); see the Supplemental Material for details.

In practice we first obtain robust initial estimates to bootstrap the coupled problem. A simple ℓ_1 -regularized regression on source-site treated data provides the starting outcome model $\hat{\alpha}_{t,s_j}^{1,(0)}$ (Hastie et al. 2009, Bühlmann & Van De Geer 2011). Let ℓ_{GLM} be the standard loss for the chosen GLM (e.g., squared error for linear outcomes or log-loss for binary outcomes):

$$\ell(\alpha_{t,s_j}^1) = \tilde{E}_{s_j} \left[I(A=1) \ell_{GLM}(Y, \psi(\phi(\mathbf{X}); \alpha_{t,s_j}^1)) \right] + \lambda_{\alpha} \|\alpha_{t,s_j}^1\|_1 \quad (11)$$

The initial site/treatment model $\hat{\gamma}_{s_j,a}^{(0)}$ is obtained from a regularized exponential-tilting loss that targets covariate balance without depending on the outcome model:

$$\ell(\gamma_{s_j,1}) = \left(\tilde{E}_t[\phi(\mathbf{X})] \right)^T \gamma_{s_j,1} + \tilde{E}_{s_j} \left[I(A=1) e^{-\phi(\mathbf{X})^T \gamma_{s_j,1}} \right] + \lambda_{\gamma} \|\gamma_{s_j,1}\|_1 \quad (12)$$

Using these initial estimates, the source site performs calibrated updates in a two-level cross-fitted way, combining the general cross-fitting idea from double/debiased machine learning with the SMMAL-style two-level calibration logic used in semi-supervised ATE estimation (Chernozhukov et al. 2018, Hou et al. 2025). For an outer evaluation fold k_1 , an initial nuisance plug-in is trained for each secondary fold $k_2 \neq k_1$ using data excluding both k_1 and k_2 . The final calibrated nuisance estimator for fold k_1 is then obtained by minimizing the average of the fold-specific calibrated losses over all $k_2 \neq k_1$: the k_2 loss contribution is evaluated on fold k_2 and uses the initial plug-in fitted on the remaining training folds. Thus calibration uses the full outer-training sample $\mathcal{D} \setminus \mathcal{D}_{k_1}$ through $K_f - 1$ fold-specific loss contributions, while the final evaluation fold $\mathcal{D}_{k_1}$ remains untouched. The plug-in order is intentional in the two-level cross-fitting construction and does not affect first-order asymptotics under the alignment condition in the Supplemental Material.

3.2 Influence Functions and Adaptive Aggregation

To construct confidence intervals and find optimal aggregation weights, we rely on the influence functions (IF) of our estimators (Tsiatis 2006). The total variance of the final aggregated estimator is a function of the variances and covariances of these IFs.

Let $\hat{\mu}_{ot}^1$ be the target-only estimator and $\hat{\mu}_{t,s_j}^1$ be the source-assisted estimator from site s_j .

For the target-only estimator, using the target-site propensity score $\hat{\pi}_{ot}(\mathbf{X})$ defined in Eq. (5) and a target-only outcome model $\hat{\alpha}_{ot}^1$, we estimate

$$\hat{\varphi}_{ot,i} = \frac{A_i Y_i}{\hat{\pi}_{ot}(\mathbf{X}_i)} - \left(\frac{A_i}{\hat{\pi}_{ot}(\mathbf{X}_i)} - 1 \right) \psi(\phi(\mathbf{X}_i); \hat{\alpha}_{ot}^1) - \hat{\mu}_{ot}^1 \quad (13)$$

The source-assisted estimator has a target-side regression component,

$$\hat{\zeta}_{t,s_j,i} = \psi(\phi(\mathbf{X}_i); \hat{\alpha}_{t,s_j}^1) - \tilde{E}_t(\psi(\phi(\mathbf{X}); \hat{\alpha}_{t,s_j}^1)) \quad (14)$$

and a source-side weighted residual component, using the calibration weight $w(\mathbf{X}; \gamma)$ from Eq. (2),

$$\hat{\xi}_{t,s_j,i} = \frac{A_i}{e^{\phi(\mathbf{X}_i)^T \hat{\gamma}_{s_j,1}}} (Y_i - \psi(\phi(\mathbf{X}_i); \hat{\alpha}_{t,s_j}^1)) - \hat{\delta}_{t,s_j}^1 \quad (15)$$

where $\hat{\delta}_{t,s_j}^1 = \tilde{E}_{s_j} \left(\frac{A}{e^{\phi(\mathbf{X})^T \hat{\gamma}_{s_j,1}}} (Y - \psi(\phi(\mathbf{X}); \hat{\alpha}_{t,s_j}^1)) \right)$.

These pieces lead directly to the variance and covariance components:

$$\begin{aligned} \hat{V}_{ot} &= \frac{1}{N_t} \sum_{i \in \mathcal{D}_t} (\hat{\varphi}_{ot,i} - \bar{\varphi}_{ot})^2 \\ \hat{V}_{t,s_j} &= \frac{1}{N_t} \sum_{i \in \mathcal{D}_t} (\hat{\zeta}_{t,s_j,i} - \bar{\zeta}_{t,s_j})^2 \\ \hat{V}_{s_j} &= \frac{1}{N_{s_j}} \sum_{i \in \mathcal{D}_{s_j}} (\hat{\xi}_{t,s_j,i} - \bar{\xi}_{t,s_j})^2 \\ \hat{C}_{ot,s_j} &= \frac{1}{N_t} \sum_{i \in \mathcal{D}_t} (\hat{\varphi}_{ot,i} - \bar{\varphi}_{ot})(\hat{\zeta}_{t,s_j,i} - \bar{\zeta}_{t,s_j}) \end{aligned}$$

One extra covariance term is easy to miss. All source-assisted estimators reuse the same target data, so the target-side components across two sources are correlated:

$$\hat{C}_{t,s_j,s_k} = \frac{1}{N_t} \sum_{i \in \mathcal{D}_t} (\hat{\zeta}_{t,s_j,i} - \bar{\zeta}_{t,s_j})(\hat{\zeta}_{t,s_k,i} - \bar{\zeta}_{t,s_k}) \quad \text{for } j \neq k$$

Ignoring these cross-site correlations would lead to a systematic underestimation of the aggregated estimator's variance and incorrect confidence intervals; see also the analogous shared-target covariance correction in FACE (Han et al. 2025).

In finite samples, the plug-in variance/covariance components (especially the shared-target covariance terms) can yield a quadratic objective in [Eq. \(16\)](#) that is only approximately convex due to estimation noise. For numerical stability, the estimated covariance inputs may be symmetrized and minimally ridge-regularized when needed to keep the weight optimization well posed; this ridge vanishes asymptotically under the regularity conditions in the Supplemental Material.

The target site combines its own estimator, $\hat{\mu}_{ot}^1$, with the estimators received from all source sites, $\{\hat{\mu}_{t,s_j}^1\}_{j=1}^K$. The final aggregated estimator is a weighted combination:

$$\hat{\mu}_{agg}^1 = (1 - \sum_{j=1}^K \eta_j) \hat{\mu}_{ot}^1 + \sum_{j=1}^K \eta_j \hat{\mu}_{t,s_j}^1$$

The estimated weights $\hat{\boldsymbol{\eta}} = (\hat{\eta}_1, \dots, \hat{\eta}_K)$ are found by minimizing the estimated variance of $\hat{\mu}_{agg}^1$, with an added penalty term to regularize the weights and perform source selection, following the adaptive borrowing strategy of FACE ([Han et al. 2025](#)). This leads to the following quadratic optimization problem:

$$\begin{aligned} \min_{\boldsymbol{\eta}} \quad & \frac{1}{N_t} \left(1 - \sum_{j=1}^K \eta_j\right)^2 \hat{V}_{ot} + \sum_{j=1}^K \frac{\eta_j^2}{N_t} \hat{V}_{t,s_j} + \sum_{j=1}^K \frac{\eta_j^2}{N_{s_j}} \hat{V}_{s_j} \\ & + \frac{2}{N_t} \sum_{j=1}^K \eta_j \left(1 - \sum_{k=1}^K \eta_k\right) \hat{C}_{ot,s_j} + \frac{2}{N_t} \sum_{j < k} \eta_j \eta_k \hat{C}_{t,s_j,s_k} \\ & + \lambda \sum_{j=1}^K |\eta_j| \left(\hat{\mu}_{ot}^1 - \hat{\mu}_{t,s_j}^1\right)^2 \end{aligned} \tag{16}$$

[Eq. \(16\)](#) is written in terms of an estimated variance $\widehat{\text{Var}}(\hat{\mu}_{agg}^1)$. For asymptotic analysis of $\sqrt{N_{all}}(\hat{\mu}_{agg}^1 - \mu^1)$, one can equivalently work with the estimated asymptotic variance $N_{all} \cdot \widehat{\text{Var}}(\hat{\mu}_{agg}^1)$, which multiplies the variance and covariance terms in [Eq. \(16\)](#) by N_{all} . This scaling can be absorbed into the tuning of λ and is used in the Supplemental Material. Define $\hat{\mu}_{agg}^0$ by applying the same aggregation construction to $\{\hat{\mu}_{t,s_j}^0\}_{j=1}^K$ (with the corresponding variance/covariance components for the control arm), and set $\hat{\tau}_{agg} := \hat{\mu}_{agg}^1 - \hat{\mu}_{agg}^0$. The Supplemental Material gives the asymptotic linearity/normality argument; inference for τ follows by standard linear combination (difference) of the two arm-specific influence function representations.

3.3 Federated Algorithm

The operational version of FACE-HD is a summary-exchange algorithm. Each site keeps its individual-level records locally, while the target and source sites pass only fitted nuisance parameters, empirical moments, sample sizes, and variance summaries, consistent with the federated-learning principle of replacing record pooling by structured communication (McMahan et al. 2017, Kairouz et al. 2021, Han et al. 2025). The default implementation in the main manuscript is a two-round algorithm with two-level cross-fitting: the two communication rounds provide source-specific initialization and calibration, while the two-level split separates the outer evaluation fold from the secondary calibration folds. The same steps are run separately for the treated and control arms; below we describe the treated arm μ^1 , with the control arm obtained by replacing $I(A = 1)$ and the corresponding nuisance parameters by their $a = 0$ analogues. A one-round version is useful when an additional communication round is operationally costly and the target-site initial outcome model is expected to be sufficiently aligned with the source-specific calibrated nuisance targets; otherwise, especially under substantial source-target heterogeneity, the two-round version is preferred because it avoids this stronger target-initialization requirement. The fold-level two-round bookkeeping, the one-round alternative, and the linear-outcome specialization are given in [Supplemental Material S1](#) and [Supplemental Material S1.3](#) and [S1.4](#).

A compact version of the theory-aligned two-round algorithm with two-level cross-fitting is given in [Algorithm 1](#). The subsequent paragraphs first spell out the summaries exchanged in a single training/evaluation split and then explain the fold-level implementation used for inference.

Local inputs. At the beginning, the target site holds $\mathcal{D}_t$ and each source site s_j holds $\mathcal{D}_{s_j}$. All sites agree on the working feature map $\phi(\mathbf{X})$, the GLM response function ψ , the fold partition used for two-level cross-fitting, and candidate grids for the nuisance penalties $(\lambda_\gamma, \lambda_\alpha)$ and aggregation penalty λ . The target also fits the target-only nuisance models needed for $\hat{\mu}_{ot}^1$ and its influence-function component $\hat{\varphi}_{ot}$.

Round 1: source initialization. Each source site first fits an initial treated-outcome model on its own treated observations:

$$\hat{\boldsymbol{\alpha}}_{t,s_j}^{1,(0)} \in \arg \min_{\boldsymbol{\alpha}} \left\{ \tilde{E}_{s_j}[I(A = 1)\ell_{GLM}\{Y, \psi(\phi(\mathbf{X}); \boldsymbol{\alpha})\}] + \lambda_\alpha \|\boldsymbol{\alpha}\|_1 \right\}.$$

The source sends only this fitted parameter vector, and not its records, to the target. For

each received $\hat{\alpha}_{t,s_j}^{1,(0)}$, the target evaluates the target-side calibration summary

$$\tilde{E}_t \left[\nabla_{\alpha} \psi \{ \phi(\mathbf{X}); \hat{\alpha}_{t,s_j}^{1,(0)} \} \right],$$

and returns this vector to the corresponding source. If the source also needs an initial site/treatment fit, the target additionally sends the target covariate mean $\tilde{E}_t \{ \phi(\mathbf{X}) \}$. These quantities are low-dimensional or p -dimensional summaries determined by the chosen feature map, rather than individual-level observations.

Round 2: source calibration. After receiving the target summaries, each source solves the two calibrated nuisance problems locally. First, it estimates the merged site/treatment tilting parameter $\hat{\gamma}_{s_j,1}$ using the calibrated exponential-tilting loss in Eq. (9). This step uses the target-side derivative summary from Round 1 and the source site's own treated observations. Second, using the resulting tilting model as the plug-in weight model, the source estimates the calibrated outcome parameter $\hat{\alpha}_{t,s_j}^1$ from Eq. (10). This ordering is the finite-sample implementation of the cross-calibration logic: the γ update targets covariate/score balance against the target, and the α update targets a weighted residual equation at the source.

The source then forms the source-side correction and variance summaries

$$\hat{\delta}_{t,s_j}^1 = \tilde{E}_{s_j} \left[w(\mathbf{X}; \hat{\gamma}_{s_j,1}) \{ Y - \psi(\phi(\mathbf{X}); \hat{\alpha}_{t,s_j}^1) \} \right],$$

and

$$\hat{V}_{s_j} = \frac{1}{N_{s_j}} \sum_{i \in \mathcal{D}_{s_j}} \left\{ w(\mathbf{X}_i; \hat{\gamma}_{s_j,1}) \left(Y_i - \psi(\phi(\mathbf{X}_i); \hat{\alpha}_{t,s_j}^1) \right) - \hat{\delta}_{t,s_j}^1 \right\}^2.$$

It sends back $\hat{\alpha}_{t,s_j}^1$, $\hat{\delta}_{t,s_j}^1$, $\hat{V}_{s_j}$, and N_{s_j} . The site/treatment parameter $\hat{\gamma}_{s_j,1}$ can remain local because the target does not need to reconstruct source-level residuals once the correction and variance summaries have been computed.

Target-side assembly. With the returned source summaries, the target builds each source-assisted estimator by combining its target-side regression average with the source correction:

$$\hat{\mu}_{t,s_j}^1 = \tilde{E}_t \left[\psi \{ \phi(\mathbf{X}); \hat{\alpha}_{t,s_j}^1 \} \right] + \hat{\delta}_{t,s_j}^1.$$

Because all source-assisted estimators use the same target data, the target also computes the target-side variance terms $\hat{V}_{t,s_j}$, the target-only/source covariance terms $\hat{C}_{ot,s_j}$, and the cross-source target covariance terms $\hat{C}_{t,s_j,s_k}$ locally from $\mathcal{D}_t$ and the received outcome

parameters. These pieces, together with $\widehat{V}_{ot}$ and the source-side variances, are exactly the inputs to the aggregation objective in [Eq. \(16\)](#). The target then selects $\widehat{\boldsymbol{\eta}}$, forms $\widehat{\mu}_{\text{agg}}^1$, repeats the same construction for μ^0 , and reports

$$\widehat{\tau}_{\text{agg}} = \widehat{\mu}_{\text{agg}}^1 - \widehat{\mu}_{\text{agg}}^0.$$

Two-level cross-fitted implementation. The description above gives the information-flow skeleton for one training/evaluation split. The estimator analyzed in the theory uses a fold-specific, two-level cross-fitted implementation following the SMMAL-style calibration construction of [Hou et al. \(2025\)](#). For an outer evaluation fold k_1 , all nuisance models and aggregation weights are learned only on the outer-training data $\mathcal{T}_{k_1} = \mathcal{D} \setminus \mathcal{D}_{k_1}$. Inside $\mathcal{T}_{k_1}$, each secondary fold $k_2 \neq k_1$ supplies a calibration fold, while the initial nuisance plug-ins for that calibration fold are trained on $\mathcal{D} \setminus (\mathcal{D}_{k_1} \cup \mathcal{D}_{k_2})$. The calibrated nuisance estimator for evaluating $\mathcal{D}_{k_1}$ is obtained by minimizing the average of these secondary-fold calibrated losses. Following the FACE validation strategy for adaptive aggregation ([Han et al. 2025](#)), the aggregation penalty λ is also selected inside $\mathcal{T}_{k_1}$, and the final weights $\widehat{\boldsymbol{\eta}}^{(-k_1)}$ are applied only to observations in the held-out outer fold. [Supplemental Material S1](#) gives the full fold-level bookkeeping, [Supplemental Material S1.3](#) records the communication-saving one-round version under the stronger initialization-alignment condition described above, and [Supplemental Material S1.4](#) gives the corresponding simplification for linear outcome models.

3.4 Cross-Validation and Tuning Parameters

The framework involves two types of regularization parameters (λ) that must be tuned.

For each nuisance model (parameters $\boldsymbol{\gamma}$ and $\boldsymbol{\alpha}$), the regularization parameters $(\lambda_{\boldsymbol{\gamma}}, \lambda_{\boldsymbol{\alpha}})$ are selected via K_f -fold cross-validation, where K_f is a fixed constant. The criterion for selection is the minimization of the corresponding calibrated loss function ([Eq. \(9\)](#) or [Eq. \(10\)](#)) on the validation folds. This is performed locally at each site during the federated step where the model is estimated.

The regularization parameter λ in the aggregation objective [Eq. \(16\)](#) controls how aggressively FACE-HD borrows from source sites. Small λ allows more borrowing and can reduce variance when source-assisted estimators are transportable, but it also increases exposure to

negative transfer. Large λ shrinks source weights toward zero, moving the estimator toward the target-only anchor and reducing transfer-bias risk at the cost of weaker precision gains.

A fixed grid Λ (e.g. $[10^{-3}, 1]$) is not robust across problem scales because the useful range of λ depends on the magnitudes of the variance components and the site discrepancies. To construct a data-adaptive grid, we identify the *natural scale* λ_{nat} at which the penalty and variance terms are comparable.

For source site s_j , the unpenalized variance contribution is

$$\sigma_j^2 := \frac{\widehat{V}_{t,s_j}}{N_t} + \frac{\widehat{V}_{s_j}}{N_{s_j}},$$

and the penalty scale is $d_j^2 := (\widehat{\mu}_{ot}^1 - \widehat{\mu}_{t,s_j}^1)^2$. Setting $\lambda d_j^2 = \sigma_j^2$ gives the transition point $\lambda_j = \sigma_j^2 / d_j^2$. When d_j^2 is very small (i.e. the source and target estimates nearly agree, suggesting low bias), we floor d_j^2 at σ_j^2 to avoid $\lambda_j \rightarrow \infty$:

$$\lambda_j = \frac{\sigma_j^2}{\max(d_j^2, \sigma_j^2)}.$$

The natural scale is defined as

$$\lambda_{\text{nat}} = \text{median}_{j=1,\dots,K} \lambda_j,$$

and the search grid spans two orders of magnitude on each side:

$$\Lambda = \left\{ \lambda : \lambda \in [10^{-2} \lambda_{\text{nat}}, 10^2 \lambda_{\text{nat}}] \right\},$$

with $|\Lambda|$ evenly spaced on the log scale (e.g. 100 points). The grid is recomputed inside each outer training set and spans the transition from weakly regularized weights to strong shrinkage toward the target-only estimator.

We select λ by the sample-splitting validation idea used in FACE (Han et al. 2025). For an outer evaluation fold k_1 , let $\mathcal{T}_{k_1} = \mathcal{D} \setminus \mathcal{D}_{k_1}$ be the data available for nuisance fitting and weight tuning. Split $\mathcal{T}_{k_1}$ into M inner folds. For inner fold m , let $\mathcal{T}_{k_1,-m}$ be the inner-training data and $\mathcal{I}_m$ the inner-validation data, keeping site proportions approximately fixed.

For each candidate $\lambda \in \Lambda$, all nuisance parameters and variance/discrepancy summaries are computed on the inner-training data $\mathcal{T}_{k_1,-m}$, and the target solves the penalized aggregation problem:

$$\widehat{\boldsymbol{\eta}}^{(-k_1,-m)}(\lambda) = \arg \min_{\boldsymbol{\eta}} \left\{ \widehat{\text{Var}}_{\text{tr},m}(\boldsymbol{\eta}) + \lambda \sum_{j=1}^K |\eta_j| \left(\widehat{\mu}_{ot,m}^{1,\text{tr}} - \widehat{\mu}_{t,s_j,m}^{1,\text{tr}} \right)^2 \right\}.$$

In parallel, each site computes validation summaries on $\mathcal{I}_m$ using nuisance parameters trained on $\mathcal{T}_{k_1, -m}$; no individual-level validation records are shared. The candidate λ is evaluated by plugging the training-set weights into the validation-set penalized criterion:

$$Q_{\text{val},m}(\lambda) = N_{\text{val},m} \widehat{\text{Var}}_{\text{val},m} \left\{ \widehat{\boldsymbol{\eta}}^{(-k_1, -m)}(\lambda) \right\} + \lambda \sum_{j=1}^K \left| \widehat{\eta}_j^{(-k_1, -m)}(\lambda) \right| \left(\widehat{\mu}_{ot,m}^{1,\text{val}} - \widehat{\mu}_{t,s_j,m}^{1,\text{val}} \right)^2, \quad (17)$$

where $N_{\text{val},m}$ is the total validation sample size across sites in $\mathcal{I}_m$ and $\widehat{\text{Var}}_{\text{val},m} \{\eta\}$ is the FACE-HD variance expression in Eq. (16) evaluated with validation-fold variance/covariance summaries and the fixed weights η . The selected aggregation tuning parameter is

$$\widehat{\lambda}^{(-k_1)} = \arg \min_{\lambda \in \Lambda} \frac{1}{M} \sum_{m=1}^M Q_{\text{val},m}(\lambda).$$

As in FACE, the split can be repeated over several inner folds (e.g., five-fold validation), and the selected value minimizes the average validation criterion. As a conservative finite-sample variant, one may choose the largest λ whose average validation criterion is within one estimated standard error, or within a small relative tolerance, of the minimum. After $\widehat{\lambda}^{(-k_1)}$ is selected, the aggregation weights $\widehat{\boldsymbol{\eta}}^{(-k_1)}$ are recomputed on all of $\mathcal{T}_{k_1}$ and then applied only to the outer evaluation fold $\mathcal{D}_{k_1}$.

4 Theory

This section states the main theoretical guarantees in the same order as the method is built: first the conditions under which each target-source estimator is valid, then the aggregation result that protects against negative transfer, and finally the implication for the TATE. Detailed proofs, including empirical-process and selection arguments, are provided in the Supplemental Material. Throughout this section K and K_f are fixed, $N_{\text{all}} \rightarrow \infty$, the site fractions N_t/N_{all} and N_{s_j}/N_{all} stay bounded away from zero, and the covariate dimension p may grow with sample size.

4.1 High-Level Conditions

The first group of conditions is causal and sampling-based. We assume consistency, standard within-site treatment ignorability given $\mathbf{X}$ for the target estimand, and that any source site treated as informative satisfies the corresponding transportability condition after adjustment

for $\mathbf{X}$. The target and source samples are independent, observations are i.i.d. within each site, and the working covariates $\phi(\mathbf{X})$ have sub-Gaussian coordinates with an intercept included. We also assume overlap: the relevant treatment and site/treatment probabilities are bounded away from zero on the target support, and the resulting exponential-tilting weights have uniformly bounded moments of order larger than two.

The second group of conditions concerns high-dimensional nuisance estimation. Let $s_{\alpha,j}$ and $s_{\gamma,j}$ denote the sparsities of the calibrated outcome and site/treatment tilting targets for source s_j , and let $s_{\text{tot},j} = s_{\alpha,j} + s_{\gamma,j}$. The population calibrated losses satisfy restricted strong convexity on the sparse cones induced by these targets, and the fold-specific nuisance estimators are trained on data independent of their evaluation folds. The two-level cross-fitting construction aligns the population limits of the initial plug-in estimators with the final calibrated targets. Under these conditions, the calibrated ℓ_1 estimators obey the usual high-dimensional rates,

$$\|\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}^*\|_2 + \|\hat{\boldsymbol{\gamma}} - \boldsymbol{\gamma}^*\|_2 = O_p\left(\sqrt{s_{\text{tot},j} \log(p)/N_{t,s_j}}\right),$$

with analogous ℓ_1 rates of order $s_{\text{tot},j} \sqrt{\log(p)/N_{t,s_j}}$, where $N_{t,s_j} = N_t + N_{s_j}$. The sparsity scaling required for the first-order theory is $s_{\text{tot},j}^2 \log^2(p)/N_{t,s_j} \rightarrow 0$ for each informative source.

4.2 Pairwise Source-Assisted Estimation

For a fixed source site s_j and treatment arm $a \in \{0, 1\}$, the calibrated source-assisted estimator has the same first-order structure as an orthogonal doubly robust estimator. The calibration losses are designed so that the derivative of the estimating equation with respect to one nuisance component vanishes when the other nuisance component is evaluated at its calibrated target. As a result, the first-order nuisance error cancels and the remainder is second order, of size

$$O_p(\|\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}^*\|_2 \|\hat{\boldsymbol{\gamma}} - \boldsymbol{\gamma}^*\|_2) = o_p(N_{t,s_j}^{-1/2})$$

under the sparsity scaling above.

Theorem 1 (Pairwise doubly robust asymptotic normality). *Under the high-level conditions above, suppose that for source s_j at least one of the two nuisance models is correctly specified for arm a : either the calibrated outcome regression equals the target-site conditional mean,*

or the exponential-tilting weight satisfies the target-source balancing identity for the relevant function class. If $s_{\text{tot},j}^2 \log^2(p)/N_{t,s_j} \rightarrow 0$ and the asymptotic variance V_{t,s_j}^a is positive, then

$$\sqrt{N_{t,s_j}}(\hat{\mu}_{t,s_j}^a - \mu^a) \xrightarrow{d} \mathcal{N}(0, V_{t,s_j}^a).$$

The plug-in variance estimator based on the cross-fitted influence-function components is consistent for V_{t,s_j}^a .

The theorem gives the pairwise protection that FACE-HD needs before aggregation: a source can be useful even if one nuisance model is misspecified, provided the other nuisance model identifies the target mean and the high-dimensional nuisance rates are fast enough. Applying the same statement to $a = 1$ and $a = 0$ yields the pairwise TATE estimator $\hat{\tau}_{t,s_j} = \hat{\mu}_{t,s_j}^1 - \hat{\mu}_{t,s_j}^0$, whose variance is obtained from the difference of the two arm-specific influence functions.

4.3 Adaptive Aggregation and Source Selection

The aggregation step decides how much each source-assisted estimator should contribute after accounting for both variance and possible bias. For arm a , define the informative source set as $\mathcal{S}^{a,*} = \{j : \text{plim } \hat{\mu}_{t,s_j}^a = \mu^a\}$. Let $\eta^{a,*}$ be the oracle minimizer of the population asymptotic variance over weights supported only on $\mathcal{S}^{a,*}$, with the target-only estimator included through the weight $1 - \sum_j \eta_j$.

The aggregation theory assumes that the population quadratic variance is locally strictly convex on the oracle subspace, that non-informative sources are separated from the target by a beta-min condition, and that the aggregation penalty satisfies $\lambda \rightarrow \infty$ while $\lambda/\sqrt{N_{\text{all}}} \rightarrow 0$. Under fixed separation, the penalty dominates the discrepancy term for biased sources and sets their weights to zero with probability tending to one; for informative sources, the penalty is asymptotically negligible at the $N_{\text{all}}^{-1/2}$ scale.

Theorem 2 (FACE-HD aggregation). *Suppose the pairwise conditions above hold for every informative source, the aggregation regularity conditions hold, and $\max_{j \in \mathcal{S}^{a,*}} s_{\text{tot},j}^2 \log^2(p)/N_{t,s_j} \rightarrow 0$. Then the adaptive aggregation weights satisfy $\|\hat{\boldsymbol{\eta}}^a - \boldsymbol{\eta}^{a,*}\|_2 = O_p(N_{\text{all}}^{-1/2})$ on the oracle subspace and $\hat{\eta}_j^a = 0$ for all $j \notin \mathcal{S}^{a,*}$ with probability tending to one. Consequently,*

$$\sqrt{N_{\text{all}}}(\hat{\mu}_{\text{agg}}^a - \mu^a) \xrightarrow{d} \mathcal{N}\{0, L_a^*(\eta^{a,*})\},$$

where $L_a^*(\eta^{a,*})$ is the oracle minimum asymptotic variance. The estimated aggregation variance $\hat{L}_a(\hat{\eta}^a)$ is consistent for $L_a^*(\eta^{a,*})$.

This result is the high-dimensional analogue of the oracle property used in FACE. The data-driven procedure behaves, to first order, as if the set of transportable sources were known in advance and the optimal variance-minimizing weights were computed on that oracle set. If some biased sources are local alternatives whose bias is $o(N_{\text{all}}^{-1/2})$, exact selection may fail, but their contribution is still negligible for first-order inference.

4.4 Inference for the TATE and Efficiency

The final estimand is the TATE $\tau = \mu^1 - \mu^0$. Applying the aggregation theorem to the treated and control arms and taking the difference gives

$$\sqrt{N_{\text{all}}}(\hat{\tau}_{\text{agg}} - \tau) \xrightarrow{d} \mathcal{N}(0, V_{\tau}^*),$$

where V_{τ}^* is the variance of the difference between the treated- and control-arm oracle influence functions. The corresponding Wald interval is

$$\hat{\tau}_{\text{agg}} \pm z_{1-\alpha/2} \sqrt{\hat{V}_{\tau}/N_{\text{all}}},$$

with $\hat{V}_{\tau}$ computed from the same cross-fitted influence-function representation used for the point estimator.

Because the target-only estimator corresponds to the feasible aggregation choice $\eta = 0$, the oracle aggregation variance is no larger than the target-only asymptotic variance. It is strictly smaller whenever at least one informative source provides a direction in the aggregation objective that lowers the population variance. Thus FACE-HD preserves target-only validity while allowing precision gains from source sites that are sufficiently transportable to the target population.

5 Simulation Study

We evaluate FACE-HD on the negative-transfer data-generating process of [Han et al. \(2025\)](#) (Section 5.1), adapted to the high-dimensional, sparse, federated regime targeted here. Each replicate has one target site ($R = t$) with N_t observations and K source sites ($R = s_1, \dots, s_K$)

with sample sizes $N_{s_1}, \dots, N_{s_K}$ (total N_{all}); the main line uses $K = 2$ with balanced site sizes. Cross-site heterogeneity enters only through a site-specific covariate law (covariate shift) and the site-specific treatment effect defined below; site membership is fixed by block rather than drawn from a site-assignment model. We report bias, root mean square error (RMSE), and 95% confidence-interval coverage for the target estimand.

Covariates. Each of the p coordinates of $\mathbf{X} = (X_1, \dots, X_p)$ is drawn independently from a skewed-normal law $\text{SN}(\kappa, 1, \nu)$ with shared location $\kappa = 0.125$ and unit scale, using the representation

$$X_j = \kappa + \delta |Z_1| + \sqrt{1 - \delta^2} Z_2, \quad Z_1, Z_2 \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1), \quad \delta = \nu / \sqrt{1 + \nu^2}.$$

The target site uses $\nu = 0$, so its covariates are symmetric, $X_j \sim \mathcal{N}(\kappa, 1)$; source site s_j uses skewness ν_{s_j} equally spaced over $[\nu_{\max}/K, \nu_{\max}]$ with $\nu_{\max} = 0.2$, keeping every source bounded away from the target so that the calibration model faces varying covariate shift. The design is sparse: only the first $d = \min(4, p)$ coordinates are active and the remaining $p - d$ are pure noise, which scales the low-dimensional benchmark of [Han et al. \(2025\)](#) to high dimension.

Propensity and outcome models. Let $\psi(u) = \{1 + \exp(-u)\}^{-1}$ and let $\mathbf{X}^{\circ 2}$ denote the element-wise square of $\mathbf{X}$. Treatment is assigned by $\pi(\mathbf{X}) = \psi\{\mathbf{X}^T \boldsymbol{\alpha}_1 + (\mathbf{X}^{\circ 2})^T \boldsymbol{\alpha}_2\}$ and $A \sim \text{Bernoulli}(\pi(\mathbf{X}))$, with the generated propensity clipped to $[0.1, 0.9]$ to enforce positivity; over the d active coordinates $\boldsymbol{\alpha}_1$ is equally spaced from 0.5 to -0.5 and $\boldsymbol{\alpha}_2 = (-0.5, 0, \dots, 0)$, with zeros elsewhere. For a unit in site $r \in \{t, s_1, \dots, s_K\}$, both potential-outcome arms share the covariate coefficients $\boldsymbol{\beta}_{\text{lin}} = \boldsymbol{\beta}_{\text{sq}}$, so the treatment effect is carried entirely by a site-specific constant Δ_r (no within-site effect modification). With $g(\mathbf{X}) = (\mathbf{X} - \kappa \mathbf{1}_p)^T \boldsymbol{\beta}_{\text{lin}} + (\mathbf{X}^{\circ 2})^T \boldsymbol{\beta}_{\text{sq}}$ denoting the covariate term, the arm- a linear predictor for a unit in site r is

$$g(\mathbf{X}) + \Delta_r I(a = 1), \quad a \in \{0, 1\}.$$

The d active coefficient entries of $\boldsymbol{\beta}_{\text{lin}} = \boldsymbol{\beta}_{\text{sq}}$ are equally spaced from 0.4 to 1.2. In the main line (matching the EHR application) outcomes are binary, $Y(a) \sim \text{Bernoulli}(\psi\{g(\mathbf{X}) + \Delta_r I(a = 1)\})$; an optional continuous variant sets $Y(a) = g(\mathbf{X}) + \Delta_r I(a = 1) + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ and $\sigma = 2\sqrt{5}$. Consistency gives $Y = AY(1) + (1 - A)Y(0)$.

Negative transfer. The target effect is fixed at $\Delta_t = 3.0$. The first n_{dev} source sites are made non-informative by setting $\Delta_{s_j} = \Delta_t + \rho$, while the remaining sources stay informative at $\Delta_{s_j} = \Delta_t$; $n_{\text{dev}} = 0$ recovers the all-informative case (deviation level 1 of Han et al. 2025). A deviated source carries an effect different from the target, so its source-assisted estimator $\hat{\mu}_{t,s_j}^a$ is biased and the adaptive aggregation must shrink its weight—precisely the negative transfer that FACE-HD is designed to control. Table 1 lists the deviation gradient we run for $K = 2$, an analogue of Table 1 of Han et al. (2025) over the operating range exercised here.

Table 1: Source-effect deviation gradient ($K = 2$, target effect $\Delta_t = 3.0$). Level 1 is all-informative; higher levels increase both the number of deviated sources n_{dev} and the additive deviation ρ .

Level	n_{dev}	ρ	Source effects ($\Delta_{s_1}, \Delta_{s_2}$)
1	0	0	(3.0, 3.0)
2	1	1	(4.0, 3.0)
3	2	2	(5.0, 5.0)
4	2	3	(6.0, 6.0)

Misspecification configurations. Misspecification corresponds to omitting the quadratic block $\mathbf{X}^{\circ 2}$. In C(1) both nuisance models are correct: the propensity design uses $[\mathbf{X}, \mathbf{X}^{\circ 2}]$ and the outcome design uses $[\mathbf{X} - \kappa \mathbf{1}_p, \mathbf{X}^{\circ 2}]$. In C(2) the outcome model is misspecified, using $\mathbf{X}$ only while the propensity design keeps $[\mathbf{X}, \mathbf{X}^{\circ 2}]$. In C(3) the propensity model is misspecified, using $\mathbf{X}$ only while the outcome design keeps $[\mathbf{X} - \kappa \mathbf{1}_p, \mathbf{X}^{\circ 2}]$. In C(4) both drop the quadratic block and use $\mathbf{X}$ only; this fully-misspecified boundary case is defined here but not exercised by the reported runs.

Estimand and operating range. The estimand is the target treated-arm mean $\mu^1 = \mathbb{E}_t[Y(1)]$, with the TATE following as $\tau = \mu^1 - \mu^0$; the target means $\mathbb{E}_t[Y(a)]$ are evaluated by Monte Carlo over the target covariate law $\mathbf{X} \sim \mathcal{N}(\kappa \mathbf{1}_p, \mathbf{I}_p)$. The site effect Δ_r enters the linear predictor, so for continuous outcomes it is additive and $\tau = \Delta_t = 3.0$, whereas for the binary main line $\Delta_t = 3.0$ is the conditional log-odds effect and the target risk difference $\tau = \mathbb{E}_t[\psi(g(\mathbf{X}) + \Delta_t)] - \mathbb{E}_t[\psi(g(\mathbf{X}))]$ is a probability-scale contrast in $(0, 1)$

rather than 3.0, likewise obtained by Monte Carlo. The main grid fixes $K = 2$ and varies $N_{\text{all}} \in \{3000, 6000\}$ (per-site $N_{s_j} \in \{1000, 2000\}$), $p \in \{50, 100\}$, cross-fitting folds $K_f \in \{5, 10\}$, and configurations C(1)–C(3), with binary outcomes over 200 Monte Carlo replicates. We compare FACE-HD with target-only estimation, sample-size weighting (SS), inverse-variance weighting (IVW), and an exponentially-tilted AIPW baseline (Tilted-AIPW) that aggregates density-ratio-weighted site-specific AIPW estimators by SS. The SS, IVW, and Tilted-AIPW baselines report standard errors from a multiplier (wild) bootstrap, while FACE-HD and the target-only estimator use the closed-form cross-fitted influence-function variance.

6 Discussion

FACE-HD is designed for the setting in which source sites can help estimate a target-site causal effect but cannot share individual-level records. The method combines federated calibration, doubly robust estimation, and adaptive aggregation in a single workflow. The calibration losses are chosen to enforce orthogonality equations, the aggregation objective accounts for covariance induced by shared target data, and the penalty discourages borrowing from sources whose source-assisted estimates disagree with the target-only estimate.

The theory treats a fixed number of source sites and relies on standard high-dimensional nuisance rates, overlap/moment conditions, and a separation condition for exact source selection. These assumptions clarify the intended operating regime: FACE-HD can gain efficiency from transportable sources, remains protected by doubly robust identification for each pairwise estimator, and falls back toward target-only behavior when source discrepancies are large. Extensions to a growing number of sources, weaker local alternatives, and more flexible nuisance learners require additional concentration and empirical-process arguments.

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Algorithm 1 Two-Round Algorithm with Two-Level Cross-Fitting

Require: Target data $\mathcal{D}_t$, source data $\{\mathcal{D}_{s_j}\}_{j=1}^K$, feature map ϕ , nuisance penalty grids, aggregation penalty grid, and K_f two-level cross-fitting folds.

Ensure: Aggregated TATE estimate $\hat{\tau}_{\text{agg}}$ and its variance estimate.

- 1: All sites keep individual-level records local and agree on ϕ , fold indices, and tuning grids.
 - 2: **for** each outer evaluation fold $k_1 = 1, \dots, K_f$ **do**
 - 3: Keep $\mathcal{D}_{k_1}$ held out. Learn nuisance models, tune λ , and estimate aggregation weights only inside $\mathcal{T}_{k_1} = \mathcal{D} \setminus \mathcal{D}_{k_1}$.
 - 4: Target fits the target-only nuisance models using the target propensity model in Eq. (5), computes the target-only IF in Eq. (13), and keeps target records local for later covariance calculations.
 - 5: **for** each treatment arm $a \in \{0, 1\}$ **do**
 - 6: **for** each source site s_j in parallel **do**
 - 7: **Round 1.** For every secondary fold $k_2 \neq k_1$, source s_j fits an initial outcome model on $\mathcal{D}_{s_j} \setminus (\mathcal{D}_{s_j, k_1} \cup \mathcal{D}_{s_j, k_2})$ using the arm-specific version of Eq. (11), and sends only the fitted parameter vector to the target.
 - 8: Target evaluates the fold-specific derivative and covariate summaries entering Eqs. (9) and (12) and returns those summaries to source s_j .
 - 9: **Round 2.** Source s_j computes the initial site/treatment model using Eq. (12), solves the fold-summed calibrated losses in Eqs. (9) and (10), and returns $\hat{\alpha}_{t, s_j}^{a, (-k_1)}$, the source correction and variance based on Eq. (15), and N_{s_j} .
 - 10: **end for**
 - 11: Target forms fold-specific source-assisted estimates using the arm-specific version of Eq. (4), estimates the variance/covariance components in Eqs. (13) to (15), selects aggregation weights by Eq. (16) inside $\mathcal{T}_{k_1}$, and applies those weights to $\mathcal{D}_{k_1}$ to compute $\hat{\mu}_{\text{agg}, k_1}^a$.
 - 12: **end for**
 - 13: Compute $\hat{\tau}_{\text{agg}, k_1} = \hat{\mu}_{\text{agg}, k_1}^1 - \hat{\mu}_{\text{agg}, k_1}^0$ on the held-out fold.
 - 14: **end for**
 - 15: Average fold-specific estimates and report $\hat{\tau}_{\text{agg}} = K_f^{-1} \sum_{k_1=1}^{K_f} \hat{\tau}_{\text{agg}, k_1}$.
-

Supplemental Material

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S1 Fold-Level Details for Two-Level Cross-Fitting

[Algorithm 1](#) in the main Method section already gives the recommended two-round FACE-HD estimator with two-level cross-fitting. This Supplemental Material section is not a separate robustness add-on; its purpose is to spell out the fold-level bookkeeping compressed in [algorithm 1](#) and to record the corresponding communication-saving one-round variant.

The two-level construction adapts the cross-fitting strategy of [Hou et al. \(2025\)](#). It uses a nested sample-splitting scheme to ensure that the data used to train an initial nuisance plug-in for an observation is disjoint from the data used for its calibrated loss contribution. As in the SMMAL calibration construction, calibration is carried out over all secondary folds rather than over one held-out fold: for each outer fold k_1 , every $k_2 \neq k_1$ supplies one calibration-fold loss, the corresponding initial fits use the out-of-two-folds data $\mathcal{D} \setminus (\mathcal{D}_{k_1} \cup \mathcal{D}_{k_2})$, and a single calibrated nuisance estimator is obtained by minimizing the average of these fold-specific losses. This separation mitigates overfitting bias in the sequential nuisance updates and supports the Neyman-orthogonal expansion used in the theory.

S1.1 Calibrated Loss Functions

The core of the two-level cross-fitting method is the use of specially designed "calibrated" loss functions for refining the nuisance models. These losses are constructed to make the final TATE estimator insensitive to small estimation errors in the nuisance parameters. Following the principles in ([Smucler et al. 2019](#)) and ([Tan 2020](#)), we introduce a calibration step after obtaining robust initial estimates.

The truncation function, GLM notation, and calibrated losses are defined earlier in [Eqs. \(9\)](#) and [\(10\)](#). The same loss definitions are used foldwise in the two-level cross-fitting construction below. For a fixed outer fold k_1 , let $\ell_{\gamma, \text{cal}}^{(k_2)}$ and $\ell_{\alpha, \text{cal}}^{(k_2)}$ denote the fold-specific versions of [Eqs. \(9\)](#) and [\(10\)](#), with $\tilde{E}_t$ and $\tilde{E}_{s_j}$ replaced by $\tilde{E}_{t, k_2}$ and $\tilde{E}_{s_j, k_2}$ and with plug-in initial estimators trained on $\mathcal{D} \setminus (\mathcal{D}_{k_1} \cup \mathcal{D}_{k_2})$. The calibrated nuisance estimators for evaluating fold k_1 are

$$\hat{\gamma}_{s_j, 1}^{(-k_1)} = \arg \min_{\gamma} \frac{1}{K_f - 1} \sum_{k_2 \neq k_1} \ell_{\gamma, \text{cal}}^{(k_2)} \left(\gamma; \hat{\alpha}_{\text{init}, (k_1, k_2)}^1 \right),$$

and

$$\hat{\alpha}_{t,s_j}^{1,(-k_1)} = \arg \min_{\alpha} \frac{1}{K_f - 1} \sum_{k_2 \neq k_1} \ell_{\alpha, \text{cal}}^{(k_2)}(\alpha; \hat{\gamma}_{\text{init},(k_1,k_2)}).$$

Thus the secondary folds enter as loss contributions in one calibrated optimization problem, rather than as separate calibrated parameter estimates to be averaged. In the two-level algorithm, the calibrated γ -loss uses $\hat{\alpha}_{\text{init}}$ while the calibrated α -loss uses $\hat{\gamma}_{\text{init}}$. In the Supplemental Material, we assume these initial estimators converge to the same population nuisance targets as the final calibrated estimators, so this plug-in order does not change the first-order asymptotic results.

S1.2 Expanded Two-Round Algorithm with Two-Level Cross-Fitting

This is the expanded fold-level version of [algorithm 1](#). It makes explicit the secondary calibration folds, the training samples used for the initial plug-ins, and the summaries exchanged in each communication round.

- 1: **Initialization:** At each site $R \in \{t, s_1, \dots, s_K\}$, randomly partition the local data $\mathcal{D}_R$ into K_f disjoint, equally sized folds: $\{\mathcal{D}_{R,k}\}_{k=1}^{K_f}$. The following steps describe the computation for a single main estimation fold $k_1 \in \{1, \dots, K_f\}$. The SMMAL-style calibration layer constructs one fold-specific initial plug-in for every secondary fold $k_2 \neq k_1$.

- 2: **Round 1: Initial Nuisance Models**

1. **Source sites s_j compute and send:** For each secondary fold $k_2 \in \{1, \dots, K_f\} \setminus \{k_1\}$, source site s_j computes:

- An initial outcome model parameter $\hat{\alpha}_{\text{init},(k_1,k_2)}^1$ by minimizing the standard GLM loss from [Eq. \(11\)](#) using all its local data **except** folds k_1 and k_2 , i.e., on data $\mathcal{D}_{s_j} \setminus (\mathcal{D}_{s_j,k_1} \cup \mathcal{D}_{s_j,k_2})$.

Note: The initial site/treatment model $\hat{\gamma}_{\text{init}}$ is **not** computed here as it requires target site information.

It sends the set of $K_f - 1$ initial outcome models $\{\hat{\alpha}_{\text{init},(k_1,k_2)}^1\}_{k_2 \neq k_1}$ to the target site.

2. **Target site t receives and sends back:** For each source s_j and each secondary fold $k_2 \neq k_1$, the target computes two summary statistics, separating the

calibration fold from the training folds:

- For the *calibrated* site/treatment model, a calibration-fold summary $\tilde{E}_{t,k_2}[\nabla_{\alpha}\psi(\phi(\mathbf{X}); \hat{\alpha}_{\text{init},(k_1,k_2)}^1)]$.
- For the *initial* site/treatment model, a training-fold summary $\tilde{E}_{t,-(k_1,k_2)}[\phi(\mathbf{X})]$ computed on $\mathcal{D}_t \setminus (\mathcal{D}_{t,k_1} \cup \mathcal{D}_{t,k_2})$.

It sends the corresponding set of $K_f - 1$ pairs of summaries back to source site s_j .

3: Round 2: Calibrated Nuisance Models and Estimation

1. Source sites s_j receive target info and compute:

- For each secondary fold $k_2 \neq k_1$, the source computes the fold-specific initial site/treatment model:
 - **First**, compute the initial site/treatment model $\hat{\gamma}_{\text{init},(k_1,k_2)}$ by minimizing its standard loss (Eq. (12)) using the received training-fold summary $\tilde{E}_{t,-(k_1,k_2)}[\phi(\mathbf{X})]$ and its own data from $\mathcal{D}_{s_j} \setminus (\mathcal{D}_{s_j,k_1} \cup \mathcal{D}_{s_j,k_2})$.
- After all fold-specific initial plug-ins are available, it solves the fold-summed calibrated objectives over the local calibration folds $\{\mathcal{D}_{s_j,k_2} : k_2 \neq k_1\}$:

$$\hat{\gamma}_{s_j,(k_1)} = \arg \min_{\gamma} \frac{1}{K_f - 1} \sum_{k_2 \neq k_1} \ell_{\gamma,\text{cal}}^{(k_2)}(\gamma; \hat{\alpha}_{\text{init},(k_1,k_2)}^1),$$

$$\hat{\alpha}_{t,s_j,(k_1)}^1 = \arg \min_{\alpha} \frac{1}{K_f - 1} \sum_{k_2 \neq k_1} \ell_{\alpha,\text{cal}}^{(k_2)}(\alpha; \hat{\gamma}_{\text{init},(k_1,k_2)}).$$

- Using these fold-summed calibrated parameters, it computes the correction term $\hat{\delta}_{t,s_j,(k_1)}^1$ and source variance $\hat{V}_{s_j,(k_1)}$ on its own main estimation data from fold $\mathcal{D}_{s_j,k_1}$.

2. **Source sites s_j send back:** The fold-summed calibrated parameters $\hat{\alpha}_{t,s_j,(k_1)}^1$, the correction term $\hat{\delta}_{t,s_j,(k_1)}^1$, and variance component $\hat{V}_{s_j,(k_1)}$.

3. **Additional source summaries for aggregation-penalty validation:** If the aggregation penalty λ is selected by the inner validation rule rather than fixed in advance, each source also prepares split-specific summaries for every inner validation fold $k_2 \neq k_1$. Let

$$\mathcal{T}_{k_1,-k_2} := \mathcal{D} \setminus (\mathcal{D}_{k_1} \cup \mathcal{D}_{k_2})$$

denote the inner-training data. Using nuisance estimates trained only on $\mathcal{T}_{k_1,-k_2}$, source s_j evaluates weighted residual summaries on both the inner-training portion and the held-out inner-validation fold:

$$\hat{\delta}_{t,s_j,(k_1,k_2)}^{1,\text{tr}}, \quad \hat{V}_{s_j,(k_1,k_2)}^{\text{tr}}, \quad N_{s_j,(k_1,k_2)}^{\text{tr}}$$

and

$$\hat{\delta}_{t,s_j,(k_1,k_2)}^{1,\text{val}}, \quad \hat{V}_{s_j,(k_1,k_2)}^{\text{val}}, \quad N_{s_j,(k_1,k_2)}^{\text{val}}.$$

It sends these fold-level summaries, together with the corresponding inner-training outcome parameter $\hat{\alpha}_{t,s_j}^{1,(-k_1,-k_2)}$, to the target. The site/treatment parameter $\hat{\gamma}_{s_j,1}^{(-k_1,-k_2)}$ is used locally to form the weighted residual summaries and need not be sent unless one wants the target to reconstruct source-side influence values explicitly. No individual-level validation records are shared.

4. **Target site t computes fold-specific estimate:** Using the received information and its own data, the target computes:

- *Inner-fold tuning of the aggregation penalty:* Within $\mathcal{T}_{k_1} = \mathcal{D} \setminus \mathcal{D}_{k_1}$, apply the FACE-style validation rule: for each secondary fold $k_2 \neq k_1$, solve [Eq. \(16\)](#) on training summaries computed from $\mathcal{T}_{k_1} \setminus \mathcal{D}_{k_2}$, using the source-side training summaries $\{\hat{\delta}_{t,s_j,(k_1,k_2)}^{1,\text{tr}}, \hat{V}_{s_j,(k_1,k_2)}^{\text{tr}}, N_{s_j,(k_1,k_2)}^{\text{tr}}\}_{j=1}^K$ and target-side training summaries computed locally. Then plug the resulting weights into the validation penalized criterion computed on $\mathcal{D}_{k_2}$, using the source-side validation summaries $\{\hat{\delta}_{t,s_j,(k_1,k_2)}^{1,\text{val}}, \hat{V}_{s_j,(k_1,k_2)}^{\text{val}}, N_{s_j,(k_1,k_2)}^{\text{val}}\}_{j=1}^K$ and target-side validation summaries computed locally. Average these validation criteria over k_2 to select $\hat{\lambda}^{(-k_1)}$.
- *Weight optimization:* Recompute the variance-covariance components on the full outer-training set $\mathcal{T}_{k_1}$ using out-of-inner-fold nuisance estimates, and solve [Eq. \(16\)](#) with $\hat{\lambda}^{(-k_1)}$ to get $\hat{\eta}^{(-k_1)}$.
- *Estimation:* Using the fold-summed calibrated nuisances $\hat{\vartheta}^{(-k_1)}$ and the weights $\hat{\eta}^{(-k_1)}$, compute the fold-specific aggregated estimate $\hat{\mu}_{\text{agg},k_1}^1$ and per-sample IF values on the evaluation fold $\mathcal{D}_{k_1}$.

4: **Final Aggregation:** The entire process is repeated for each main fold $k_1 = 1, \dots, K_f$.

The final point estimate is the average: $\hat{\mu}_{\text{agg}}^1 = \frac{1}{K_f} \sum_{k_1=1}^{K_f} \hat{\mu}_{\text{agg},k_1}^1$.

To ensure that the aggregation weights $\hat{\eta}^{(-k_1)}$ are independent of the evaluation data $\mathcal{D}_{k_1}$, both the tuning of λ and the final weight optimization are carried out inside the training set $\mathcal{T}_{k_1} = \mathcal{D} \setminus \mathcal{D}_{k_1}$. Specifically, set $M = K_f - 1$ inner folds that coincide with the secondary folds $\{k_2 : k_2 \neq k_1\}$. For each inner validation fold k_2 :

1. Use inner-training summaries from $\mathcal{T}_{k_1} \setminus \mathcal{D}_{k_2}$ to solve the penalized aggregation problem over the candidate grid Λ , producing $\hat{\eta}^{(-k_1,-k_2)}(\lambda)$ for each λ . On the source side these summaries consist of the fold-level correction terms $\hat{\delta}_{t,s_j,(k_1,k_2)}^{1,\text{tr}}$,

source variance components $\hat{V}_{s_j, (k_1, k_2)}^{\text{tr}}$, source sample sizes $N_{s_j, (k_1, k_2)}^{\text{tr}}$, and the inner-training outcome parameters $\hat{\alpha}_{t, s_j}^{1, (-k_1, -k_2)}$ needed by the target to compute target-side means and covariances.

2. Use the same nuisance estimates, trained without $\mathcal{D}_{k_1}$ and $\mathcal{D}_{k_2}$, to compute validation summaries on $\mathcal{D}_{k_2}$. On the source side this gives $\hat{\delta}_{t, s_j, (k_1, k_2)}^{1, \text{val}}$, $\hat{V}_{s_j, (k_1, k_2)}^{\text{val}}$, and $N_{s_j, (k_1, k_2)}^{\text{val}}$; on the target side, the target computes the corresponding target-only IF, target-side regression components, and shared-target covariance terms locally.
3. Evaluate $Q_{\text{val}, k_2}(\lambda)$ by plugging $\hat{\eta}^{(-k_1, -k_2)}(\lambda)$ into the validation penalized criterion in Eq. (17).

The validation criteria are averaged across the $K_f - 1$ inner folds to select $\hat{\lambda}^{(-k_1)}$. The final outer-training components are then averaged as

$$\hat{V}_{\cdot}^{(-k_1)} := \frac{1}{K_f - 1} \sum_{k_2 \neq k_1} \hat{V}_{\cdot, (k_1, k_2)},$$

and similarly for $\hat{C}_{ot}^{(-k_1)}$ and $\hat{C}_{\text{cross}}^{(-k_1)}$. The weights $\hat{\eta}^{(-k_1)}$ are obtained by solving Eq. (16) with these components and $\hat{\lambda}^{(-k_1)}$. Because the entire tuning and weight-estimation procedure uses only $\mathcal{T}_{k_1}$, $\hat{\eta}^{(-k_1)}$ is independent of the evaluation fold $\mathcal{D}_{k_1}$.

The point estimate and variance are derived from the *same* set of per-observation uncentered influence functions. Under the standard multi-site sampling regime (independent samples per site with fixed sampling fractions), the variance of the overall mean depends on *within-site* variation only. For each observation i in evaluation fold $\mathcal{D}_{k(i)}$, define

$$\hat{\Phi}_i := \begin{cases} \frac{N_{\text{all}}}{N_t} \left[(1 - \sum_j \hat{\eta}_j^{(-k(i))}) \tilde{Y}_{ot, i} + \sum_j \hat{\eta}_j^{(-k(i))} \psi(\phi(\mathbf{X}_i); \hat{\alpha}_{t, s_j}^{1, (-k(i))}) \right], & i \in \text{target}, \\ \frac{N_{\text{all}}}{N_{s_j}} \hat{\eta}_j^{(-k(i))} w(\mathbf{X}_i; \hat{\gamma}_{s_j, 1}^{(-k(i))}) (Y_i - \psi(\phi(\mathbf{X}_i); \hat{\alpha}_{t, s_j}^{1, (-k(i))})), & i \in \text{source } s_j, \end{cases}$$

Here $N_{\text{all}} = N_t + \sum_j N_{s_j}$, $\tilde{Y}_{ot, i}$ is the *uncentered* AIPW pseudo-outcome for the target-only estimator,

$$\tilde{Y}_{ot, i} := \frac{A_i Y_i}{\hat{\pi}_{ot}(\mathbf{X}_i)} - \left(\frac{A_i}{\hat{\pi}_{ot}(\mathbf{X}_i)} - 1 \right) \psi(\phi(\mathbf{X}_i); \hat{\alpha}_{ot}^{1, (-k(i))}),$$

so that the centered IF defined in Eq. (13) satisfies $\hat{\varphi}_{ot, i} = \tilde{Y}_{ot, i} - \hat{\mu}_{ot, k(i)}^1$ (the fold-specific target-only point estimate). Likewise, the centered target-site component Eq. (14) satisfies $\hat{\zeta}_{t, s_j, i} = \psi(\phi(\mathbf{X}_i); \hat{\alpha}_{t, s_j}^{1, (-k(i))}) - \tilde{E}_{t, k(i)}[\psi(\cdot)]$. All nuisance parameters $\hat{\alpha}_{t, s_j}^{1, (-k)}$, $\hat{\gamma}_{s_j, 1}^{(-k)}$, and $\hat{\pi}_{ot}^{(-k)}$ are trained on data excluding the evaluation fold $\mathcal{D}_k$ (with the fold-summed calibration over secondary folds as described above). Note that the source

contribution is written in uncentered form; subtracting the fold-specific correction term $\widehat{\delta}_{t,s_j}^{(-k)} = \widetilde{E}_{s_j,k}[w(Y - \psi)]$ yields the centered influence component $\widehat{\xi}_{t,s_j,i}$ defined in Eq. (15). The point estimate and variance are then derived from the *same* set $\{\widehat{\Phi}_i\}_{i=1}^{N_{\text{all}}}$:

$$\begin{aligned}\widehat{\mu}_{\text{agg}}^1 &= \bar{\Phi} := \frac{1}{N_{\text{all}}} \sum_{i=1}^{N_{\text{all}}} \widehat{\Phi}_i, \\ \widehat{\text{Var}}(\widehat{\mu}_{\text{agg}}^1) &= \frac{1}{N_{\text{all}}^2} \sum_{g \in \{t, s_1, \dots, s_K\}} \sum_{i \in \mathcal{D}_g} (\widehat{\Phi}_i - \bar{\Phi}_g)^2, \quad \bar{\Phi}_g := \frac{1}{N_g} \sum_{i \in \mathcal{D}_g} \widehat{\Phi}_i.\end{aligned}$$

This within-site centered form is the natural plug-in estimator for the variance of a mean under independent multi-sample (site-stratified) sampling; global centering would add an ANOVA between-site term that is not sampling variability in this regime. With balanced folds, $\bar{\Phi}$ coincides with the fold-average $(1/K_f) \sum_k \widehat{\mu}_{\text{agg},k}^1$.

S1.3 One-Round Algorithm with Two-Level Cross-Fitting

This algorithm adapts the two-level split to the more communication-efficient one-round protocol. The target site bootstraps the entire cross-fitting process by computing and sending all necessary initial models and summary statistics in a single round.

1: **Initialization:** All sites partition their local data into K_f folds: $\{\mathcal{D}_{R,k}\}_{k=1}^{K_f}$. The following describes the computation for a single main estimation fold k_1 .

2: **Round 1: Single Communication Round**

1. **Target site t computes and sends:**

- For each secondary fold k_2 where $k_2 \neq k_1$:
 - Computes a preliminary target-only outcome model $\widehat{\alpha}_{\text{init},(k_1,k_2)}^1$ on its local data from all folds **except** k_1 and k_2 .
 - Computes the corresponding summary statistic for the *calibrated* model:

$$\widetilde{E}_{t,k_2}[\nabla_{\alpha} \psi(\phi(\mathbf{X}); \widehat{\alpha}_{\text{init},(k_1,k_2)}^1)]$$

on its local fold $\mathcal{D}_{t,k_2}$.

- Computes the summary statistic for the *initial* model: $\widetilde{E}_{t,-(k_1,k_2)}[\phi(\mathbf{X})]$ on $\mathcal{D}_t \setminus (\mathcal{D}_{t,k_1} \cup \mathcal{D}_{t,k_2})$.
- It sends the full set of $(K_f - 1)$ initial models and their corresponding $(K_f - 1)$ pairs of summary statistics (all associated with main fold k_1) to all source sites.

2. **Source sites s_j receive info, compute, and send back:**

- For each secondary fold $k_2 \neq k_1$, the source site computes the fold-specific initial site/treatment model:
 - **First**, compute the initial site/treatment model $\hat{\gamma}_{\text{init},(k_1,k_2)}$ on its local data from $\mathcal{D}_{s_j} \setminus (\mathcal{D}_{s_j,k_1} \cup \mathcal{D}_{s_j,k_2})$ using the received training-fold summary $\tilde{E}_{t,-(k_1,k_2)}[\phi(\mathbf{X})]$.
- It then solves the SMMAL-style fold-summed calibrated objectives over all secondary folds $k_2 \neq k_1$ to get the final nuisance models for the main fold k_1 : $\hat{\alpha}_{t,s_j,(k_1)}^1$ and $\hat{\gamma}_{s_j,(k_1)}$.
- It computes the correction term $\hat{\delta}_{t,s_j,(k_1)}^1$ and variance $\hat{V}_{s_j,(k_1)}$ on its main estimation data from fold $\mathcal{D}_{s_j,k_1}$.
- Each source sends back the fold-summed calibrated parameters $\hat{\alpha}_{t,s_j,(k_1)}^1$ along with $\hat{\delta}_{t,s_j,(k_1)}^1$ and $\hat{V}_{s_j,(k_1)}$.

3. **Target site t receives info and computes fold-specific estimate:**

- Using the received fold-summed calibrated parameters and inner-fold target-only models, the target applies the FACE-style inner validation rule to select $\hat{\lambda}^{(-k_1)}$ without using $\mathcal{D}_{k_1}$.
- It solves the final weight optimization on the outer-training summaries with $\hat{\lambda}^{(-k_1)}$ and computes the fold-specific aggregated estimate $\hat{\mu}_{\text{agg},k_1}^1$ and per-sample IF values on the evaluation fold $\mathcal{D}_{k_1}$.

3: **Final Aggregation:** As before, the process is repeated for each main fold $k_1 = 1, \dots, K_f$. The point estimate is $\hat{\mu}_{\text{agg}}^1 = \frac{1}{K_f} \sum_{k_1} \hat{\mu}_{\text{agg},k_1}^1$, and the variance is computed using the sample-level DML formula with inner M -fold weights, as described in the two-round algorithm above.

S1.4 Linear Outcome One-Round Algorithm with Two-Level Cross-Fitting

When the outcome model is linear, $\psi\{\phi(\mathbf{X}); \alpha\} = \alpha^T \phi(\mathbf{X})$, the one-round algorithm above simplifies because

$$\nabla_{\alpha} \psi\{\phi(\mathbf{X}); \alpha\} = \phi(\mathbf{X})$$

does not depend on an initial outcome parameter. Thus the target does not need to send fold-specific initial outcome models to the sources. It only sends fold-specific covariate means needed for calibration, while the source sites compute the linear calibration and weighted least-squares updates locally. The outer fold k_1 and secondary fold k_2 are still kept separate, so this is the cross-fitted linear specialization of the one-round estimator, not the non-cross-fitted sufficient-statistics shortcut.

For a fixed target-source pair and the treated arm, define the secondary-fold linear calibration loss

$$\ell_{\gamma, \text{lin}}^{(k_1, k_2)}(\gamma) = \tilde{E}_{t, k_2} \{ \phi(\mathbf{X}) \}^T \gamma + \tilde{E}_{s_j, k_2} \left[I(A = 1) \exp \{ -\phi(\mathbf{X})^T \gamma \} \right] + \lambda_\gamma \|\gamma\|_1. \quad (18)$$

This is exactly the GLM calibrated γ -loss with $\psi'(\cdot) = 1$. For the outcome update, each secondary fold uses an initial tilting fit $\hat{\gamma}_{\text{init}, (k_1, k_2)}$ trained on $\mathcal{D}_{s_j} \setminus (\mathcal{D}_{s_j, k_1} \cup \mathcal{D}_{s_j, k_2})$ with the corresponding target training-fold mean $\tilde{E}_{t, -(k_1, k_2)} \{ \phi(\mathbf{X}) \}$. The fold-specific weighted least-squares loss is

$$\ell_{\alpha, \text{lin}}^{(k_1, k_2)}(\alpha) = \tilde{E}_{s_j, k_2} \left[I(A = 1) \exp \{ -\phi(\mathbf{X})^T \hat{\gamma}_{\text{init}, (k_1, k_2)} \} \{ Y - \alpha^T \phi(\mathbf{X}) \}^2 \right] + \lambda_\alpha \|\alpha\|_1. \quad (19)$$

The nuisance estimators used on the held-out outer fold are obtained by minimizing the fold-summed versions of these two losses:

$$\hat{\gamma}_{s_j, 1}^{(-k_1)} \in \arg \min_{\gamma} \frac{1}{K_f - 1} \sum_{k_2 \neq k_1} \ell_{\gamma, \text{lin}}^{(k_1, k_2)}(\gamma), \quad \hat{\alpha}_{t, s_j}^{1, (-k_1)} \in \arg \min_{\alpha} \frac{1}{K_f - 1} \sum_{k_2 \neq k_1} \ell_{\alpha, \text{lin}}^{(k_1, k_2)}(\alpha).$$

1: **Initialization:** All sites use the same outer and secondary folds as in [Supplemental Material S1.3](#). The steps below describe one outer evaluation fold k_1 and the treated arm; the control arm is obtained by replacing $I(A = 1)$ with $I(A = 0)$ and using the corresponding nuisance parameters.

2: **Round 1: Single Communication Round**

1. **Target site t computes and sends:**

- For each secondary fold $k_2 \neq k_1$, the calibration-fold mean $\tilde{E}_{t, k_2} \{ \phi(\mathbf{X}) \}$ entering [Eq. \(18\)](#).
- For each secondary fold $k_2 \neq k_1$, the training-fold mean $\tilde{E}_{t, -(k_1, k_2)} \{ \phi(\mathbf{X}) \}$ used to compute the initial tilting fit for [Eq. \(19\)](#).

2. **Source site s_j receives summaries, computes, and sends back:**

- For each $k_2 \neq k_1$, compute $\hat{\gamma}_{\text{init}, (k_1, k_2)}$ on $\mathcal{D}_{s_j} \setminus (\mathcal{D}_{s_j, k_1} \cup \mathcal{D}_{s_j, k_2})$ using the received training-fold target mean.

- Solve the fold-summed linear calibration and weighted least-squares problems in Eqs. (18) and (19) to obtain $\hat{\gamma}_{s_j,1}^{(-k_1)}$ and $\hat{\alpha}_{t,s_j}^{1,(-k_1)}$.
 - On the held-out source fold $\mathcal{D}_{s_j,k_1}$, compute the source correction $\hat{\delta}_{t,s_j,k_1}^1$ and source-side variance using the cross-fitted nuisance pair.
 - Send $\hat{\alpha}_{t,s_j}^{1,(-k_1)}$, $\hat{\delta}_{t,s_j,k_1}^1$, the source-side variance, and N_{s_j,k_1} to the target.
3. **Target site t receives info and computes fold-specific aggregation:**
- Form $\hat{\mu}_{t,s_j,k_1}^1 = \tilde{E}_{t,k_1} \{ \hat{\alpha}_{t,s_j}^{1,(-k_1)T} \phi(\mathbf{X}) \} + \hat{\delta}_{t,s_j,k_1}^1$.
 - Compute target-side variances and cross-source covariances locally on $\mathcal{D}_{t,k_1}$. If $\hat{\Sigma}_{t,k_1}$ is the covariance matrix of $\phi(\mathbf{X})$ on the target evaluation fold, then the target-side source covariance has the linear form $\hat{\alpha}_{t,s_j}^{1,(-k_1)T} \hat{\Sigma}_{t,k_1} \hat{\alpha}_{t,s_\ell}^{1,(-k_1)}$.
 - Select aggregation weights inside the outer-training sample as in the FACE-style validation rule, apply them to the held-out fold, and repeat across $k_1 = 1, \dots, K_f$.

[Marquis: A few general comments:

1. The main ideas and techniques are ready from existing literatures, so they look correct enough.
2. Please survey all definitions made inside your theory. Decide if you want to disband them or move them to the “Setup and Notation” (should be a separate section).
3. Need to clean up the assumptions. Quite a few seem redundant and obscure. Perhaps organize all assumptions in one place and provide a comprehensive discussion.
4. Reconsider the organization of the document. Even if you applied existing theorems for proving the oracle inequality, it should be useful to state the conclusion in a separate section before the asymptotic normality.
5. Add the theory for selective aggregation.

]

We use the same mathematical notation as the main manuscript throughout this Supplemental Material. The theory is organized in a theorem-first format: setup and notation, assumptions, main limit statements, variance estimation, and then the supporting lemmas. Aggregation is treated in its own section, where the covariance structure, penalized weight estimation, source selection, and final oracle-limit theorem are stated before the proof details.

S2 Asymptotic Normality of the Cross-Fitted Estimator

S2.1 Setup and Notation

This section is organized in the standard theorem-driven order used in high-dimensional inference: we first state the setup and assumptions, then give the main asymptotic-normality result and variance estimator, and finally collect the supporting lemmas used in the proof. Readers interested only in the main pairwise limit theory may read [Theorem 3](#), [Lemma 2](#), and [Corollary 1](#) first, and return to the technical lemmas later.

We analyze the asymptotic properties of the K_f -fold cross-fitted estimator for a single target-source pair (t, s) , derived using the two-level cross-fitting strategy described in [Supplemental Material S1](#). Let K_f be a fixed integer (e.g., 5 or 10). The data $\mathcal{D} = \mathcal{D}_t \cup \mathcal{D}_s$ is partitioned into K_f folds, $\mathcal{D} = \cup_{k=1}^{K_f} \mathcal{D}_k$, where $\mathcal{D}_k = \mathcal{D}_{t,k} \cup \mathcal{D}_{s,k}$. Let $N_k = |\mathcal{D}_k| \approx N/K_f$. For notational simplicity, we assume the folds are balanced within each site (e.g., N_t and N_s are divisible by K_f) so that $|\mathcal{D}_{t,k}| = N_t/K_f$ and $|\mathcal{D}_{s,k}| = N_s/K_f$ for all k ; the minor rounding effects in practice are $o_p(N^{-1/2})$ and do not affect the first-order results. For $r \in \{t, s\}$, $\mathbb{E}_r[\cdot]$ denotes expectation conditional on site membership $R = r$. Define the target-site potential outcome means $\mu^1 := \mathbb{E}_t[Y(1)]$ and $\mu^0 := \mathbb{E}_t[Y(0)]$, and the target average treatment effect (TATE) $\tau := \mu^1 - \mu^0$. For any function $f(D)$ of a single observation $D = (R, \mathbf{X}, A, Y)$, we write $\tilde{E}_k[f(D)] := \frac{1}{N_k} \sum_{i \in \mathcal{D}_k} f(D_i)$ for the empirical average over the combined k -th fold. Let $\mathcal{D}^{(-k)} = \mathcal{D} \setminus \mathcal{D}_k$ denote the out-of-fold data used for training nuisance models evaluated on fold k . The crucial aspect of cross-fitting is that the data used for training, $\mathcal{D}^{(-k)}$, is independent of the data used for evaluation, $\mathcal{D}_k$.

Let $\vartheta = (\boldsymbol{\alpha}^1, \boldsymbol{\gamma}_1)$ be the vector of nuisance parameters. For this fixed pair (t, s) we write $\boldsymbol{\alpha}^1 \equiv \boldsymbol{\alpha}_{t,s}^1$ and $\boldsymbol{\gamma}_1 \equiv \boldsymbol{\gamma}_{s,1}$. Let $\vartheta^* = (\boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)$ denote the population targets of the calibrated estimation step ([Eq. \(9\)](#) and [Eq. \(10\)](#) in the main manuscript), and let $\hat{\vartheta}^{(-k)} = (\hat{\boldsymbol{\alpha}}_{t,s}^{1,(-k)}, \hat{\boldsymbol{\gamma}}_{s,1}^{(-k)})$ be the nuisance estimators trained on data $\mathcal{D}^{(-k)}$ using the two-level cross-fitting procedure. Here the superscript $(-k)$ denotes the *final* two-level cross-fitted nuisance estimators used for evaluating fold k (including the fold-summed calibration over secondary folds), and hence $\hat{\vartheta}^{(-k)}$ is independent of $\mathcal{D}_k$. The inner construction follows the SMMAL two-level calibration device: for each outer fold k_1 , every secondary fold $k_2 \neq k_1$ supplies one calibrated-loss contribution, and the initial nuisance fits entering that contribution are trained on $\mathcal{D} \setminus (\mathcal{D}_{k_1} \cup \mathcal{D}_{k_2})$. The final $\hat{\vartheta}^{(-k_1)}$ is obtained by minimizing the average of these fold-specific calibrated losses, so it uses all outer-training folds while keeping the initial plug-in for each contribution disjoint from that contribution's calibration fold.

To ensure numerical stability and theoretical regularity (bounded weights), we define a truncation function $\mathcal{T}(x) = \text{sign}(x) \min(|x|, M_\tau)$, where $M_\tau = M_{\tau,N}$ is allowed to depend on N (notation suppressed). (We avoid the symbol $\tau(\cdot)$ to prevent collision with the TATE τ in the main manuscript.) [Marquis: If we always use $\phi(\mathbf{X})$ when involve the covariate $\mathbf{X}$, let us just drop ϕ and claim $\mathbf{X}$ is the covariate after proper transformation. This is methodologically trivial.] Equivalently, one may define the working covariate vector

$Z = \phi(\mathbf{X})$ and write all nuisance models directly in terms of Z . We keep the notation $\phi(\mathbf{X})$ only to match the main manuscript and to distinguish raw covariates from the fixed feature/basis expansion. All tail, sparsity, and restricted-curvature conditions below are imposed on this working covariate vector. The truncated weighting function is defined as:

$$w_\tau(\mathbf{X}; \boldsymbol{\gamma}) := \frac{I(A = 1)}{e^{\mathcal{T}(\phi(\mathbf{X})^T \boldsymbol{\gamma})}}.$$

Define also the (untruncated) weight

$$w(\mathbf{X}; \boldsymbol{\gamma}) := \frac{I(A = 1)}{e^{\phi(\mathbf{X})^T \boldsymbol{\gamma}}}.$$

The truncation function $\mathcal{T}$ is applied *only in the calibrated loss functions* (Eq. (9) and Eq. (10) in the main manuscript) to ensure numerical stability during nuisance estimation; the estimators and influence functions below use the untruncated weight $w(\mathbf{X}; \boldsymbol{\gamma})$, consistent with the main manuscript. Under Assumption 2(c), truncation is asymptotically inactive at the population targets (so $w_\tau(\mathbf{X}; \boldsymbol{\gamma}_1^*) = w(\mathbf{X}; \boldsymbol{\gamma}_1^*)$ and $\psi'(\mathcal{T}(\phi(\mathbf{X})^T \boldsymbol{\alpha}^{1,*})) = \psi'(\phi(\mathbf{X})^T \boldsymbol{\alpha}^{1,*})$ w.p.a.1), and hence the population first-order conditions induced by the calibrated losses are unchanged up to an $o(1)$ term.

This corresponds to using a finite M_τ during nuisance optimization, while inference is based on the untruncated final DR estimating equation, equivalently $M_{\tau, \text{inference}} = \infty$. Setting a finite $M_{\tau, \text{inference}}$ would instead define a clipped estimating equation, as discussed below.

If, instead, one *also* clips the weights in the final AIPW/DR estimating equation (i.e., replaces w by w_τ inside $\hat{\mu}_{t,s}^1$), then the resulting estimator has a different influence function in finite samples. Under Assumption 2(c) (clipping probability $o(N^{-1})$ at the population targets), the difference between the clipped and unclipped estimators is $o_p(N^{-1/2})$ and the first-order limit theory is unchanged; otherwise, the clipping-induced bias must be accounted for (see, e.g., Tan (2020)).

The proofs below use three conventions that keep the asymptotic argument aligned with the estimator analyzed: foldwise IF components are centered when forming variance and covariance estimators; clipping is used only during nuisance fitting and is asymptotically inactive for the final estimating equation under Assumption 2(c); and any finite-sample symmetrization or ridge stabilization used in the aggregation quadratic program is treated as a numerical device that vanishes asymptotically. [Marquis: The table by itself is not very informative.] Rather than relying on a standalone table, the Supplemental Material

now states the role of each block in prose. [Lemma 1](#) identifies the target mean under either nuisance-model correctness; [Assumptions 1 to 3](#) collect the sampling, overlap/moment, and nuisance-rate requirements; [Theorem 3](#) gives the pairwise cross-fitted CLT; [Lemma 2](#) proves variance consistency; and [Supplemental Material S3](#) extends these ingredients to adaptive aggregation and source selection.

We assume $\phi(\mathbf{X})$ includes an intercept term. Let $p := \dim(\phi(\mathbf{X}))$. When the feature expansion is fixed in advance, one may equivalently regard $\phi(\mathbf{X})$ as the working covariate vector; we keep the notation $\phi(\mathbf{X})$ only to match the main manuscript. For a real-valued random variable Z , $\|Z\|_{\psi_2}$ denotes the sub-Gaussian Orlicz norm. For a vector $v \in \mathbb{R}^p$, $\|v\|_0$ denotes its sparsity level. For canonical exponential-family GLMs, we use the standard notation that the negative log-likelihood takes the form $-Y\theta + \Psi(\theta)$ with linear predictor $\theta = \phi(\mathbf{X})^T \boldsymbol{\alpha}$, and that $\psi(\theta) = \Psi'(\theta)$; $\psi'(\theta)$ denotes the derivative with respect to θ . We also use the shorthand $\psi(\phi(\mathbf{X}); \boldsymbol{\alpha}) \equiv \psi(\phi(\mathbf{X})^T \boldsymbol{\alpha})$. To avoid confusion with the notation used in the reference PDFs (in the `proof ref` folder), we consistently follow the main manuscript's notation: $\phi(\mathbf{X})$ denotes the feature/basis expansion and $\psi(\cdot)$ denotes the GLM conditional mean. (For example, some references use $\phi(\cdot)$ and $\Phi(\cdot)$ for a Normal density/CDF; we do *not* use that convention here.)

As in the main manuscript, the target estimand is the target-site treated potential outcome mean $\mu^1 \equiv \mathbb{E}_t[Y(1)]$. Define the target-site treated regression function by

$$m_t^1(\mathbf{X}) := \mathbb{E}_t[Y \mid A = 1, \mathbf{X}].$$

The K_f -fold cross-fitted estimator for μ^1 is:

$$\hat{\mu}_{t,s}^{1,\text{CF}} = \frac{1}{K_f} \sum_{k=1}^{K_f} \left\{ \tilde{E}_{t,k} \left[\psi(\phi(\mathbf{X}); \hat{\boldsymbol{\alpha}}_{t,s}^{1,(-k)}) \right] + \tilde{E}_{s,k} \left[w(\mathbf{X}; \hat{\boldsymbol{\gamma}}_{s,1}^{(-k)}) \left(Y - \psi(\phi(\mathbf{X}); \hat{\boldsymbol{\alpha}}_{t,s}^{1,(-k)}) \right) \right] \right\}.$$

Here,

$$\tilde{E}_{t,k}[g(D)] := \frac{1}{N_{t,k}} \sum_{i \in \mathcal{D}_{t,k}} g(D_i), \quad \tilde{E}_{s,k}[g(D)] := \frac{1}{N_{s,k}} \sum_{i \in \mathcal{D}_{s,k}} g(D_i),$$

denote sample averages over the k -th fold of the target and source data, with $N_{t,k} = |\mathcal{D}_{t,k}|$, $N_{s,k} = |\mathcal{D}_{s,k}|$, and $N_k = N_{t,k} + N_{s,k}$. Let $N = N_t + N_s$ be the total sample size, and let $\rho_t = N_t/N$ and $\rho_s = N_s/N$. We write $\mathbb{E}[\cdot]$ for expectation under the pooled distribution, i.e., $\mathbb{E}[g(D)] = \rho_t \mathbb{E}_t[g(D)] + \rho_s \mathbb{E}_s[g(D)]$. Throughout the asymptotics we consider the standard two-sample (multi-sample) regime with sampling fractions bounded away from 0 and 1, i.e., $N_t/N \rightarrow \rho_t \in (0, 1)$ and $N_s/N \rightarrow \rho_s \in (0, 1)$. Under overlap/positivity for

the relevant treatment arm, the effective sample sizes within each arm also scale linearly with N (e.g., $\sum_{i \in \mathcal{D}_s} I(A_i = 1) = \Theta_p(N_s)$), so writing high-dimensional rates in terms of N is equivalent (up to constants) to writing them in terms of the arm-specific sample sizes. Under the balanced-fold assumption above, the estimator can equivalently be written using the combined-fold empirical average as

$$\hat{\mu}_{t,s}^{1,\text{CF}} = \frac{1}{K_f} \sum_{k=1}^{K_f} \tilde{E}_k [\tilde{\varphi}_{t,s}(D; \hat{\vartheta}^{(-k)})],$$

which matches the $\tilde{E}_k$ notation used in the remainder decomposition below.

Define the (uncentered) score contribution at a generic nuisance parameter $\vartheta = (\alpha^1, \gamma_1)$:

$$\tilde{\varphi}_{t,s}(D_i; \vartheta) := \frac{I(R_i = t)}{\rho_t} \psi(\phi(\mathbf{X}_i); \alpha^1) + \frac{I(R_i = s)}{\rho_s} w(\mathbf{X}_i; \gamma_1) \{Y_i - \psi(\phi(\mathbf{X}_i); \alpha^1)\}.$$

Define the corresponding population estimand at ϑ as

$$\mu^1(\vartheta) := \mathbb{E}[\tilde{\varphi}_{t,s}(D; \vartheta)] = \mathbb{E}_t[\psi(\phi(\mathbf{X}); \alpha^1)] + \mathbb{E}_s[w(\mathbf{X}; \gamma_1) \{Y - \psi(\phi(\mathbf{X}); \alpha^1)\}].$$

[Marquis: Defining the notations inside the Lemma is confusing. Either use $\mathbb{E}_t[Y \mid A = 1, \mathbf{X}]$ (not very long) or define m_t^1 in notation.]

Lemma 1 (Identification / double robustness for $\mu^1(\vartheta^*)$). *Assume the target causal parameter is identified as $\mu^1 = \mathbb{E}_t[Y(1)]$ (e.g., by consistency and ignorability/overlap on the target site), so that $\mu^1 = \mathbb{E}_t[m_t^1(\mathbf{X})]$. Recall $\phi(\mathbf{X})$ includes an intercept. Let $\vartheta^* = (\alpha^{1,*}, \gamma_1^*)$ denote the population targets of the calibrated losses. Then $\mu^1(\vartheta^*) = \mu^1$ holds under either of the following conditions.*

1. **Outcome-model correctness.** *If $\psi(\phi(\mathbf{X}); \alpha^{1,*}) = m_t^1(\mathbf{X})$ almost surely, then the calibrated outcome loss implies the population moment condition $\mathbb{E}_s[w(\mathbf{X}; \gamma_1^*) \phi(\mathbf{X}) \{Y - \psi(\phi(\mathbf{X}); \alpha^{1,*})\}] = \mathbf{0}$ (see Lemma 5(ii) below). In particular, the intercept component yields $\mathbb{E}_s[w(\mathbf{X}; \gamma_1^*) \{Y - \psi(\phi(\mathbf{X}); \alpha^{1,*})\}] = 0$, and therefore $\mu^1(\vartheta^*) = \mathbb{E}_t[\psi(\phi(\mathbf{X}); \alpha^{1,*})] = \mathbb{E}_t[m_t^1(\mathbf{X})] = \mu^1$.*
2. **Weight-model correctness.** *Suppose the weighting model is correct in the sense that for any square-integrable measurable function $h(\mathbf{X})$,*

$$\mathbb{E}_s[w(\mathbf{X}; \gamma_1^*) h(\mathbf{X})] = \mathbb{E}_t[h(\mathbf{X})],$$

and, for the treated potential outcome, the usual covariate-shift/transportability condition holds for this pair (t, s) : $\mathbb{E}_s[Y(1) \mid \mathbf{X}] = \mathbb{E}_t[Y(1) \mid \mathbf{X}]$ almost surely. Then for

any outcome model $m(\mathbf{X})$,

$$\mathbb{E}_s[w(\mathbf{X}; \gamma_1^*)\{Y(1) - m(\mathbf{X})\}] = \mathbb{E}_t[Y(1) - m(\mathbf{X})],$$

and hence $\mu^1(\boldsymbol{\alpha}, \gamma_1^*) = \mu^1$ for any $\boldsymbol{\alpha}$, in particular $\mu^1(\vartheta^*) = \mu^1$ even if the outcome model is misspecified.

Define the centered influence function at the population parameters ϑ^* :

$$\begin{aligned} \varphi_{t,s}(D_i; \vartheta^*) &:= \tilde{\varphi}_{t,s}(D_i; \vartheta^*) - \mu^1(\vartheta^*) \\ &= \frac{I(R_i = t)}{\rho_t} \left(\psi(\phi(\mathbf{X}_i); \boldsymbol{\alpha}^{1,*}) - \mu_t^1(\boldsymbol{\alpha}^{1,*}) \right) \\ &\quad + \frac{I(R_i = s)}{\rho_s} \left(w(\mathbf{X}_i; \gamma_1^*) \{Y_i - \psi(\phi(\mathbf{X}_i); \boldsymbol{\alpha}^{1,*})\} - \delta_s^1(\vartheta^*) \right). \end{aligned}$$

where $\mu_t^1(\boldsymbol{\alpha}^{1,*}) = \mathbb{E}_t[\psi(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})]$ and $\delta_s^1(\vartheta^*) = \mathbb{E}_s[w(\mathbf{X}; \gamma_1^*)\{Y - \psi(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})\}]$. More generally, for any ϑ we define $\varphi_{t,s}(D; \vartheta) := \tilde{\varphi}_{t,s}(D; \vartheta) - \mu^1(\vartheta)$. Let $V_{t,s}^* := \text{Var}(\varphi_{t,s}(D; \vartheta^*))$.

Definition 1 (Sparsity bookkeeping). Let

$$s_\alpha := \|\boldsymbol{\alpha}^{1,*}\|_0, \quad s_\gamma := \|\gamma_1^*\|_0.$$

When the initial nuisance estimators have distinct population targets, define $s_{\alpha,\text{init}} := \|\boldsymbol{\alpha}_{\text{init}}^{1,*}\|_0$ and $s_{\gamma,\text{init}} := \|\gamma_{1,\text{init}}^*\|_0$. These are bookkeeping quantities for the high-dimensional rates rather than separate modeling assumptions. The growth restrictions are imposed explicitly in the theorems and, when initial nuisance rates are needed, we assume $s_{\alpha,\text{init}} + s_{\gamma,\text{init}} = O(s_\alpha + s_\gamma)$.

S2.2 Assumptions

The assumptions below are stated in the style commonly used in high-dimensional doubly robust inference: [Assumption 1](#) collects sampling and working-covariate tail conditions; [Assumption 2](#) collects overlap, moment, and smoothness conditions ensuring a valid influence-function expansion; and [Assumption 3](#) collects the nuisance-estimation conditions needed for high-dimensional rates. Sparsity levels are defined in [Definition 1](#), and their allowed growth is stated directly in the theorems. This organization is closest in spirit to [Tan \(2020\)](#), [Hou et al. \(2025\)](#), and [Chernozhukov et al. \(2018\)](#).

[Marquis: Define Orlicz norm in notation not in the assumptions. Sparsity is a definition not assumption.]

Assumption 1 (Sampling and Working-Covariate Tails). (a) The target and source samples are independent, each sample is i.i.d. within site, and the site fractions satisfy $N_t/N \rightarrow \rho_t \in (0, 1)$ and $N_s/N \rightarrow \rho_s \in (0, 1)$.

(b) The working covariates $\phi(\mathbf{X})$ include an intercept and satisfy $\max_{1 \leq j \leq p} \|\phi_j(\mathbf{X})\|_{\psi_2} \leq K_\phi$ under both $R = t$ and $R = s$.

Assumption 2 (Overlap, Smoothness, and Moments). (a) **Overlap and weight regularity.** The ideal density-ratio weight is well-defined on the relevant target support: $P_s\{A = 1 \mid \mathbf{X}\} > 0$ almost surely on that support. For the main proofs we require the following local weight envelope condition: for some $\delta > 0$,

$$\sup_{\gamma \in \mathcal{N}(\gamma_1^*)} \mathbb{E}_s[w(\mathbf{X}; \gamma)^{2+\delta}] < \infty,$$

and, coordinatewise,

$$\max_{1 \leq j \leq p} \sup_{\gamma \in \mathcal{N}(\gamma_1^*)} \mathbb{E}_s[w(\mathbf{X}; \gamma)^{2+\delta} |\phi_j(\mathbf{X})|^{2+\delta}] < \infty.$$

A simple sufficient condition is an upper-overlap bound $w(\mathbf{X}; \gamma) \leq B_w < \infty$ almost surely for all $\gamma \in \mathcal{N}(\gamma_1^*)$, together with coordinatewise $(2 + \delta)$ moments of $\phi(\mathbf{X})$. No uniform bound on $\|\phi(\mathbf{X})\|_2$ is imposed; in high dimensions this norm may scale as $\sqrt{p}$. [Marquis: Need to be specific. I would consider the weights need an upper bound (but not necessarily a lower bound)?] [Marquis: The first assumption is essentially an overlapping condition. The second assumption needs clarification: the $\|\phi(\mathbf{X})\|_2 \propto \sqrt{p}$, so there would not be an upper bound. Otherwise, we probably just need 4-th moments? Probably just follow the standard overlapping assumption.]

(b) **Smoothness and moments.** The GLM mean function ψ is continuously differentiable with bounded derivative and Lipschitz slope:

$$\sup_{u \in \mathbb{R}} |\psi'(u)| \leq L_\psi < \infty, \quad |\psi'(u_1) - \psi'(u_2)| \leq L_{\psi'} |u_1 - u_2|.$$

Moreover, under both $R = t$ and $R = s$, the outcome has finite $(2 + \delta)$ -moment and the uncentered estimating-function contribution is locally $(2 + \delta)$ -integrable:

$$\mathbb{E}[|Y|^{2+\delta} \mid R = r] < \infty, \quad \sup_{\vartheta \in \mathcal{N}(\vartheta^*)} \mathbb{E}[\tilde{\varphi}_{t,s}(D; \vartheta)^{2+\delta}] < \infty.$$

[Marquis: IF? Try to use the consistent notation. If you assume sub-Gaussian covariates and outcomes, you should be able to derive the bound for moments] A

sufficient condition, used in the concentration arguments below, is the upper-overlap bound above plus sub-Gaussian tails for $Y \mid (A = 1, R = r)$ and for the coordinates of $\phi(\mathbf{X})$; then products such as $w\{Y - \psi(\phi(\mathbf{X}); \boldsymbol{\alpha})\}$ have the required finite $(2 + \delta)$ moments in a local neighborhood of ϑ^* .

(c) **Asymptotically inactive truncation.** With $M_{\tau,N}$ as in the definition of $\mathcal{T}(\cdot)$,

$$\begin{aligned} P\left(\left|\phi(\mathbf{X})^T \boldsymbol{\alpha}^{1,*}\right| > M_{\tau,N} \mid R = r\right) &= o(N^{-1}) \quad \text{for } r \in \{t, s\}, \\ P\left(\left|\phi(\mathbf{X})^T \boldsymbol{\gamma}_1^*\right| > M_{\tau,N} \mid R = s\right) &= o(N^{-1}). \end{aligned}$$

This condition is stronger than strictly necessary, but it cleanly guarantees that clipping has no first-order effect on the final estimating equation.

[Marquis: Use “enumerate” to list assumptions (customizable via “enumitem” package)]

Assumption 3 (Nuisance Model Properties). (a) **Restricted curvature.** The population versions of the calibrated losses in [Eq. \(9\)](#) and [Eq. \(10\)](#) are convex and satisfy restricted strong convexity in neighborhoods of ϑ^* compatible with the relevant sparse cones.

(b) **Cross-fitting.** For each fold k , the nuisance estimator $\hat{\vartheta}^{(-k)}$ is trained on $\mathcal{D}^{(-k)}$ only and is independent of the evaluation fold $\mathcal{D}_k$.

(c) **Two-level alignment.** Let $(\boldsymbol{\alpha}_{\text{init}}^{1,*}, \boldsymbol{\gamma}_{1,\text{init}}^*)$ denote the population limits of the initial nuisance estimators used inside the calibrated losses. These limits coincide with the calibrated targets:

$$\boldsymbol{\alpha}_{\text{init}}^{1,*} = \boldsymbol{\alpha}^{1,*}, \quad \boldsymbol{\gamma}_{1,\text{init}}^* = \boldsymbol{\gamma}_1^*.$$

(d) **Optional foldwise score-centering.** If one wants to invoke the sharpened **Special** rate in [Lemma 4](#), then the foldwise conditional score-centering condition there is assumed for the calibrated losses. This is analogous to the foldwise score-centering condition used in [Hou et al. \(2025\)](#). Without this strengthening, the theorem uses the **General** rate in [Lemma 4](#).

Remark 2 (On the alignment condition). [Assumption 3\(c\)](#) is a technical device to avoid tracking multiple population limits induced by the nested/coupled optimization. It is satisfied, for example, when (i) the population calibrated losses have unique minimizers $(\boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)$ (by strict convexity in the relevant restricted set), and (ii) the mapping from an initial plug-in value to the population minimizer of the next calibrated update is continuous at

$(\boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)$, so that any consistent initialization converges to the same fixed point. In practice, the same calibrated estimating equations are solved throughout the nuisance-estimation pipeline, making the alignment natural under uniqueness.

S2.3 Main Theorem and Proof

Theorem 3 (Asymptotic Normality of $\hat{\mu}_{t,s}^{1,\text{CF}}$). *Suppose [Assumption 1](#), [Assumption 2](#), and [Assumption 3\(a\)–Assumption 3\(c\)](#) hold. ([Assumption 3\(d\)](#) is an optional strengthening used only to invoke the **Special** rate in [Lemma 4](#); the theorem also holds under the **General** rate condition there.) Assume ϑ^* is the unique population target pair induced by the calibrated losses under the alignment condition in [Assumption 3\(c\)](#), implying [Lemma 5](#) holds. If further, at least one nuisance model (outcome or weighting) is correctly specified in the sense of [Lemma 1](#) (so $\mu^1(\vartheta^*) = \mu^1$), the calibrated nuisance sparsities satisfy*

$$\frac{(s_\alpha + s_\gamma)^2 \log^2(p)}{N} \rightarrow 0,$$

and $V_{t,s}^* > 0$, then the cross-fitted estimator is asymptotically normal: *[Marquis: no need to define this notation. Also, do you specify the sparsities of initial and calibrated coefficients?]* Here s_α and s_γ are the calibrated population sparsities from [Definition 1](#). If the initial nuisance estimators are estimated by separate penalized fits, their population sparsities are $s_{\alpha,\text{init}}$ and $s_{\gamma,\text{init}}$ and are assumed to be of the same order, so the initialization error is covered by [Assumption 3\(c\)](#) and [Lemma 4](#).

$$\sqrt{N} \left(\hat{\mu}_{t,s}^{1,\text{CF}} - \mu^1 \right) \xrightarrow{d} \mathcal{N}(0, V_{t,s}^*).$$

If the initial nuisance estimators enter the two-level algorithm separately from the calibrated update, their stochastic error is required only through [Assumption 3\(c\)](#), which aligns their population limits with $(\boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)$. Moreover, the finite-sample KKT perturbation from the ℓ_1 penalties is absorbed in the remainder term (via [Lemma 4](#)) and is $o_p(N^{-1/2})$ under the stated sparsity/rate condition.

Proof. Suppose in addition that at least one nuisance model is correctly specified in the sense of [Lemma 1](#), so that $\mu^1(\vartheta^*) = \mu^1$. We may then center the expansion at μ^1 . *[Marquis: Good to define “Term I” and “Term II” this way. However, please use a math notation*

instead of text if you want to use them in equations.]

$$S_N := \frac{1}{N} \sum_{i=1}^N \varphi_{t,s}(D_i; \vartheta^*), \quad \text{Rem}_N := \hat{\mu}_{t,s}^{1,\text{CF}} - \mu^1 - S_N.$$

Then

$$\hat{\mu}_{t,s}^{1,\text{CF}} - \mu^1 = S_N + \text{Rem}_N,$$

where $\mathbb{E}[\varphi_{t,s}(D; \vartheta^*)] = 0$ by construction, and $\mu^1(\vartheta^*) = \mu_t^1(\boldsymbol{\alpha}^{1,*}) + \delta_s^1(\vartheta^*)$. By [Lemma 3](#),

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N \varphi_{t,s}(D_i; \vartheta^*) \xrightarrow{d} \mathcal{N}(0, V_{t,s}^*),$$

hence $\sqrt{N} S_N \xrightarrow{d} \mathcal{N}(0, V_{t,s}^*)$. Using the foldwise representation of $\hat{\mu}_{t,s}^{1,\text{CF}}$ and adding/subtracting $\tilde{\varphi}_{t,s}(D; \vartheta^*)$ inside each fold average,

$$\text{Rem}_N = \frac{1}{K_f} \sum_{k=1}^{K_f} (R_{1,k} + R_{2,k}),$$

where

$$R_{1,k} = \left(\tilde{E}_k - \mathbb{E}[\cdot \mid \mathcal{D}^{(-k)}] \right) \left[\tilde{\varphi}_{t,s}(D; \hat{\vartheta}^{(-k)}) - \tilde{\varphi}_{t,s}(D; \vartheta^*) \right]$$

and

$$R_{2,k} = \mathbb{E}[\tilde{\varphi}_{t,s}(D; \hat{\vartheta}^{(-k)}) - \tilde{\varphi}_{t,s}(D; \vartheta^*) \mid \mathcal{D}^{(-k)}].$$

[\[Marquis: Expand the details\]](#). $R_{1,k}$ is the conditional empirical process fluctuation of the *difference score*:

$$R_{1,k} = \left(\tilde{E}_k - \mathbb{E}[\cdot \mid \mathcal{D}^{(-k)}] \right) \left[\tilde{\varphi}_{t,s}(D; \hat{\vartheta}^{(-k)}) - \tilde{\varphi}_{t,s}(D; \vartheta^*) \right].$$

By cross-fitting, conditional on $\mathcal{D}^{(-k)}$, the summands are independent (with a two-sample/triangular-array structure) on fold k with mean zero. Using [Lemma 6](#) (Part 2),

$$\mathbb{E} \left[\left(\tilde{\varphi}_{t,s}(D; \hat{\vartheta}^{(-k)}) - \tilde{\varphi}_{t,s}(D; \vartheta^*) \right)^2 \mid \mathcal{D}^{(-k)} \right] \leq C \left(\|\hat{\boldsymbol{\alpha}}_{t,s}^{1,(-k)} - \boldsymbol{\alpha}^{1,*}\|_2^2 + \|\hat{\boldsymbol{\gamma}}_{s,1}^{(-k)} - \boldsymbol{\gamma}_1^*\|_2^2 \right),$$

and therefore the conditional variance of $R_{1,k}$ obeys

$$\text{Var}(R_{1,k} \mid \mathcal{D}^{(-k)}) = \text{Var}(\tilde{E}_k[\Delta_k(D)] \mid \mathcal{D}^{(-k)}) \leq \frac{1}{N_k} \mathbb{E}[\Delta_k(D)^2 \mid \mathcal{D}^{(-k)}],$$

where $\Delta_k(D) := \tilde{\varphi}_{t,s}(D; \hat{\vartheta}^{(-k)}) - \tilde{\varphi}_{t,s}(D; \vartheta^*)$. Here we used that, conditional on $\mathcal{D}^{(-k)}$, the $\{\Delta_k(D_i) : i \in \mathcal{D}_k\}$ are independent, so

$$\begin{aligned} \text{Var}(\tilde{E}_k[\Delta_k(D)] \mid \mathcal{D}^{(-k)}) &= \text{Var}(\tilde{E}_k[\Delta_k(D)] - \mathbb{E}[\Delta_k(D) \mid \mathcal{D}^{(-k)}] \mid \mathcal{D}^{(-k)}) \\ &\leq \frac{1}{N_k} \mathbb{E}[(\Delta_k(D) - \mathbb{E}[\Delta_k(D) \mid \mathcal{D}^{(-k)}])^2 \mid \mathcal{D}^{(-k)}] \\ &\leq \frac{1}{N_k} \mathbb{E}[\Delta_k(D)^2 \mid \mathcal{D}^{(-k)}]. \end{aligned}$$

Since $N_k \asymp N$ for fixed K_f , Chebyshev's inequality yields

$$|R_{1,k}| = O_p \left(\frac{\|\hat{\alpha}_{t,s}^{1,(-k)} - \alpha^{1,*}\|_2 + \|\hat{\gamma}_{s,1}^{(-k)} - \gamma_1^*\|_2}{\sqrt{N}} \right) = O_p \left(\frac{\sqrt{(s_\alpha + s_\gamma) \log(p)}}{N} \right) = o_p(N^{-1/2}).$$

$R_{2,k}$ is the first-order bias remainder, conditional on $\mathcal{D}^{(-k)}$:

$$R_{2,k} = \mathbb{E}[\tilde{\varphi}_{t,s}(D; \hat{\vartheta}^{(-k)}) - \tilde{\varphi}_{t,s}(D; \vartheta^*) \mid \mathcal{D}^{(-k)}]$$

By [Lemma 6](#) (Part 1), which relies on the orthogonality conditions derived in [Lemma 5](#), this bias is bounded by a second-order term:

$$|R_{2,k}| \leq C \cdot \left(\|\hat{\alpha}_{t,s}^{1,(-k)} - \alpha^{1,*}\|_2^2 + \|\hat{\gamma}_{s,1}^{(-k)} - \gamma_1^*\|_2^2 + \|\hat{\alpha}_{t,s}^{1,(-k)} - \alpha^{1,*}\|_2 \|\hat{\gamma}_{s,1}^{(-k)} - \gamma_1^*\|_2 \right)$$

Using the nuisance rates from [Lemma 4](#):

$$|R_{2,k}| = O_p \left(\frac{s_\alpha \log(p)}{N} \right) + O_p \left(\frac{s_\gamma \log(p)}{N} \right) + O_p \left(\frac{\sqrt{s_\alpha s_\gamma} \log(p)}{N} \right) = O_p \left(\frac{(s_\alpha + s_\gamma) \log(p)}{N} \right).$$

Averaging over k gives the total bias remainder:

$$R_2 = \frac{1}{K_f} \sum_{k=1}^{K_f} R_{2,k} = O_p \left(\frac{(s_\alpha + s_\gamma) \log(p)}{N} \right)$$

We require $\sqrt{N} \cdot R_2 \rightarrow 0$, i.e., $O_p((s_\alpha + s_\gamma) \log(p) / \sqrt{N}) \rightarrow 0$. Squaring both sides, this is implied by the theorem's condition $\frac{(s_\alpha + s_\gamma)^2 \log^2(p)}{N} \rightarrow 0$. Thus, the total remainder Rem_N satisfies $\sqrt{N} \text{Rem}_N = o_p(1)$, and hence

$$\sqrt{N} (\hat{\mu}_{t,s}^{1,\text{CF}} - \mu^1) = \frac{1}{\sqrt{N}} \sum_{i=1}^N \varphi_{t,s}(D_i; \vartheta^*) + o_p(1).$$

Together with [Lemma 3](#), this yields $\sqrt{N} (\hat{\mu}_{t,s}^{1,\text{CF}} - \mu^1) \xrightarrow{d} \mathcal{N}(0, V_{t,s}^*)$ by Slutsky's theorem. $\square$

Lemma 2 (Variance Estimation). *[Marquis: Should be lemma not remark. Also, you might need the variance bounded away from zero.] The asymptotic variance $V_{t,s}^*$ can be consistently estimated using a plug-in estimator based on the empirical influence function evaluated within each fold k using the out-of-fold nuisance estimates $\hat{\vartheta}^{(-k)}$:*

$$\hat{V}_{t,s} = \frac{1}{K_f} \sum_{k=1}^{K_f} \tilde{E}_k \left[\left(\tilde{\varphi}_{t,s}(D; \hat{\vartheta}^{(-k)}) - \hat{\mu}_{t,s}^{1,\text{CF}} \right)^2 \right] \xrightarrow{p} V_{t,s}^*.$$

For consistency of the variance estimator itself no lower bound is needed beyond finiteness. For Wald inference and studentization we assume, as in [Theorem 3](#), that $V_{t,s}^*$ is bounded away from zero. **Consistency Argument:** The consistency of $\hat{V}_{t,s}$ relies on three key ingredients:

1. *Nuisance Consistency:* By [Lemma 4](#), $\|\hat{\vartheta}^{(-k)} - \vartheta^*\|_2 = o_p(1)$.
2. *L_2 -Lipschitz Continuity:* Under [Assumption 2\(b\)](#), the score contribution is locally Lipschitz in ϑ in mean-square, i.e., there exists $C < \infty$ such that for ϑ_1, ϑ_2 in a neighborhood of ϑ^* ,

$$\mathbb{E}[(\tilde{\varphi}_{t,s}(D; \vartheta_1) - \tilde{\varphi}_{t,s}(D; \vartheta_2))^2] \leq C\|\vartheta_1 - \vartheta_2\|_2^2.$$

3. *Uniform Integrability:* By the $(2 + \delta)$ moment condition in [Assumption 2\(b\)](#), $\{\tilde{\varphi}_{t,s}(D; \hat{\vartheta}^{(-k)})^2\}_{k,N}$ is uniformly integrable.

We now give a detailed proof (not just a sketch).

Step 1 (Reduce to within-fold centering). For each fold k , define the foldwise score

$$Z_i^{(k)} := \tilde{\varphi}_{t,s}(D_i; \hat{\vartheta}^{(-k)}), \quad i \in \mathcal{D}_k,$$

and the foldwise mean $\hat{\mu}_k := \tilde{E}_k[Z^{(k)}]$. Since $\hat{\mu}_{t,s}^{1,\text{CF}} = \frac{1}{K_f} \sum_{k=1}^{K_f} \hat{\mu}_k$, we have the identity

$$\tilde{E}_k[(Z^{(k)} - \hat{\mu}_{t,s}^{1,\text{CF}})^2] = \tilde{E}_k[(Z^{(k)} - \hat{\mu}_k)^2] + (\hat{\mu}_k - \hat{\mu}_{t,s}^{1,\text{CF}})^2,$$

where the cross term vanishes because $\tilde{E}_k[Z^{(k)} - \hat{\mu}_k] = 0$ by definition of $\hat{\mu}_k$. Therefore,

$$\hat{V}_{t,s} = \frac{1}{K_f} \sum_{k=1}^{K_f} \tilde{E}_k[(Z^{(k)} - \hat{\mu}_k)^2] + \frac{1}{K_f} \sum_{k=1}^{K_f} (\hat{\mu}_k - \hat{\mu}_{t,s}^{1,\text{CF}})^2.$$

By [Lemma 4](#), $\|\hat{\vartheta}^{(-k)} - \vartheta^*\|_2 = o_p(1)$, and by the conditional LLN below we have $\hat{\mu}_k = \mu^1(\vartheta^*) + o_p(1)$ for each fixed k . Since K_f is fixed, this implies $\max_{1 \leq k \leq K_f} |\hat{\mu}_k - \hat{\mu}_{t,s}^{1,\text{CF}}| = o_p(1)$, and hence the second term is $o_p(1)$. Thus it suffices to show

$$\frac{1}{K_f} \sum_{k=1}^{K_f} \tilde{E}_k[(Z^{(k)} - \hat{\mu}_k)^2] \xrightarrow{p} V_{t,s}^*.$$

Step 2 (Conditional LLN for first and second moments). Fix a fold k . Conditional on the training sample $\mathcal{D}^{(-k)}$, the nuisance estimate $\hat{\vartheta}^{(-k)}$ is fixed and independent of $\mathcal{D}_k$. Therefore, conditional on $\mathcal{D}^{(-k)}$, the random variables $\{Z_i^{(k)} : i \in \mathcal{D}_k\}$ are independent with finite $(2 + \delta)$ moments uniformly in N by [Assumption 2\(b\)](#). By the (conditional) weak law of large numbers,

$$\hat{\mu}_k - \mathbb{E}[Z^{(k)} \mid \mathcal{D}^{(-k)}] = o_p(1), \quad \tilde{E}_k[(Z^{(k)})^2] - \mathbb{E}[(Z^{(k)})^2 \mid \mathcal{D}^{(-k)}] = o_p(1).$$

Since $\tilde{E}_k[(Z^{(k)} - \hat{\mu}_k)^2] = \tilde{E}_k[(Z^{(k)})^2] - \hat{\mu}_k^2$, combining the two displays yields

$$\tilde{E}_k[(Z^{(k)} - \hat{\mu}_k)^2] - \left\{ \mathbb{E}[(Z^{(k)})^2 \mid \mathcal{D}^{(-k)}] - \mathbb{E}[Z^{(k)} \mid \mathcal{D}^{(-k)}]^2 \right\} = o_p(1).$$

Step 3 (Plug-in stability: conditional expectations converge to population targets). It remains to show that the bracketed conditional variance converges in probability to $V_{t,s}^*$. First, by nuisance consistency (Ingredient 1) and the L_2 -Lipschitz property (Ingredient 2),

$$\mathbb{E}[(Z^{(k)} - \tilde{\varphi}_{t,s}(D; \vartheta^*))^2] = \mathbb{E}[(\tilde{\varphi}_{t,s}(D; \hat{\vartheta}^{(-k)}) - \tilde{\varphi}_{t,s}(D; \vartheta^*))^2] \leq C \|\hat{\vartheta}^{(-k)} - \vartheta^*\|_2^2 \xrightarrow{p} 0.$$

Next we establish convergence of the conditional first and second moments. Let

$$m_k := \mathbb{E}[Z^{(k)} \mid \mathcal{D}^{(-k)}], \quad q_k := \mathbb{E}[(Z^{(k)})^2 \mid \mathcal{D}^{(-k)}], \quad \mu^* := \mu^1(\vartheta^*) = \mathbb{E}[\tilde{\varphi}_{t,s}(D; \vartheta^*)].$$

Then

$$m_k - \mu^* = \mathbb{E}[\tilde{\varphi}_{t,s}(D; \hat{\vartheta}^{(-k)}) - \tilde{\varphi}_{t,s}(D; \vartheta^*) \mid \mathcal{D}^{(-k)}],$$

so by conditional Jensen,

$$(m_k - \mu^*)^2 \leq \mathbb{E}[(\tilde{\varphi}_{t,s}(D; \hat{\vartheta}^{(-k)}) - \tilde{\varphi}_{t,s}(D; \vartheta^*))^2 \mid \mathcal{D}^{(-k)}].$$

Taking expectations and using the L_2 -Lipschitz bound in Ingredient 2 gives $\mathbb{E}[(m_k - \mu^*)^2] \rightarrow 0$, hence $m_k \rightarrow_p \mu^*$. Similarly, write

$$q_k - \mathbb{E}[\tilde{\varphi}_{t,s}(D; \vartheta^*)^2] = \mathbb{E}[(Z^{(k)})^2 - \tilde{\varphi}_{t,s}(D; \vartheta^*)^2 \mid \mathcal{D}^{(-k)}].$$

Using $(a^2 - b^2) = (a - b)(a + b)$ and conditional Cauchy-Schwarz,

$$\begin{aligned} |q_k - \mathbb{E}[\tilde{\varphi}_{t,s}(D; \vartheta^*)^2]| &= |\mathbb{E}[(Z^{(k)})^2 - \tilde{\varphi}_{t,s}(D; \vartheta^*)^2 \mid \mathcal{D}^{(-k)}]| \\ &\leq \mathbb{E}[|Z^{(k)} - \tilde{\varphi}_{t,s}(D; \vartheta^*)| \cdot |Z^{(k)} + \tilde{\varphi}_{t,s}(D; \vartheta^*)| \mid \mathcal{D}^{(-k)}] \\ &\leq \left(\mathbb{E}[(Z^{(k)} - \tilde{\varphi}_{t,s}(D; \vartheta^*))^2 \mid \mathcal{D}^{(-k)}] \right)^{1/2} \left(\mathbb{E}[(Z^{(k)} + \tilde{\varphi}_{t,s}(D; \vartheta^*))^2 \mid \mathcal{D}^{(-k)}] \right)^{1/2}. \end{aligned}$$

The first factor converges to 0 in probability by the L_2 -Lipschitz bound in Ingredient 2 and nuisance consistency (Ingredient 1). The second factor is $O_p(1)$ by the $(2 + \delta)$ moment assumption in [Assumption 2\(b\)](#) (uniform integrability). Therefore $q_k \rightarrow_p \mathbb{E}[\tilde{\varphi}_{t,s}(D; \vartheta^*)^2]$. Consequently, since $m_k \rightarrow_p \mu^*$,

$$\mathbb{E}[(Z^{(k)})^2 \mid \mathcal{D}^{(-k)}] - \mathbb{E}[Z^{(k)} \mid \mathcal{D}^{(-k)}]^2 \xrightarrow{p} \mathbb{E}[\tilde{\varphi}_{t,s}(D; \vartheta^*)^2] - \mu^1(\vartheta^*)^2 = \text{Var}(\tilde{\varphi}_{t,s}(D; \vartheta^*)) = V_{t,s}^*.$$

Step 4 (Average over folds). Since K_f is fixed, combining Step 2 and Step 3 and averaging over k yields $\hat{V}_{t,s} \xrightarrow{p} V_{t,s}^*$.

Extension to the aggregated estimator. For the weighted aggregated estimator with inner M -fold cross-fitted weights $\hat{\eta}^{(-k)}$, we define per-observation uncentered scaled influence functions that serve as the unified estimating equation for both the point estimate and the variance.

Unified estimating equation. For each observation i evaluated on fold $\mathcal{D}_{k(i)}$, define the uncentered IF

$$\hat{\Phi}_i := \begin{cases} \frac{N_{\text{all}}}{N_t} \left[(1 - \sum_j \hat{\eta}_j^{(-k(i))}) \tilde{Y}_{ot,i} + \sum_j \hat{\eta}_j^{(-k(i))} \psi(\phi(\mathbf{X}_i); \hat{\alpha}_{t,s_j}^{1,(-k(i))}) \right], & i \in \mathcal{D}_t, \\ \frac{N_{\text{all}}}{N_{s_j}} \hat{\eta}_j^{(-k(i))} w(\mathbf{X}_i; \hat{\gamma}_{s_j,1}^{(-k(i))}) (Y_i - \psi(\phi(\mathbf{X}_i); \hat{\alpha}_{t,s_j}^{1,(-k(i))})), & i \in \mathcal{D}_{s_j}, \end{cases}$$

Here $\tilde{Y}_{ot,i}$ is the uncentered AIPW pseudo-outcome for the target-only estimator,

$$\tilde{Y}_{ot,i} := \frac{A_i Y_i}{\hat{\pi}_{ot}(\mathbf{X}_i)} - \left(\frac{A_i}{\hat{\pi}_{ot}(\mathbf{X}_i)} - 1 \right) \psi(\phi(\mathbf{X}_i); \hat{\alpha}_{ot}^{1,(-k(i))}),$$

so that the centered IF satisfies $\hat{\varphi}_{ot,i} = \tilde{Y}_{ot,i} - \hat{\mu}_{ot,k(i)}^1$. Likewise, $\hat{\zeta}_{t,s_j,i} = \psi(\phi(\mathbf{X}_i); \hat{\alpha}_{t,s_j}^{1,(-k(i))}) - \tilde{E}_{t,k(i)}[\psi(\cdot)]$. All nuisance parameters $\hat{\alpha}_{t,s_j}^{1,(-k)}$, $\hat{\gamma}_{s_j,1}^{(-k)}$, and $\hat{\pi}_{ot}^{(-k)}$ are trained on data excluding the evaluation fold $\mathcal{D}_k$ (with the fold-summed calibration over secondary folds described above), and $\hat{\eta}^{(-k)}$ is estimated via the FACE-style inner validation loop within $\mathcal{T}_k = \mathcal{D} \setminus \mathcal{D}_k$. Note that the source contribution is written in uncentered form; subtracting the fold-specific correction $\hat{\delta}_{t,s_j}^{(-k)} = \tilde{E}_{s_j,k}[w(Y - \psi)]$ yields the centered influence component $\hat{\xi}_{t,s_j,i}$ used in the variance-component derivations above.

The point estimate and variance are then derived from the same set $\{\hat{\Phi}_i\}_{i=1}^{N_{\text{all}}}$:

$$\begin{aligned} \hat{\mu}_{\text{agg}}^1 &= \bar{\Phi} := \frac{1}{N_{\text{all}}} \sum_{i=1}^{N_{\text{all}}} \hat{\Phi}_i, \\ \hat{V}_{\text{agg}} &= \frac{1}{N_{\text{all}}^2} \sum_{g \in \{t, s_1, \dots, s_K\}} \sum_{i \in \mathcal{D}_g} (\hat{\Phi}_i - \bar{\Phi}_g)^2, \end{aligned}$$

where $\bar{\Phi}_g := \frac{1}{N_g} \sum_{i \in \mathcal{D}_g} \hat{\Phi}_i$ is the within-group mean for group $g \in \{t, s_1, \dots, s_K\}$. In the ANOVA decomposition $TSS = WSS + BSS$, the estimator $\hat{V}_{\text{agg}}$ uses only the within-group sum of squares WSS/N_{all}^2 ; the between-group sum of squares BSS/N_{all}^2 is excluded.

Why within-group centering is necessary. Because $\hat{\Phi}_i$ is scaled by N_{all}/N_g (i.e., N_{all}/N_t for target, N_{all}/N_{s_j} for source), the conditional expectations differ across groups:

$\mathbb{E}[\widehat{\Phi}_i \mid i \in \mathcal{D}_t] \approx (N_{\text{all}}/N_t)\mu^1$ and $\mathbb{E}[\widehat{\Phi}_i \mid i \in \mathcal{D}_{s_j}] \approx 0$ (when $\delta_{t,s_j}^* = 0$). The inferential target here is

$$\frac{1}{N_{\text{all}}^2} \sum_{i=1}^{N_{\text{all}}} \text{Var}(\Phi_i),$$

rather than $\text{Var}(\Phi)$ under a pooled-mixture distribution. Global centering at $\bar{\Phi}$ estimates the latter object and introduces an additional between-group term

$$\sum_g \rho_g (\mu_g - \mu)^2,$$

equivalently the ANOVA component BSS/N_{all}^2 , which is of the same order as $V_{\text{agg}}^*/N_{\text{all}}$ and therefore does not vanish. By contrast, WSS/N_{all}^2 expands as

$$\widehat{V}_{\text{agg}} = \frac{1}{N_t} \widehat{\sigma}_T^2 + \sum_{j=1}^K \frac{1}{N_{s_j}} \widehat{\eta}_j^2 \widehat{\sigma}_{s_j}^2,$$

where

$$\widehat{\sigma}_T^2 = \widehat{\text{Var}}_t \left((1 - \sum_j \widehat{\eta}_j) \widehat{\varphi}_{ot,i} + \sum_j \widehat{\eta}_j \widehat{\zeta}_{t,s_j,i} \right).$$

Hence $\widehat{\sigma}_T^2$ is not a target-only variance term; it already contains the shared-target cross-covariances (including terms corresponding to $\text{Cov}(\widehat{\varphi}_{ot}, \widehat{\zeta}_{t,s_j})$ and $\text{Cov}(\widehat{\zeta}_{t,s_j}, \widehat{\zeta}_{t,s_k})$ for $j \neq k$), fully consistent with the $\widehat{C}_{ot,s_j}$ and $\widehat{C}_{t,s_j,s_k}$ terms in [Eq. \(16\)](#).

With balanced folds ($|\mathcal{D}_{t,k}| = N_t/K_f$, $|\mathcal{D}_{s_j,k}| = N_{s_j}/K_f$), $\bar{\Phi}$ coincides with the fold-average $(1/K_f) \sum_k \widehat{\mu}_{\text{agg},k}^1$; with minor fold imbalance the difference is $O(N_{\text{all}}^{-1})$.

This unified form ensures $\widehat{\mu}_{\text{agg}}^1 - \mu^1 = \frac{1}{N_{\text{all}}} \sum_{i=1}^{N_{\text{all}}} (\widehat{\Phi}_i - \mu^1)$, so the DML representation of the estimation error as an IF average is preserved. By the same arguments as above (nuisance consistency, L_2 -Lipschitz continuity, and $\widehat{\boldsymbol{\eta}}^{(-k)} \rightarrow_p \boldsymbol{\eta}^*$), $\widehat{V}_{\text{agg}} \xrightarrow{p} V_{\text{agg}}^*$.

S2.4 Technical Lemmas

Lemma 3 (CLT for IF Average). *Under [Assumption 1](#) and [Assumption 2](#), the influence function average is asymptotically normal:*

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N \varphi_{t,s}(D_i; \vartheta^*) \xrightarrow{d} \mathcal{N}(0, V_{t,s}^*).$$

Proof. The observations consist of two independent samples: $\{D_i : i \in \mathcal{D}_t\}$ drawn i.i.d. from the target distribution and $\{D_i : i \in \mathcal{D}_s\}$ drawn i.i.d. from the source distribution, with

$N_t/N \rightarrow \rho_t \in (0, 1)$ and $N_s/N \rightarrow \rho_s$. Thus the summands $\{\varphi_{t,s}(D_i; \vartheta^*)\}_{i=1}^N$ are independent (but not identically distributed), have mean zero, and have finite $(2 + \delta)$ moments by [Assumption 2\(b\)](#). In particular, the Lindeberg condition holds. We verify a Lyapunov condition, which implies the Lindeberg condition for this independent triangular array. Let $X_{i,N} := \varphi_{t,s}(D_i; \vartheta^*)$ so that $\mathbb{E}[X_{i,N}] = 0$ and $\sum_{i=1}^N \text{Var}(X_{i,N}) = NV_{t,s}^* + o(N)$. By [Assumption 2\(b\)](#), $\sup_{i,N} \mathbb{E}[|X_{i,N}|^{2+\delta}] < \infty$ for some $\delta > 0$. Therefore,

$$\frac{1}{(NV_{t,s}^*)^{1+\delta/2}} \sum_{i=1}^N \mathbb{E}[|X_{i,N}|^{2+\delta}] \leq \frac{CN}{N^{1+\delta/2}} = CN^{-\delta/2} \rightarrow 0,$$

which is the Lyapunov condition. Hence the Lindeberg–Feller CLT yields $\frac{1}{\sqrt{N}} \sum_{i=1}^N X_{i,N} \xrightarrow{d} \mathcal{N}(0, V_{t,s}^*)$, proving the lemma. $\square$

S2.5 Extension to the Control Mean and the TATE

The results above were stated for the treated potential outcome mean $\mu^1 = \mathbb{E}_t[Y(1)]$ for notational simplicity. The same proof strategy applies to the control mean μ^0 by replacing the treatment level 1 by 0 throughout (including the corresponding nuisance parameters (α^0, γ_0) and weights). In particular, define the control-arm weight

$$w^0(\mathbf{X}; \gamma_0) := \frac{I(A = 0)}{e^{\phi(\mathbf{X})^T \gamma_0}},$$

and the cross-fitted estimator

$$\hat{\mu}_{t,s}^{0,\text{CF}} = \frac{1}{K_f} \sum_{k=1}^{K_f} \left\{ \tilde{E}_{t,k} \left[\psi(\phi(\mathbf{X}); \hat{\alpha}_{t,s}^{0,(-k)}) \right] + \tilde{E}_{s,k} \left[w^0(\mathbf{X}; \hat{\gamma}_{s,0}^{(-k)}) \left(Y - \psi(\phi(\mathbf{X}); \hat{\alpha}_{t,s}^{0,(-k)}) \right) \right] \right\}.$$

Let $\vartheta^{0,*} = (\alpha^{0,*}, \gamma_0^*)$ denote the population targets of the control-arm calibrated losses (the same construction as for μ^1 but with $I(A = 0)$), and define the corresponding centered influence function $\varphi_{t,s}^0(D; \vartheta^{0,*})$ analogously.

Theorem 4 (Asymptotic Normality of $\hat{\mu}_{t,s}^{0,\text{CF}}$). *Under the analogues of [Assumptions 1 to 3](#) for the control arm, and assuming identification of $\mu^0(\vartheta^{0,*}) = \mu^0$ under outcome-model or weight-model correctness (control-arm analogue of [Lemma 1](#)), if the total sparsity satisfies $\frac{s_{\text{tot}}^2 \log^2(p)}{N} \rightarrow 0$, then*

$$\sqrt{N} \left(\hat{\mu}_{t,s}^{0,\text{CF}} - \mu^0 \right) \xrightarrow{d} \mathcal{N}(0, V_{t,s}^{0,*}), \quad V_{t,s}^{0,*} := \text{Var}(\varphi_{t,s}^0(D; \vartheta^{0,*})).$$

Proof. The proof is identical to that of [Theorem 3](#) after replacing the treatment level 1 by 0 in the definition of the score contribution, the calibrated moment conditions, and the corresponding nuisance estimators. $\square$

Corollary 1 (Asymptotic Normality of the TATE). *With $\tau = \mu^1 - \mu^0$ as defined above, define the cross-fitted TATE estimator*

$$\hat{\tau}_{t,s}^{\text{CF}} := \hat{\mu}_{t,s}^{1,\text{CF}} - \hat{\mu}_{t,s}^{0,\text{CF}}.$$

Under the conditions of [Theorem 3](#) and [Theorem 4](#), we have the asymptotic linear representation

$$\sqrt{N} \left(\hat{\tau}_{t,s}^{\text{CF}} - \tau \right) = \frac{1}{\sqrt{N}} \sum_{i=1}^N \left(\varphi_{t,s}(D_i; \vartheta^*) - \varphi_{t,s}^0(D_i; \vartheta^{0,*}) \right) + o_p(1),$$

and hence

$$\sqrt{N} \left(\hat{\tau}_{t,s}^{\text{CF}} - \tau \right) \xrightarrow{d} \mathcal{N}(0, V_{t,s}^{\tau,*}), \quad V_{t,s}^{\tau,*} := \text{Var} \left(\varphi_{t,s}(D; \vartheta^*) - \varphi_{t,s}^0(D; \vartheta^{0,*}) \right).$$

Lemma 4 (Cross-fitted ℓ_1 -penalized nuisance rates (adapted from [Hou et al. \(2025\)](#), Lemma A21)). *Fix a constant number of folds K_f . Let the full sample $\mathcal{D} = \{D_1, \dots, D_N\}$ be partitioned into folds $\mathcal{D} = \bigcup_{k=1}^{K_f} \mathcal{D}_k$, and let $\mathcal{D}^{(-k)} = \mathcal{D} \setminus \mathcal{D}_k$ denote the out-of-fold data. For each fold k , let $\hat{\gamma}^{(-k)}$ be an estimator trained on $\mathcal{D}^{(-k)}$ (hence independent of $\mathcal{D}_k$). For an observation index $i \in \{1, \dots, N\}$, let $\text{fold}(i) \in \{1, \dots, K_f\}$ denote the fold index such that $D_i \in \mathcal{D}_{\text{fold}(i)}$; we use $\hat{\gamma}^{(-\text{fold}(i))}$ to denote the out-of-fold nuisance estimate used when evaluating the loss contribution from D_i . Consider the cross-fitted ℓ_1 -penalized estimator*

$$\hat{\beta} \in \arg \min_{\beta \in \mathbb{R}^p} \left\{ \frac{1}{K_f} \sum_{k=1}^{K_f} \ell_k(\beta; \hat{\gamma}^{(-k)}) + \lambda \|\beta\|_1 \right\}, \quad \ell_k(\beta; \gamma) := \tilde{E}_k[\ell(D; \beta, \gamma)],$$

where $\ell(D; \beta, \gamma)$ is convex in β . Define also the fold-averaged empirical risk

$$L_N(\beta) := \frac{1}{K_f} \sum_{k=1}^{K_f} \ell_k(\beta; \hat{\gamma}^{(-k)}).$$

Let $\bar{\beta}$ be the population minimizer at a population nuisance target $\bar{\gamma}$,

$$\bar{\beta} \in \arg \min_{\beta \in \mathbb{R}^p} \mathbb{E}[\ell(D; \beta, \bar{\gamma})], \quad s_{\beta} := \|\bar{\beta}\|_0.$$

Assume the following high-dimensional regularity conditions (cf. Assumption 6 of [Hou et al. \(2025\)](#) and the RSC framework of [Negahban et al. \(2012\)](#)):

- (a) (*Sub-Gaussian covariates*) Each coordinate of $\phi(\mathbf{X})$ is sub-Gaussian with $\max_{1 \leq j \leq p} \|\phi_j(\mathbf{X})\|_{\psi_2} \leq K_\phi$ (allowing $\|\phi(\mathbf{X})\|_\infty$ to grow at the usual $\sqrt{\log p}$ rate).
- (b) (*Response/link tails and smoothness*) Y is uniformly sub-Gaussian (or bounded), and the GLM mean function and its derivative are uniformly bounded on a neighborhood of $\bar{\beta}$.
- (c) (*Identifiability*) The population Gram matrix is non-degenerate: $\lambda_{\min}(\mathbb{E}[\phi(\mathbf{X})\phi(\mathbf{X})^T]) \geq c > 0$.
- (d) (*Weight regularity*) Any weights induced by γ (e.g., density-ratio type weights) have finite moments and are locally Lipschitz in γ on a neighborhood of $\bar{\gamma}$ (sufficient for the score stability bound used below).
- (e) (*Restricted strong convexity*) With probability tending to 1, $L_N(\cdot)$ satisfies an RSC inequality on the standard cone associated with $\text{supp}(\bar{\beta})$.

In our FACE-HD setting, we impose these regularity conditions in addition to [Assumption 1](#), [Assumption 2](#), and [Assumption 3](#) (noting that (a) matches [Assumption 1\(a\)](#), while (d) is ensured by the overlap/weight-moment assumptions and local smoothness). Choose $\lambda \asymp \sqrt{\log(p)/N}$. Then:

Remark 3 (On the choice of N in λ). The scaling $\lambda \asymp \sqrt{\log(p)/N}$ is stated in terms of the total sample size N for notational simplicity. In applications where the nuisance loss for a given arm/site pair is effectively averaged over an arm-specific subsample (e.g., treated units in the source), one may equivalently write $\lambda \asymp \sqrt{\log(p)/N_{\text{eff}}}$ where N_{eff} is the corresponding effective sample size. Under the sampling-fraction regime and overlap/positivity conditions stated above, $N_{\text{eff}} \asymp N$, so the rate statements are unchanged.

1. (**General**)

$$\|\hat{\beta} - \bar{\beta}\|_2 = O_p\left(\sqrt{\frac{s_\beta \log(p)}{N}}\right) + O_p\left(\sup_{1 \leq k \leq K_f} \|\hat{\gamma}^{(-k)} - \bar{\gamma}\|_2\right).$$

2. (**Special**) If, for each fold k , the conditional foldwise score at $\bar{\beta}$ satisfies

$$\mathbb{E}\left[\nabla_{\beta} \ell(D; \bar{\beta}, \hat{\gamma}^{(-k)}) \mid \mathcal{D}^{(-k)}\right] = \mathbf{0},$$

then the plug-in score term disappears and

$$\|\hat{\beta} - \bar{\beta}\|_2 = O_p\left(\sqrt{\frac{s_\beta \log(p)}{N}}\right).$$

Moreover, we can convert the ℓ_2 -rate to an ℓ_1 -rate by an explicit cone argument. Let $\Delta := \hat{\beta} - \bar{\beta}$ and $\mathcal{J} := \text{supp}(\bar{\beta})$ with $|\mathcal{J}| = s_\beta$. From the proof below (see the KKT-based cone inequality leading to [Eq. \(21\)](#)), on a high-probability event we have

$$\|\Delta_{\mathcal{J}^c}\|_1 \leq 3\|\Delta_{\mathcal{J}}\|_1 + \frac{2}{\lambda}\|B\|_\infty\|\Delta\|_1,$$

where $\|B\|_\infty$ is the plug-in score term induced by cross-fitting. With $\lambda \asymp \sqrt{\log(p)/N}$ and $\|B\|_\infty = O_p(\sqrt{\log(p)/N})$ (proved below), we have $(2/\lambda)\|B\|_\infty \leq 1/2$ w.p.a.1, and hence

$$\|\Delta\|_1 = \|\Delta_{\mathcal{J}}\|_1 + \|\Delta_{\mathcal{J}^c}\|_1 \leq 4\|\Delta_{\mathcal{J}}\|_1 \leq 4\sqrt{s_\beta}\|\Delta\|_2.$$

Combining this with the ℓ_2 bounds above yields the corresponding ℓ_1 -rate:

$$\|\hat{\beta} - \bar{\beta}\|_1 = O_p\left(s_\beta \sqrt{\frac{\log(p)}{N}}\right) + O_p\left(\sqrt{s_\beta} \sup_{1 \leq k \leq K_f} \|\hat{\gamma}^{(-k)} - \bar{\gamma}\|_2\right),$$

with the second term dropping in the special case.

Instantiation for FACE-HD calibrated losses. In our application, β is instantiated as either α^1 or γ_1 . If [Assumption 3\(d\)](#) holds (as in the foldwise score-centering condition used by [Hou et al. \(2025\)](#)), then we may invoke the (**Special**) case above and obtain

$$\|\hat{\alpha}_{t,s}^{1,(-k)} - \alpha^{1,*}\|_q = O_p\left(s_\alpha^{1-1/q} \sqrt{\frac{\log(p)}{N}}\right), \quad \|\hat{\gamma}_{s,1}^{(-k)} - \gamma_1^*\|_q = O_p\left(s_\gamma^{1-1/q} \sqrt{\frac{\log(p)}{N}}\right), \quad q \in \{1, 2\}.$$

Proof. We give a detailed argument in our notation; it follows the structure of [Hou et al. \(2025\)](#) (proof of Lemma A21). Write the (random) fold-averaged empirical loss as

$$L_N(\beta) := \frac{1}{K_f} \sum_{k=1}^{K_f} \ell_k(\beta; \hat{\gamma}^{(-k)}), \quad \hat{\beta} \in \arg \min_{\beta} \{L_N(\beta) + \lambda \|\beta\|_1\}.$$

Let $\Delta := \hat{\beta} - \bar{\beta}$, and let $\mathcal{J} := \text{supp}(\bar{\beta})$ with $|\mathcal{J}| = s_\beta$.

Step 1 (Basic inequality). By optimality,

$$L_N(\bar{\beta} + \Delta) + \lambda \|\bar{\beta} + \Delta\|_1 \leq L_N(\bar{\beta}) + \lambda \|\bar{\beta}\|_1,$$

so

$$L_N(\bar{\beta} + \Delta) - L_N(\bar{\beta}) \leq \lambda(\|\bar{\beta}\|_1 - \|\bar{\beta} + \Delta\|_1) \leq \lambda(\|\Delta_{\mathcal{J}}\|_1 - \|\Delta_{\mathcal{J}^c}\|_1).$$

Step 2 (RSC expansion). By convexity,

$$L_N(\bar{\beta} + \Delta) - L_N(\bar{\beta}) \geq \nabla L_N(\bar{\beta})^T \Delta + \mathcal{R}_N(\Delta),$$

where the remainder $\mathcal{R}_N(\Delta)$ satisfies the RSC lower bound (Assumption 3(a) as applied to this loss)

$$\mathcal{R}_N(\Delta) \geq \kappa_1 \|\Delta\|_2^2 - \kappa_2 \frac{\log(p)}{N} \|\Delta\|_1^2$$

on the standard restricted cone. Combining with Step 1 yields

$$\kappa_1 \|\Delta\|_2^2 \leq \|\nabla L_N(\bar{\beta})\|_\infty \|\Delta\|_1 + \lambda(\|\Delta_{\mathcal{J}}\|_1 - \|\Delta_{\mathcal{J}^c}\|_1) + \kappa_2 \frac{\log(p)}{N} \|\Delta\|_1^2. \quad (20)$$

Step 3 (Score decomposition and choice of λ). Decompose the score

$$\nabla L_N(\bar{\beta}) = \underbrace{(\nabla L_N(\bar{\beta}) - \mathbb{E}[\nabla L_N(\bar{\beta}) \mid \{\mathcal{D}^{(-k)}\}_{k=1}^{K_f}])}_{\text{stochastic term } A} + \underbrace{\mathbb{E}[\nabla L_N(\bar{\beta}) \mid \{\mathcal{D}^{(-k)}\}_{k=1}^{K_f}]}_{\text{plug-in term } B}.$$

By cross-fitting, conditional on $\{\mathcal{D}^{(-k)}\}$, each fold average $\tilde{E}_k[\nabla_{\beta} \ell(D; \bar{\beta}, \hat{\gamma}^{(-k)})]$ is an average of independent sub-exponential random vectors with uniformly controlled coordinates, so a union bound with Bernstein/Hoeffding inequalities gives

$$\|A\|_\infty = O_p\left(\sqrt{\frac{\log(p)}{N}}\right).$$

Choose $\lambda \asymp \sqrt{\log(p)/N}$ so that $\lambda \geq 2\|A\|_\infty$ w.p.a.1.

For the plug-in term, note that

$$B - \mathbb{E}[\nabla_{\beta} \ell(D; \bar{\beta}, \bar{\gamma})] = \mathbb{E}[\nabla_{\beta} \ell(D; \bar{\beta}, \hat{\gamma}^{(-\text{fold}(D))}) - \nabla_{\beta} \ell(D; \bar{\beta}, \bar{\gamma}) \mid \{\mathcal{D}^{(-k)}\}],$$

and the weight-regularity/Lipschitz condition implies

$$\|B - \mathbb{E}[\nabla_{\beta} \ell(D; \bar{\beta}, \bar{\gamma})]\|_2 \leq C \sup_{1 \leq k \leq K_f} \|\hat{\gamma}^{(-k)} - \bar{\gamma}\|_2.$$

In the special case, $\mathbb{E}[\nabla_{\beta} \ell(D; \bar{\beta}, \hat{\gamma}^{(-k)}) \mid \mathcal{D}^{(-k)}] = 0$ for all k , hence $B = \mathbf{0}$.

Step 4 (Cone property and ℓ_2 rate). Plugging $\|\nabla L_N(\bar{\beta})\|_\infty \leq \|A\|_\infty + \|B\|_\infty$ into Eq. (20) and using $\lambda \geq 2\|A\|_\infty$ yields the cone inequality

$$\|\Delta_{\mathcal{J}^c}\|_1 \leq 3\|\Delta_{\mathcal{J}}\|_1 + \frac{2}{\lambda} \|B\|_\infty \|\Delta\|_1. \quad (21)$$

On the event $\{(2/\lambda)\|B\|_\infty \leq 1/2\}$, [Eq. \(21\)](#) implies $\|\Delta_{\mathcal{J}^c}\|_1 \leq 7\|\Delta_{\mathcal{J}}\|_1$ and hence $\|\Delta\|_1 \leq 8\|\Delta_{\mathcal{J}}\|_1 \leq 8\sqrt{s_\beta}\|\Delta\|_2$. Plugging this back into the RSC lower bound ([Assumption 3\(a\)](#) as applied to this loss) absorbs the $\kappa_2(\log(p)/N)\|\Delta\|_1^2$ term and yields

$$\|\Delta\|_2 = O_p(\sqrt{s_\beta}\lambda) + O_p\left(\sup_{1 \leq k \leq K_f} \|\hat{\gamma}^{(-k)} - \bar{\gamma}\|_2\right).$$

This is the general-case ℓ_2 rate. In the special case, $B = \mathbf{0}$ and the second term drops.

Step 5 (ℓ_1 rate). From the cone property and $\|\Delta_{\mathcal{J}}\|_1 \leq \sqrt{s_\beta}\|\Delta\|_2$,

$$\|\Delta\|_1 = O_p(s_\beta\lambda) + O_p\left(\sqrt{s_\beta} \sup_{1 \leq k \leq K_f} \|\hat{\gamma}^{(-k)} - \bar{\gamma}\|_2\right),$$

with the second term dropping in the special case.

Finally, for FACE-HD, the **(Special)** conclusion may be invoked for $\beta = \alpha^1$ and $\beta = \gamma_1$ only when the foldwise score-centering condition is assumed/verified as in [Assumption 3\(d\)](#). Otherwise, one should use the **(General)** bounds, which explicitly propagate the plug-in nuisance error. $\square$

Lemma 5 (Orthogonality from Calibrated Losses). *Under [Assumption 3\(c\)](#), let $\vartheta^* = (\alpha^{1,*}, \gamma_1^*)$ be the population targets of the calibrated losses in the two-level algorithm (see [Supplemental Material S1](#); [Eq. \(9\)](#) and [Eq. \(10\)](#)):*

$$\begin{aligned} L_\gamma(\gamma; \alpha^{1,*}) &= \mathbb{E}_t \left[\nabla_\alpha \psi(\phi(\mathbf{X}); \alpha^{1,*})^T \gamma \right] + \mathbb{E}_s \left[I(A=1) e^{-\phi(\mathbf{X})^T \gamma} \psi'(\mathcal{T}(\phi(\mathbf{X})^T \alpha^{1,*})) \right] \\ L_\alpha(\alpha; \gamma_1^*) &= \mathbb{E}_s \left[w_\tau(\mathbf{X}; \gamma_1^*) \left(-Y \cdot (\phi(\mathbf{X})^T \alpha) + \Psi(\phi(\mathbf{X})^T \alpha) \right) \right] \end{aligned}$$

(Ignoring regularization terms, which vanish in population). For canonical-link GLMs,

$$\nabla_\alpha \psi(\phi(\mathbf{X}); \alpha) = \phi(\mathbf{X}) \psi'(\phi(\mathbf{X})^T \alpha).$$

Remark 4 (Penalization and approximate orthogonality). The population moment conditions below are derived from the unpenalized population losses. In finite samples, the ℓ_1 penalty induces a KKT subgradient term of order λ in the estimating equations. Under the nuisance rates in [Lemma 4](#) and the sparsity scaling used in [Theorem 3](#), this penalty-induced perturbation contributes at most an $o_p(N^{-1/2})$ term to the final DR remainder and therefore does not affect the first-order limit theory.

Under asymptotically inactive truncation ([Assumption 2\(c\)](#)), the population first-order optimality conditions are therefore equivalent to the following moment conditions at ϑ^* :

$$\begin{aligned} (i) \quad & \mathbb{E}_t \left[\nabla_{\alpha} \psi(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*}) \right] - \mathbb{E}_s \left[w(\mathbf{X}; \boldsymbol{\gamma}_1^*) \nabla_{\alpha} \psi(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*}) \right] = \mathbf{0} \\ (ii) \quad & \mathbb{E}_s \left[w(\mathbf{X}; \boldsymbol{\gamma}_1^*) \phi(\mathbf{X}) \{Y - \psi(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})\} \right] = \mathbf{0}. \end{aligned}$$

Proof. We derive the moment conditions by taking the gradients of the population loss functions L_{γ} and L_{α} with respect to the parameter being optimized and setting them to zero at the minimum ϑ^* .

Condition (i) from FOC of $L_{\gamma}(\boldsymbol{\gamma}; \boldsymbol{\alpha}^{1,*})$ w.r.t. $\boldsymbol{\gamma}$: The gradient of $L_{\gamma}(\boldsymbol{\gamma}; \boldsymbol{\alpha}^{1,*})$ with respect to $\boldsymbol{\gamma}$ is:

$$\begin{aligned} \nabla_{\boldsymbol{\gamma}} L_{\boldsymbol{\gamma}}(\boldsymbol{\gamma}; \boldsymbol{\alpha}^{1,*}) &= \mathbb{E}_t \left[\nabla_{\boldsymbol{\alpha}} \psi(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*}) \right] \\ &\quad + \mathbb{E}_s \left[I(A=1)(-\phi(\mathbf{X})) e^{-\phi(\mathbf{X})^T \boldsymbol{\gamma}} \psi'(\mathcal{T}(\phi(\mathbf{X})^T \boldsymbol{\alpha}^{1,*})) \right] \end{aligned}$$

Setting this gradient to zero at $\boldsymbol{\gamma} = \boldsymbol{\gamma}_1^*$ gives:

$$\mathbb{E}_t \left[\nabla_{\alpha} \psi(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*}) \right] = \mathbb{E}_s \left[\phi(\mathbf{X}) w(\mathbf{X}; \boldsymbol{\gamma}_1^*) \psi'(\mathcal{T}(\phi(\mathbf{X})^T \boldsymbol{\alpha}^{1,*})) \right].$$

This equation is constructed to enforce the moment condition (i). Note that for standard GLMs with canonical links,

$$\nabla_{\alpha} \psi(\phi(\mathbf{X}); \boldsymbol{\alpha}) = \phi(\mathbf{X}) \psi'(\phi(\mathbf{X})^T \boldsymbol{\alpha}).$$

Truncation Error Analysis: Define the event $\mathcal{E}_{\alpha,N} = \{|\phi(\mathbf{X})^T \boldsymbol{\alpha}^{1,*}| \leq M_{\tau,N}\}$. By [Assumption 2\(c\)](#), $P(\mathcal{E}_{\alpha,N}^c \mid R=s) = o(N^{-1})$. On $\mathcal{E}_{\alpha,N}$, $\mathcal{T}(\phi(\mathbf{X})^T \boldsymbol{\alpha}^{1,*}) = \phi(\mathbf{X})^T \boldsymbol{\alpha}^{1,*}$, so $\psi'(\mathcal{T}(\cdot)) = \psi'(\cdot)$ exactly. Therefore, replacing $\psi'(\mathcal{T}(\phi(\mathbf{X})^T \boldsymbol{\alpha}^{1,*}))$ by $\psi'(\phi(\mathbf{X})^T \boldsymbol{\alpha}^{1,*})$ in the population FOC changes the source-side moment by at most an $o(1)$ term, and condition (i) holds asymptotically.

Condition (ii) from FOC of $L_{\alpha}(\boldsymbol{\alpha}; \boldsymbol{\gamma}_1^*)$ w.r.t. $\boldsymbol{\alpha}$: The loss L_{α} is the population weighted GLM log-likelihood. Its gradient with respect to $\boldsymbol{\alpha}$ is:

$$\nabla_{\alpha} L_{\alpha}(\boldsymbol{\alpha}; \boldsymbol{\gamma}_1^*) = \mathbb{E}_s \left[w_{\tau}(\mathbf{X}; \boldsymbol{\gamma}_1^*) \left(-Y \cdot \phi(\mathbf{X}) + \nabla_{\alpha} \Psi(\phi(\mathbf{X})^T \boldsymbol{\alpha}) \right) \right]$$

Using the GLM identity $\nabla_{\alpha} \Psi(u) = \psi(u) \phi(\mathbf{X})$ where $u = \phi(\mathbf{X})^T \boldsymbol{\alpha}$:

$$\nabla_{\alpha} L_{\alpha}(\boldsymbol{\alpha}; \boldsymbol{\gamma}_1^*) = \mathbb{E}_s \left[w_{\tau}(\mathbf{X}; \boldsymbol{\gamma}_1^*) \phi(\mathbf{X}) \left(\psi(\phi(\mathbf{X})^T \boldsymbol{\alpha}) - Y \right) \right]$$

Setting this gradient to zero at $\boldsymbol{\alpha} = \boldsymbol{\alpha}^{1,*}$ gives:

$$\mathbb{E}_s \left[w_\tau(\mathbf{X}; \boldsymbol{\gamma}_1^*) \phi(\mathbf{X}) \left(Y - \psi(\phi(\mathbf{X})^T \boldsymbol{\alpha}^{1,*}) \right) \right] = \mathbf{0}.$$

Finally, under [Assumption 2\(c\)](#), truncation is asymptotically inactive at $\boldsymbol{\gamma}_1^*$, so $w_\tau(\mathbf{X}; \boldsymbol{\gamma}_1^*) = w(\mathbf{X}; \boldsymbol{\gamma}_1^*)$ w.p.a.1. This yields condition (ii). $\square$

Lemma 6 (Bias-Variance Bounds). *Let $\vartheta_n = (\boldsymbol{\alpha}_n^1, \boldsymbol{\gamma}_{1,n})$ be a sequence of nuisance parameters converging to $\vartheta^* = (\boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)$.*

1. **(Bias Bound - Double Robustness)** *If ϑ^* minimizes the population calibrated losses ([Lemma 5](#) holds), then:*

$$\begin{aligned} |\mathbb{E}[\tilde{\varphi}_{t,s}(D; \vartheta_n)] - \mathbb{E}[\tilde{\varphi}_{t,s}(D; \vartheta^*)]| &\leq C \cdot \left(\|\boldsymbol{\alpha}_n^1 - \boldsymbol{\alpha}^{1,*}\|_2^2 + \|\boldsymbol{\gamma}_{1,n} - \boldsymbol{\gamma}_1^*\|_2^2 \right. \\ &\quad \left. + \|\boldsymbol{\alpha}_n^1 - \boldsymbol{\alpha}^{1,*}\|_2 \cdot \|\boldsymbol{\gamma}_{1,n} - \boldsymbol{\gamma}_1^*\|_2 \right) \end{aligned}$$

2. **(Variance Bound)**

$$\text{Var}(\tilde{\varphi}_{t,s}(D; \vartheta_n) - \tilde{\varphi}_{t,s}(D; \vartheta^*)) \leq C \cdot \left(\|\boldsymbol{\alpha}_n^1 - \boldsymbol{\alpha}^{1,*}\|_2^2 + \|\boldsymbol{\gamma}_{1,n} - \boldsymbol{\gamma}_1^*\|_2^2 \right)$$

Proof. We analyze the bias term $\mathbb{E}[\tilde{\varphi}_{t,s}(D; \vartheta_n) - \tilde{\varphi}_{t,s}(D; \vartheta^*)]$. Let $\Delta_\alpha = \boldsymbol{\alpha}_n^1 - \boldsymbol{\alpha}^{1,*}$ and $\Delta_\gamma = \boldsymbol{\gamma}_{1,n} - \boldsymbol{\gamma}_1^*$. We decompose the bias using an intermediate step:

$$\text{Bias} = \underbrace{\mathbb{E}[\tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}_n^1, \boldsymbol{\gamma}_{1,n}) - \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_{1,n})]}_{\text{Bias from } \boldsymbol{\alpha}} + \underbrace{\mathbb{E}[\tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_{1,n}) - \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)]}_{\text{Bias from } \boldsymbol{\gamma}}$$

Step 1 (Bias from $\boldsymbol{\alpha}$: mean-value expansion + orthogonality). Define the path $h_\alpha(t) := \mathbb{E}[\tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*} + t\Delta_\alpha, \boldsymbol{\gamma}_{1,n})]$ for $t \in [0, 1]$. By the fundamental theorem of calculus (and dominated convergence, justified by [Assumption 2\(b\)](#) and local boundedness of the derivatives), we have

$$h_\alpha(1) - h_\alpha(0) = \int_0^1 h'_\alpha(t) dt = \int_0^1 \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*} + t\Delta_\alpha, \boldsymbol{\gamma}_{1,n})]^T \Delta_\alpha dt.$$

Hence

$$\begin{aligned} \text{Bias from } \boldsymbol{\alpha} &= \left(\mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)] \right)^T \Delta_\alpha + \int_0^1 \left\{ \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*} + t\Delta_\alpha, \boldsymbol{\gamma}_{1,n})] \right. \\ &\quad \left. - \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)] \right\}^T \Delta_\alpha dt. \end{aligned}$$

The first term is zero by [Lemma 5\(i\)](#). For the second term, we use Cauchy–Schwarz:

$$|\text{Bias from } \boldsymbol{\alpha}| \leq \|\Delta_\alpha\|_2 \int_0^1 \left\| \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*} + t\Delta_\alpha, \boldsymbol{\gamma}_{1,n})] - \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)] \right\|_2 dt.$$

The mapping $\vartheta \mapsto \mathbb{E}[\nabla_{\alpha} \tilde{\varphi}_{t,s}(D; \vartheta)]$ is locally Lipschitz in ϑ by the explicit derivative formula below, the bounded-slope/Lipschitz condition on ψ' ([Assumption 2\(b\)\(i\)](#)), and the bounded-weights/moment conditions ([Assumption 2\(a,b\)](#)). Therefore, for ϑ_n sufficiently close to ϑ^* ,

$$\left\| \mathbb{E}[\nabla_{\alpha} \tilde{\varphi}_{t,s}(D; \alpha^{1,*} + t\Delta_{\alpha}, \gamma_{1,n})] - \mathbb{E}[\nabla_{\alpha} \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^*)] \right\|_2 \leq C(\|\Delta_{\alpha}\|_2 + \|\Delta_{\gamma}\|_2),$$

uniformly over $t \in [0, 1]$, and hence

$$|\text{Bias from } \alpha| \leq C(\|\Delta_{\alpha}\|_2^2 + \|\Delta_{\alpha}\|_2 \|\Delta_{\gamma}\|_2).$$

Step 2 (Bias from γ : mean-value expansion + orthogonality). Similarly define $h_{\gamma}(t) := \mathbb{E}[\tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^* + t\Delta_{\gamma})]$. Then

$$\text{Bias from } \gamma = \int_0^1 \mathbb{E}[\nabla_{\gamma} \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^* + t\Delta_{\gamma})]^T \Delta_{\gamma} dt.$$

Adding and subtracting $\mathbb{E}[\nabla_{\gamma} \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^*)]$, the leading term is zero by [Lemma 5\(ii\)](#), and the remainder is bounded (again by local Lipschitz continuity and Cauchy–Schwarz) as

$$|\text{Bias from } \gamma| \leq C\|\Delta_{\gamma}\|_2^2.$$

Step 3 (Collecting). Combining the two displays yields the stated second-order bias bound.

Derivative identities used above. For example, the gradient w.r.t. α at $(\alpha^{1,*}, \gamma_1^*)$ is:

$$\mathbb{E}[\nabla_{\alpha} \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^*)] = \mathbb{E}_t[\nabla_{\alpha} \psi(\cdot; \alpha^{1,*})] - \mathbb{E}_s[w(\mathbf{X}; \gamma_1^*) \nabla_{\alpha} \psi(\cdot; \alpha^{1,*})]$$

where $w(\mathbf{X}; \gamma) = I(A = 1)/e^{\phi(\mathbf{X})^T \gamma}$. This is exactly $\mathbf{0}$ by [Lemma 5\(i\)](#). Similarly, the gradient w.r.t. γ is:

$$\begin{aligned} \mathbb{E}[\nabla_{\gamma} \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^*)] &= \mathbb{E}_s[\nabla_{\gamma} w(\mathbf{X}; \gamma_1^*) \{Y - \psi(\phi(\mathbf{X}); \alpha^{1,*})\}] \\ &= -\mathbb{E}_s[w(\mathbf{X}; \gamma_1^*) \phi(\mathbf{X}) \{Y - \psi(\phi(\mathbf{X}); \alpha^{1,*})\}] \end{aligned}$$

This is exactly $\mathbf{0}$ by [Lemma 5\(ii\)](#). This proves part 1.

Part 2 (Variance): explicit L_2 -Lipschitz bound. Write the difference as

$$\tilde{\varphi}_{t,s}(D; \vartheta_n) - \tilde{\varphi}_{t,s}(D; \vartheta^*) = T_t(D) + T_s(D),$$

where

$$T_t(D) := \frac{I(R=t)}{\rho_t} \left\{ \psi(\phi(\mathbf{X}); \boldsymbol{\alpha}_n^1) - \psi(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*}) \right\},$$

and

$$\begin{aligned} T_s(D) &:= \frac{I(R=s)}{\rho_s} \left(w(\mathbf{X}; \boldsymbol{\gamma}_{1,n}) \{Y - \psi(\phi(\mathbf{X}); \boldsymbol{\alpha}_n^1)\} - w(\mathbf{X}; \boldsymbol{\gamma}_1^*) \{Y - \psi(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})\} \right) \\ &= \frac{I(R=s)}{\rho_s} \{T_{s,1}(D) + T_{s,2}(D)\}. \end{aligned}$$

where

$$T_{s,1}(D) := \{w(\mathbf{X}; \boldsymbol{\gamma}_{1,n}) - w(\mathbf{X}; \boldsymbol{\gamma}_1^*)\} \{Y - \psi(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})\},$$

$$T_{s,2}(D) := w(\mathbf{X}; \boldsymbol{\gamma}_{1,n}) \{\psi(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*}) - \psi(\phi(\mathbf{X}); \boldsymbol{\alpha}_n^1)\}.$$

By $(a+b+c)^2 \leq 3(a^2+b^2+c^2)$ and $\text{Var}(Z) \leq \mathbb{E}[Z^2]$,

$$\text{Var}(\tilde{\varphi}_{t,s}(D; \boldsymbol{\vartheta}_n) - \tilde{\varphi}_{t,s}(D; \boldsymbol{\vartheta}^*)) \leq 3\mathbb{E}[T_t(D)^2] + 3\mathbb{E}[T_{s,1}(D)^2] + 3\mathbb{E}[T_{s,2}(D)^2].$$

For T_t and $T_{s,2}$, the bounded-slope assumption on ψ ([Assumption 2\(b\)\(i\)](#)) implies $|\psi(u_1) - \psi(u_2)| \leq L_\psi |u_1 - u_2|$, hence

$$\mathbb{E}[T_t(D)^2] + \mathbb{E}[T_{s,2}(D)^2] \leq C \mathbb{E}[\|\phi(\mathbf{X})\|_2^2] \|\Delta_\alpha\|_2^2,$$

using the weight-moment condition ([Assumption 2\(a\)\(ii\)](#)) and finite second moments of $\phi(\mathbf{X})$ ([Assumption 1\(a\)](#)). For $T_{s,1}$, local L_2 -Lipschitz continuity of $w(\mathbf{X}; \boldsymbol{\gamma})$ in $\boldsymbol{\gamma}$ (implied by the exponential form of w , the overlap/weight-moment assumptions, and the moment condition in [Assumption 2\(b\)](#)) yields

$$\mathbb{E}[T_{s,1}(D)^2] \leq C \|\Delta_\gamma\|_2^2.$$

Combining the bounds gives the stated variance inequality. $\square$

Lemma 7 (Double Robustness of Bias Terms). *Let $\Delta_\alpha = \boldsymbol{\alpha}_n^1 - \boldsymbol{\alpha}^{1,*}$ and $\Delta_\gamma = \boldsymbol{\gamma}_{1,n} - \boldsymbol{\gamma}_1^*$. Define the two first-order bias components:*

$$\begin{aligned} \Psi_1(\boldsymbol{\alpha}_n^1, \boldsymbol{\gamma}_1^*) &:= \mathbb{E}[\tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}_n^1, \boldsymbol{\gamma}_1^*) - \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)] \\ \Psi_2(\boldsymbol{\gamma}_{1,n}, \boldsymbol{\alpha}^{1,*}) &:= \mathbb{E}[\tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_{1,n}) - \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)] \end{aligned}$$

Under the conditions:

1. If [Lemma 5\(i\)](#) holds (from calibration w.r.t. $\boldsymbol{\gamma}$), then $\Psi_1(\boldsymbol{\alpha}_n^1, \boldsymbol{\gamma}_1^*) = O(\|\Delta_\alpha\|_2^2)$.
2. If [Lemma 5\(ii\)](#) holds (from calibration w.r.t. $\boldsymbol{\alpha}$), then $\Psi_2(\boldsymbol{\gamma}_{1,n}, \boldsymbol{\alpha}^{1,*}) = O(\|\Delta_\gamma\|_2^2)$.

Note: For identification of the target parameter, [Theorem 3](#) still requires at least one nuisance model to be correctly specified so that $\mu^1(\vartheta^*) = \mu^1$. This is formalized in [Lemma 1](#).

Proof. 1. Analyzing Ψ_1 using [Lemma 5\(i\)](#) (mean-value expansion). Let $h_1(t) := \mathbb{E}[\tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*} + t\Delta_\alpha, \boldsymbol{\gamma}_1^*)]$ for $t \in [0, 1]$. Then $\Psi_1 = h_1(1) - h_1(0)$ and

$$\Psi_1 = \int_0^1 h_1'(t) dt = \int_0^1 \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*} + t\Delta_\alpha, \boldsymbol{\gamma}_1^*)]^T \Delta_\alpha dt.$$

Add and subtract $\mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)]$ inside the integral to get

$$\begin{aligned} \Psi_1 = & \left(\mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)] \right)^T \Delta_\alpha + \int_0^1 \left\{ \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*} + t\Delta_\alpha, \boldsymbol{\gamma}_1^*)] \right. \\ & \left. - \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)] \right\}^T \Delta_\alpha dt. \end{aligned}$$

The first term is zero by [Lemma 5\(i\)](#). For the second term, we bound the gradient difference. From the definition of $\tilde{\varphi}_{t,s}$ and the canonical-link GLM identity $\nabla_\alpha \psi(\phi(\mathbf{X}); \boldsymbol{\alpha}) = \phi(\mathbf{X})\psi'(\phi(\mathbf{X})^T \boldsymbol{\alpha})$,

$$\nabla_\alpha \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}, \boldsymbol{\gamma}_1^*) = \frac{I(R=t)}{\rho_t} \phi(\mathbf{X})\psi'(\phi(\mathbf{X})^T \boldsymbol{\alpha}) - \frac{I(R=s)}{\rho_s} w(\mathbf{X}; \boldsymbol{\gamma}_1^*) \phi(\mathbf{X})\psi'(\phi(\mathbf{X})^T \boldsymbol{\alpha}).$$

Using the Lipschitz property of ψ' ([Assumption 2\(b\)\(i\)](#)) and the weight-moment condition ([Assumption 2\(a\)\(ii\)](#)), we have

$$\left\| \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*} + t\Delta_\alpha, \boldsymbol{\gamma}_1^*)] - \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)] \right\|_2 \leq C \|\Delta_\alpha\|_2,$$

uniformly over $t \in [0, 1]$, where $C < \infty$ depends on ρ_t, ρ_s , the moment bound on $\phi(\mathbf{X})$ ([Assumption 1\(a\)](#)), and $L_{\psi'}$. Therefore, by Cauchy–Schwarz,

$$|\Psi_1| \leq \int_0^1 C \|\Delta_\alpha\|_2 \cdot \|\Delta_\alpha\|_2 dt = O(\|\Delta_\alpha\|_2^2).$$

2. Analyzing Ψ_2 using [Lemma 5\(ii\)](#) (mean-value expansion). Let $h_2(t) := \mathbb{E}[\tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^* + t\Delta_\gamma)]$. Then $\Psi_2 = h_2(1) - h_2(0)$ and

$$\Psi_2 = \int_0^1 \mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^* + t\Delta_\gamma)]^T \Delta_\gamma dt.$$

Add and subtract the gradient at $\boldsymbol{\gamma}_1^*$:

$$\begin{aligned} \Psi_2 = & \left(\mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)] \right)^T \Delta_\gamma + \int_0^1 \left\{ \mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^* + t\Delta_\gamma)] \right. \\ & \left. - \mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)] \right\}^T \Delta_\gamma dt. \end{aligned}$$

The first term is zero by [Lemma 5\(ii\)](#). For the remainder, recall $w(\mathbf{X}; \gamma) = I(A = 1) \exp(-\phi(\mathbf{X})^T \gamma)$ so that $\nabla_\gamma w(\mathbf{X}; \gamma) = -w(\mathbf{X}; \gamma) \phi(\mathbf{X})$. Thus

$$\nabla_\gamma \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \gamma) = \frac{I(R = s)}{\rho_s} \nabla_\gamma w(\mathbf{X}; \gamma) \{Y - \psi(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})\}.$$

Under [Assumption 2\(a,b\)](#), the mapping $\gamma \mapsto \mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \gamma)]$ is locally Lipschitz in γ , hence

$$\left\| \mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \gamma_1^* + t\Delta_\gamma)] - \mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \gamma_1^*)] \right\|_2 \leq C \|\Delta_\gamma\|_2$$

uniformly over $t \in [0, 1]$. Therefore $|\Psi_2| \leq \int_0^1 C \|\Delta_\gamma\|_2^2 dt = O(\|\Delta_\gamma\|_2^2)$.

Clarification on Double Robustness: The key insight is that the calibrated loss functions are designed such that:

- [Lemma 5\(i\)](#) holds *by construction* of L_γ at any ϑ^* that minimizes L_γ , regardless of whether the weighting model is correct.
- [Lemma 5\(ii\)](#) holds *by construction* of L_α at any ϑ^* that minimizes L_α , regardless of whether the outcome model is correct.

Thus the first-order sensitivity to each nuisance can be removed by calibration; combined with model correctness of at least one nuisance for identification, this yields the double-robust conclusion in [Theorem 3](#). $\square$

S3 Aggregated Estimator: FACE-HD

This section follows the same theorem-driven layout. The main aggregation result is [Theorem 5](#); the preceding subsections collect the covariance structure, the penalized weight-estimation problem, and the oracle-selection assumptions needed for that theorem. On a first reading, [Lemma 8](#), [Theorem 5](#), and [Corollary 2](#) contain the main statistical message, while the later lemmas provide the supporting perturbation and selection arguments.

S3.1 Setup and Notation for Aggregation

Following Section 4.3 in the main manuscript, we aggregate the source-assisted estimators $\{\hat{\mu}_{t,s_j}^1\}_{j=1}^K$ with the target-only estimator $\hat{\mu}_{ot}^1$ using an adaptive weighting scheme. Let K be

the number of source sites. Define the global sample size across all participating sites as

$$N_{\text{all}} := N_t + \sum_{j=1}^K N_{s_j},$$

and assume $N_t/N_{\text{all}} \rightarrow \rho_t^{\text{all}} \in (0, 1)$ and $N_{s_j}/N_{\text{all}} \rightarrow \rho_{s_j}^{\text{all}} \in (0, 1)$ for each fixed j .

Notation Correspondence with Main Manuscript:

- $\hat{\mu}_{ot}^1$: Target-only estimator for $\mu^1 = \mathbb{E}_t[Y(1)]$ (uses only target site data)
- $\hat{\mu}_{t,s_j}^1$: Source-assisted estimator using source site s_j , as defined in the main manuscript
- $\hat{\mu}_{\text{agg}}^1$: Aggregated FACE-HD estimator
- η_j : Weight for source site s_j
- $\varphi_{ot,i}$: Population IF contribution for the target-only estimator (population analogue of $\hat{\varphi}_{ot,i}$ in the main manuscript)

The FACE-HD aggregated estimator (Eq. (16) in main manuscript) is:

$$\hat{\mu}_{\text{agg}}^1 = \left(1 - \sum_{j=1}^K \eta_j\right) \hat{\mu}_{ot}^1 + \sum_{j=1}^K \eta_j \hat{\mu}_{t,s_j}^1$$

which can be rewritten as:

$$\hat{\mu}_{\text{agg}}^1 = \hat{\mu}_{ot}^1 + \sum_{j=1}^K \eta_j \left(\hat{\mu}_{t,s_j}^1 - \hat{\mu}_{ot}^1 \right)$$

where $\hat{\eta} = (\hat{\eta}_1, \dots, \hat{\eta}_K)^T$ are adaptive weights minimizing the estimated asymptotic variance with a selection penalty.

Lemma 8 (Variance-Covariance Structure for Aggregation). *For each source site s_j , let $\varphi_{t,s_j,i}$ denote the population influence-function contribution of the source-assisted estimator $\hat{\mu}_{t,s_j}^1$ under the common $\sqrt{N_{\text{all}}}$ scaling, and let $\varphi_{ot,i}$ denote the corresponding target-only contribution. Write*

$$\varphi_{t,s_j,i} = \zeta_{j,i} + \xi_{j,i},$$

where $\zeta_{j,i}$ is the target-sample outcome-regression component and $\xi_{j,i}$ is the source-sample weighted residual component. Then, for $j \neq k$,

$$\text{Cov}(\xi_j, \xi_k) = 0, \quad \text{Cov}(\zeta_j, \xi_k) = 0, \quad \text{Cov}(\varphi_{ot}, \xi_k) = 0,$$

because these terms are based on disjoint source samples or on target-versus-source samples that are independent across sites. Consequently, all cross-site covariance terms arise only through the shared target component:

$$\text{Cov}(\varphi_{t,s_j}, \varphi_{t,s_k}) = \text{Cov}(\zeta_j, \zeta_k), \quad \text{Cov}(\varphi_{ot}, \varphi_{t,s_j}) = \text{Cov}(\varphi_{ot}, \zeta_j).$$

In particular, the population variance of the aggregated estimator can be written as a quadratic form

$$L^*(\eta) = \eta^T H \eta + g^T \eta + c,$$

where the off-diagonal entries of H are induced entirely by the shared target sample.

[Marquis: Add a lemma on the structure of asymptotic variance covariance matrix. OR component, IPW residuals from target or source are all uncorrelated.]

Proof. The target sample $\mathcal{D}_t$ is independent of each source sample $\mathcal{D}_{s_j}$, and the source samples are mutually independent across j . By construction, $\zeta_{j,i}$ depends only on target observations, while $\xi_{j,i}$ depends only on observations from source site s_j . Therefore, whenever two terms are functions of independent site-level samples and each has mean zero, their covariance is zero. This yields $\text{Cov}(\xi_j, \xi_k) = 0$ for $j \neq k$, $\text{Cov}(\zeta_j, \xi_k) = 0$ for all j, k , and $\text{Cov}(\varphi_{ot}, \xi_k) = 0$ because φ_{ot} depends only on the target sample. The only remaining non-zero cross-site covariance terms are those involving two target-based components observed on the same target sample, namely $\text{Cov}(\zeta_j, \zeta_k)$ and $\text{Cov}(\varphi_{ot}, \zeta_j)$. Expanding the variance of

$$\varphi_{\text{agg}}(\eta) = \varphi_{ot} + \sum_{j=1}^K \eta_j (\varphi_{t,s_j} - \varphi_{ot})$$

then gives the stated quadratic representation, with off-diagonal terms coming solely from the shared target contribution. $\square$

Remark 5 (Relation to FACE as a special case). [Lemma 8](#) is compatible with the covariance decomposition underlying [Han et al. \(2025\)](#). In particular, if the source-assisted estimators are specialized to the FACE construction and the nuisance structure is simplified to the corresponding low-dimensional setting, then the same argument shows that cross-source covariance is induced only by the shared target sample. Thus the FACE variance formula can be viewed as a special case of the present decomposition, whereas the current statement is written to accommodate the more general high-dimensional calibrated nuisance estimators used in FACE-HD.

S3.2 Penalized Variance Estimation

Define the empirical loss function as the estimated asymptotic variance of the aggregated estimator:

$$\widehat{L}(\eta) = N_{\text{all}} \cdot \widehat{\text{Var}}\left(\widehat{\mu}_{\text{agg}}^1(\eta)\right)$$

which is a quadratic function of η (equivalently, N_{all} times the variance expression in [Eq. \(16\)](#) in the main manuscript):

$$\widehat{L}(\eta) = \eta^T \widehat{H} \eta + \widehat{g}^T \eta + \widehat{c}$$

where $\widehat{H}$, $\widehat{g}$, and $\widehat{c}$ are the estimated Hessian, gradient, and constant terms derived from the variance components in Section 4.2 of the main manuscript.

Lemma 9 (Consistency of Variance/Covariance Components). *Assume [Assumption 1](#) and [Assumption 2](#), and that K is fixed. Let H, g, c denote the population analogues of $\widehat{H}, \widehat{g}, \widehat{c}$ defined by replacing empirical averages with expectations and using the population influence functions (with the same cross-site covariance terms induced by the shared target sample). Then:*

$$\|\widehat{H} - H\|_{\text{op}} = O_p(N_{\text{all}}^{-1/2}), \quad \|\widehat{g} - g\|_2 = O_p(N_{\text{all}}^{-1/2}), \quad |\widehat{c} - c| = O_p(N_{\text{all}}^{-1/2}),$$

and consequently, for any fixed compact set $\mathcal{B} \subset \mathbb{R}^K$,

$$\sup_{\eta \in \mathcal{B}} |\widehat{L}(\eta) - L^*(\eta)| = O_p(N_{\text{all}}^{-1/2}).$$

Proof. We prove entrywise $\sqrt{N_{\text{all}}}$ -consistency and then convert to the stated matrix/vector norms using that K is fixed.

Step 1 (Generic moment form). Each element of $\widehat{H}$, $\widehat{g}$ and $\widehat{c}$ is a (finite) linear combination of empirical averages of the form

$$\widehat{M}_r = \frac{1}{N_r} \sum_{i \in \mathcal{D}_r} f(D_i; \widehat{v}^{(-\text{fold}(i))}), \quad r \in \{t, s_1, \dots, s_K\},$$

where f is a product of two empirical influence-function components (including the cross-site covariance terms computed on the *same* target sample), and $\widehat{v}^{(-\text{fold}(i))}$ denotes the cross-fitted nuisance estimate used to evaluate the fold containing i . Let M_r be the population analogue obtained by replacing the sample average with $\mathbb{E}_r[\cdot]$ and $\widehat{v}^{(-\text{fold}(i))}$ with the population target v^* .

Step 2 (Conditional LLN and plug-in stability). By cross-fitting, conditional on the nuisance-training samples, the summands $\{f(D_i; \hat{\vartheta}^{(-\text{fold}(i))}) : i \in \mathcal{D}_r\}$ are independent within each site r , with conditional mean $\mathbb{E}_r[f(D; \hat{\vartheta}^{(-\text{fold}(D))}) \mid \{\mathcal{D}^{(-k)}\}]$. [Assumption 2\(b\)](#) gives a finite $(1 + \delta)$ moment for $f(D; \vartheta)$ uniformly in a neighborhood of ϑ^* (since it is a product of two $(2 + \delta)$ -moment IF components), hence the conditional variance is finite and a conditional LLN/CLT implies

$$\widehat{M}_r - \mathbb{E}_r[f(D; \hat{\vartheta}^{(-\text{fold}(D))}) \mid \{\mathcal{D}^{(-k)}\}] = O_p(N_r^{-1/2}).$$

Next, by [Lemma 4](#), $\hat{\vartheta}^{(-k)} \rightarrow_p \vartheta^*$, and by [Lemma 6](#) (Part 2) the mapping $\vartheta \mapsto f(D; \vartheta)$ is locally L_2 -Lipschitz. Therefore,

$$\mathbb{E}_r[f(D; \hat{\vartheta}^{(-\text{fold}(D))})] = \mathbb{E}_r[f(D; \vartheta^*)] + o(1),$$

and combining the two displays yields $\widehat{M}_r - M_r = O_p(N_r^{-1/2})$. Under the site-ratio condition $N_r/N_{\text{all}} \rightarrow \rho_r > 0$, this is $O_p(N_{\text{all}}^{-1/2})$.

Step 3 (From entrywise to matrix/vector norms). Since $\widehat{H}$ has fixed dimension $K \times K$ and $\widehat{g}$ has fixed dimension K , the preceding bound applies to each entry, hence

$$\|\widehat{H} - H\|_{\text{op}} \leq C_K \max_{j,k} |\widehat{H}_{jk} - H_{jk}| = O_p(N_{\text{all}}^{-1/2}), \quad \|\widehat{g} - g\|_2 \leq \sqrt{K} \max_j |\widehat{g}_j - g_j| = O_p(N_{\text{all}}^{-1/2}),$$

Finally, $\widehat{c}$ is a finite linear combination of the same sample averages $\{\widehat{M}_r\}$ (coming from the constant terms in the variance expansion), so the same entrywise $O_p(N_{\text{all}}^{-1/2})$ bound implies $|\widehat{c} - c| = O_p(N_{\text{all}}^{-1/2})$.

Step 4 (Uniform convergence of the quadratic form). For η in a fixed compact set $\mathcal{B} \subset \mathbb{R}^K$,

$$|\widehat{L}(\eta) - L^*(\eta)| \leq \|\widehat{H} - H\|_{\text{op}} \|\eta\|_2^2 + \|\widehat{g} - g\|_2 \|\eta\|_2 + |\widehat{c} - c|,$$

and the right-hand side is $O_p(N_{\text{all}}^{-1/2})$ uniformly over $\eta \in \mathcal{B}$ because $\sup_{\eta \in \mathcal{B}} \|\eta\|_2 < \infty$. $\square$

The penalized objective from [Eq. \(16\)](#) is:

$$\widehat{\eta} = \arg \min_{\eta} \left\{ \widehat{L}(\eta) + \lambda \sum_{j=1}^K |\eta_j| \left(\widehat{\mu}_{ot}^1 - \widehat{\mu}_{t,s_j}^1 \right)^2 \right\} \quad (22)$$

where $\lambda > 0$ is a tuning parameter that depends on N_{all} .

The penalty term $(\widehat{\mu}_{ot}^1 - \widehat{\mu}_{t,s_j}^1)^2$ measures the discrepancy between the target-only and source-assisted estimators. Large discrepancies indicate potential covariate shift or model misspecification, and the penalty drives the corresponding η_j toward zero.

Remark 6 (FACE-style validation for λ). The main manuscript selects the aggregation tuning parameter by the FACE sample-splitting validation rule: for each outer evaluation fold, candidate weights are learned on inner-training summaries by solving the penalized aggregation problem, and these fixed weights are then evaluated on held-out inner-validation summaries through the validation penalized criterion. This validation step is a finite-sample rule for choosing a stable point along the penalized aggregation path; it is not the same object as the nuisance-model cross-validation used for α and γ . For the asymptotic arguments below, λ may be deterministic or data-adaptive, provided it is measurable with respect to the outer training data and satisfies [Assumption 4](#)(b)–(c) with probability tending to one. The inner validation construction preserves the cross-fitting independence needed for the influence-function expansion because the training summaries used to compute $\hat{\eta}^{(-k,-m)}(\lambda)$, the validation summaries used to choose $\hat{\lambda}^{(-k)}$, and the final outer evaluation fold $\mathcal{D}_k$ are separated. Consequently neither $\hat{\lambda}^{(-k)}$ nor $\hat{\eta}^{(-k)}$ uses the outer evaluation fold $\mathcal{D}_k$.

Remark 7 (Soft-thresholding interpretation). The penalty in [Eq. \(22\)](#) is an adaptive ℓ_1 penalty, so its selection mechanism is the same as soft-thresholding in penalized quadratic optimization. To see this, consider the special case in which $\widehat{H}$ is diagonal, say $\widehat{H} = \text{diag}(\hat{h}_1, \dots, \hat{h}_K)$ with $\hat{h}_j > 0$. Then the objective separates coordinatewise, and each $\hat{\eta}_j$ minimizes

$$\hat{h}_j \eta_j^2 + \hat{g}_j \eta_j + \lambda \hat{d}_j |\eta_j|, \quad \hat{d}_j := (\hat{\mu}_{ot}^1 - \hat{\mu}_{t,s_j}^1)^2.$$

Completing the square yields the explicit solution

$$\hat{\eta}_j = -\frac{1}{2\hat{h}_j} \text{sign}(\hat{g}_j) \left(|\hat{g}_j| - \lambda \hat{d}_j \right)_+,$$

which is the usual soft-thresholding rule: if $|\hat{g}_j| \leq \lambda \hat{d}_j$, then $\hat{\eta}_j = 0$; otherwise the unconstrained quadratic solution is shrunk toward zero by the amount $\lambda \hat{d}_j / (2\hat{h}_j)$. In the general non-diagonal case, the coordinates are coupled through $\widehat{H}$, so one no longer has a closed-form coordinatewise solution, but the same ℓ_1 -induced shrinkage-and-selection interpretation remains valid.

S3.3 Informative Source Sites

Let $\mu^1 = \mathbb{E}_t[Y(1)]$ be the true target-site potential outcome mean.

Define the set of *informative* source sites as:

$$\mathcal{S}^* = \{j \in \{1, \dots, K\} : \text{plim } \hat{\mu}_{t,s_j}^1 = \mu^1\}$$

i.e., sites where the source-assisted estimator is consistent for the target parameter μ^1 .

Define the oracle-restricted space:

$$\mathbb{R}_{\mathcal{S}^*} := \{\eta \in \mathbb{R}^K : \eta_j = 0 \text{ for all } j \notin \mathcal{S}^*\}.$$

Let η^* be the oracle optimal weights that minimize the population asymptotic variance over $\mathbb{R}_{\mathcal{S}^*}$:

$$\eta^* = \arg \min_{\eta \in \mathbb{R}_{\mathcal{S}^*}} L^*(\eta), \quad L^*(\eta) := \text{Var} \left(\sqrt{N_{\text{all}}} (\hat{\mu}_{\text{agg}}^1(\eta) - \mu^1) \right).$$

Assumption 4 (Aggregation Regularity). (a) (Sufficient Informative Sources + Local Strong Convexity) The set $\mathcal{S}^*$ is non-empty, i.e., there exists at least one informative source site. Moreover, the population aggregation objective is locally strictly convex on the oracle subspace: the Hessian of $L^*(\eta)$ restricted to $\mathbb{R}_{\mathcal{S}^*}$, denoted $H_{\mathcal{S}^*}$, is positive definite (equivalently, η^* is locally unique). (b) (Separation / “ β -min” for biased sites) Define

$$b_{0,N} := \min_{j \notin \mathcal{S}^*} |\text{plim}(\hat{\mu}_{t,s_j}^1) - \mu^1|.$$

Assume $\lambda b_{0,N}^2 \rightarrow \infty$. A special case is *fixed separation*: $b_{0,N} \geq b_0 > 0$. (c) (Penalty Growth) The penalty parameter satisfies $\lambda \rightarrow \infty$ and $\lambda/\sqrt{N_{\text{all}}} \rightarrow 0$. (d) (Fixed K) The number of source sites K is fixed.

When λ is selected by the validation rule in the main manuscript, [Assumption 4](#)(b)–(c) should be read as high-probability conditions on the selected sequence $\hat{\lambda}$. Equivalently, the theorem below is conditional on the event that the validation-selected penalty lies in the oracle-selection window. This separates the finite-sample tuning rule from the asymptotic support-recovery requirement.

Remark 8 (When separation fails / local alternatives). [Assumption 4](#)(b) is used only for *support recovery* (forcing $\hat{\eta}_j = 0$ for biased sites). If some biased sites are only *weakly* biased in the sense that $b_{0,N} \rightarrow 0$, then selection consistency may fail. In that regime, the aggregated estimator remains asymptotically linear for any deterministic η ([Lemma 10](#)), and the additional first-order bias is controlled by

$$\left| \sum_{j \notin \mathcal{S}^*} \eta_j \{\text{plim}(\hat{\mu}_{t,s_j}^1) - \mu^1\} \right|,$$

so if any included weakly biased sites satisfy $b_{j,N} = o(N_{\text{all}}^{-1/2})$, their contribution is still negligible at $\sqrt{N_{\text{all}}}$ scale.

S3.4 Main Results for Aggregation

For later use in the aggregation proofs, we restate the pairwise input result that is repeatedly invoked below.

Theorem (restated). *Suppose [Assumption 1](#), [Assumption 2](#), and [Assumption 3\(a\)–Assumption 3\(c\)](#) hold. ([Assumption 3\(d\)](#) is an optional strengthening used only to invoke the **Special** rate in [Lemma 4](#); the theorem also holds under the **General** rate condition there.) Assume ϑ^* is the unique population target pair induced by the calibrated losses under the alignment condition in [Assumption 3\(c\)](#), implying [Lemma 5](#) holds. If further, at least one nuisance model (outcome or weighting) is correctly specified in the sense of [Lemma 1](#) (so $\mu^1(\vartheta^*) = \mu^1$), the calibrated nuisance sparsities satisfy*

$$\frac{(s_\alpha + s_\gamma)^2 \log^2(p)}{N} \rightarrow 0,$$

and $V_{t,s}^* > 0$, then the cross-fitted estimator is asymptotically normal. Here s_α and s_γ denote the calibrated sparsities from [Definition 1](#):

$$\sqrt{N} \left(\hat{\mu}_{t,s}^{1,\text{CF}} - \mu^1 \right) \xrightarrow{d} \mathcal{N}(0, V_{t,s}^*).$$

If the initial nuisance estimators enter the two-level algorithm separately from the calibrated update, their stochastic error is required only through [Assumption 3\(c\)](#), which aligns their population limits with $(\alpha^{1,*}, \gamma_1^*)$. Moreover, the finite-sample KKT perturbation from the ℓ_1 penalties is absorbed in the remainder term (via [Lemma 4](#)) and is $o_p(N^{-1/2})$ under the stated sparsity/rate condition.

Theorem 5 (Asymptotic Normality of FACE-HD). *Suppose [Assumptions 1 to 3](#) and [Assumption 4](#) hold. For each informative source $s_j \in \mathcal{S}^*$, let $N_{t,s_j} = N_t + N_{s_j}$ and $s_{\text{tot},j} = s_{\alpha,j} + s_{\gamma,j}$ denote the pairwise total sparsity. If*

$$\max_{j \in \mathcal{S}^*} \frac{s_{\text{tot},j}^2 \log^2(p)}{N_{t,s_j}} \rightarrow 0,$$

(equivalently under fixed site ratios, the same condition with N_{all}), and $L^*(\eta^*) > 0$, then:

$$\sqrt{N_{\text{all}}} \left(\hat{\mu}_{\text{agg}}^1 - \mu^1 \right) \xrightarrow{d} \mathcal{N}(0, L^*(\eta^*))$$

where $L^*(\eta^*)$ is the minimum asymptotic variance achieved by the oracle aggregation.

Moreover, the variance estimator is consistent:

$$\widehat{V} = \widehat{L}(\widehat{\eta}) \xrightarrow{P} L^*(\eta^*)$$

and the standardized statistic is asymptotically standard normal:

$$\sqrt{N_{\text{all}}/\widehat{V}} \left(\widehat{\mu}_{\text{agg}}^1 - \mu^1 \right) \xrightarrow{d} \mathcal{N}(0, 1).$$

Corollary 2 (Asymptotic Normality of the Aggregated TATE). *Define the aggregated TATE estimator as*

$$\widehat{\tau}_{\text{agg}} := \widehat{\mu}_{\text{agg}}^1 - \widehat{\mu}_{\text{agg}}^0,$$

where $\widehat{\mu}_{\text{agg}}^0$ is constructed by applying the same aggregation procedure to the control-arm estimators $\{\widehat{\mu}_{t,s_j}^0\}_{j=1}^K$ (with the corresponding nuisance estimates and variance/covariance components). Assume the analogues of [Assumptions 1](#) to [4](#) hold for both arms, so that both $\widehat{\mu}_{\text{agg}}^1$ and $\widehat{\mu}_{\text{agg}}^0$ admit asymptotic linear representations under $\sqrt{N_{\text{all}}}$ scaling. Then

$$\sqrt{N_{\text{all}}}(\widehat{\tau}_{\text{agg}} - \tau) \xrightarrow{d} \mathcal{N}(0, V_{\tau}^*), \quad V_{\tau}^* := \text{Var}\left(\varphi_{\text{agg}}^1(D) - \varphi_{\text{agg}}^0(D)\right),$$

where φ_{agg}^1 and φ_{agg}^0 denote the aggregated influence functions (oracle-level) for the treated and control aggregated estimators, respectively.

Proposition 1 (Efficiency Gain). *Under [Assumptions 1](#) to [3](#) and [Assumption 4](#), the FACE-HD estimator is asymptotically at least as efficient as the target-only estimator:*

$$L^*(\eta^*) \leq L^*(0)$$

with strict inequality whenever there exists $j \in \mathcal{S}^*$ such that

$$\text{Var}(\varphi_{t,s_j} - \varphi_{ot}) > 0 \quad \text{and} \quad \text{Cov}(\varphi_{ot}, \varphi_{t,s_j} - \varphi_{ot}) \neq 0.$$

S3.5 Supporting Lemmas for Aggregation

Lemma 10 (Influence Function Representation). *Under [Assumptions 1](#) to [3](#) and [Assumption 4\(a\)](#), the aggregated estimator admits the asymptotic linear representation:*

$$\sqrt{N_{\text{all}}} \left(\widehat{\mu}_{\text{agg}}^1(\eta^*) - \mu^1 \right) = \frac{1}{\sqrt{N_{\text{all}}}} \sum_{i=1}^{N_{\text{all}}} \varphi_{\text{agg},i}(\eta^*) + o_p(1)$$

where $\varphi_{\text{agg},i}(\eta^*)$ is the aggregated influence function:

$$\varphi_{\text{agg},i}(\eta^*) = \left(1 - \sum_{j \in \mathcal{S}^*} \eta_j^*\right) \varphi_{ot,i} + \sum_{j \in \mathcal{S}^*} \eta_j^* \varphi_{t,s_j,i},$$

with $\varphi_{t,s_j,i}$ denoting the (rescaled) IF contribution of source-assisted estimator j under $\sqrt{N_{\text{all}}}$ scaling. **Convention:** Each $\varphi_{ot,i}$ (resp. $\varphi_{t,s_j,i}$) is understood as zero for observations not in the target data $\mathcal{D}_t$ (resp. not in $\mathcal{D}_t \cup \mathcal{D}_{s_j}$), and includes the constant rescaling needed to convert its natural $\sqrt{N_t}$ (resp. $\sqrt{N_{t,s_j}}$) linear representation into the common $\sqrt{N_{\text{all}}}$ scaling.

Proof. By [Theorem 3](#), for each fixed $j \in \mathcal{S}^*$, there is an asymptotic linear representation under pairwise scaling $N_{t,s_j} := N_t + N_{s_j}$:

$$\sqrt{N_{t,s_j}}(\hat{\mu}_{t,s_j}^1 - \text{plim}(\hat{\mu}_{t,s_j}^1)) = \frac{1}{\sqrt{N_{t,s_j}}} \sum_{i=1}^{N_{\text{all}}} \bar{\varphi}_{t,s_j,i} + o_p(1),$$

where $\bar{\varphi}_{t,s_j,i}$ is understood as zero for observations not in $\mathcal{D}_t \cup \mathcal{D}_{s_j}$. Since $N_{t,s_j}/N_{\text{all}} \rightarrow c_j \in (0, 1)$, the above is equivalent to

$$\sqrt{N_{\text{all}}}(\hat{\mu}_{t,s_j}^1 - \text{plim}(\hat{\mu}_{t,s_j}^1)) = \frac{1}{\sqrt{N_{\text{all}}}} \sum_{i=1}^{N_{\text{all}}} \varphi_{t,s_j,i} + o_p(1)$$

for a rescaled IF $\varphi_{t,s_j,i}$. The target-only estimator has the analogous form:

$$\sqrt{N_{\text{all}}}(\hat{\mu}_{ot}^1 - \mu^1) = \frac{1}{\sqrt{N_{\text{all}}}} \sum_{i=1}^{N_{\text{all}}} \varphi_{ot,i} + o_p(1).$$

For $j \in \mathcal{S}^*$, $\text{plim}(\hat{\mu}_{t,s_j}^1) = \mu^1$ by definition.

By linearity, the aggregated estimator with fixed $\eta = \eta^*$ satisfies:

$$\sqrt{N_{\text{all}}}(\hat{\mu}_{\text{agg}}^1(\eta^*) - \mu^1) = \frac{1}{\sqrt{N_{\text{all}}}} \sum_{i=1}^{N_{\text{all}}} \varphi_{\text{agg},i}(\eta^*) + o_p(1).$$

The remainder remains $o_p(1)$ because K is fixed. □

Lemma 11 (Selection Consistency). *Under [Assumptions 1 to 3](#) and [Assumption 4](#), the penalized estimator $\hat{\eta}$ from [Eq. \(22\)](#) satisfies:*

1. **(Rate)** $\|\hat{\eta} - \eta^*\|_2 = O_p(N_{\text{all}}^{-1/2})$
2. **(Selection)** $\hat{\eta}_j = 0$ for all $j \notin \mathcal{S}^*$ with probability approaching 1.

Proof. We give a self-contained argument; it parallels the FACE proof in Han et al. (2025) (Lemma S5) but the objective here is an explicit quadratic form. Recall $\widehat{L}(\eta) = \eta^T \widehat{H} \eta + \widehat{g}^T \eta + \widehat{c}$ and its population analogue $L^*(\eta) = \eta^T H \eta + g^T \eta + c$.

Part 1 (Rate on the oracle subspace). Define the oracle-restricted estimator

$$\tilde{\eta} = \arg \min_{\eta \in \mathbb{R}_{\mathcal{S}^*}} \left\{ \widehat{L}(\eta) + \lambda \sum_{j=1}^K |\eta_j| (\widehat{\mu}_{ot}^1 - \widehat{\mu}_{t,s_j}^1)^2 \right\}.$$

On $\mathbb{R}_{\mathcal{S}^*}$, only coordinates $j \in \mathcal{S}^*$ can be nonzero, so the penalty term reduces to $\sum_{j \in \mathcal{S}^*} |\eta_j| (\widehat{\mu}_{ot}^1 - \widehat{\mu}_{t,s_j}^1)^2$. By Theorem 3 and the site-ratio regularity, for each fixed $j \in \mathcal{S}^*$, $\widehat{\mu}_{ot}^1 - \widehat{\mu}_{t,s_j}^1 = O_p(N_{\text{all}}^{-1/2})$, hence $(\widehat{\mu}_{ot}^1 - \widehat{\mu}_{t,s_j}^1)^2 = O_p(N_{\text{all}}^{-1})$ and therefore the penalty contribution on $\mathbb{R}_{\mathcal{S}^*}$ is $O_p(\lambda/N_{\text{all}}) = o_p(N_{\text{all}}^{-1/2})$ under Assumption 4(c).

Next, by Lemma 9, the quadratic coefficients satisfy $\|\widehat{H} - H\|_{\text{op}} = O_p(N_{\text{all}}^{-1/2})$ and $\|\widehat{g} - g\|_2 = O_p(N_{\text{all}}^{-1/2})$. Let $H_{\mathcal{S}^*}$ and $\widehat{H}_{\mathcal{S}^*}$ denote the principal submatrices corresponding to $\mathcal{S}^*$, and similarly for g . Because K is fixed, $H_{\mathcal{S}^*}$ is a fixed-dimensional positive definite matrix (local strong convexity of L^* around η^*), hence $\|H_{\mathcal{S}^*}^{-1}\|_{\text{op}} < \infty$ and $\|\widehat{H}_{\mathcal{S}^*}^{-1}\|_{\text{op}} = O_p(1)$. More explicitly, by Weyl's inequality and Lemma 9, $\lambda_{\min}(\widehat{H}_{\mathcal{S}^*}) \geq \lambda_{\min}(H_{\mathcal{S}^*}) - \|\widehat{H}_{\mathcal{S}^*} - H_{\mathcal{S}^*}\|_{\text{op}}$, which is bounded away from 0 w.p.a.1, so $\widehat{H}_{\mathcal{S}^*}$ is invertible and $\|\widehat{H}_{\mathcal{S}^*}^{-1}\|_{\text{op}} = O_p(1)$. Ignoring the $o_p(N_{\text{all}}^{-1/2})$ penalty perturbation, the minimizers of $\widehat{L}$ and L^* on $\mathbb{R}_{\mathcal{S}^*}$ satisfy the first-order conditions $2\widehat{H}_{\mathcal{S}^*}\eta + \widehat{g}_{\mathcal{S}^*} = 0$ and $2H_{\mathcal{S}^*}\eta + g_{\mathcal{S}^*} = 0$, so

$$\eta_{\mathcal{S}^*}^* = -\frac{1}{2}H_{\mathcal{S}^*}^{-1}g_{\mathcal{S}^*}, \quad \tilde{\eta}_{\mathcal{S}^*} = -\frac{1}{2}\widehat{H}_{\mathcal{S}^*}^{-1}\widehat{g}_{\mathcal{S}^*} + o_p(N_{\text{all}}^{-1/2}).$$

To make the perturbation step explicit, write

$$\tilde{\eta}_{\mathcal{S}^*} - \eta_{\mathcal{S}^*}^* = -\frac{1}{2} \left(\widehat{H}_{\mathcal{S}^*}^{-1}\widehat{g}_{\mathcal{S}^*} - H_{\mathcal{S}^*}^{-1}g_{\mathcal{S}^*} \right) + o_p(N_{\text{all}}^{-1/2}).$$

Then

$$\widehat{H}_{\mathcal{S}^*}^{-1}\widehat{g}_{\mathcal{S}^*} - H_{\mathcal{S}^*}^{-1}g_{\mathcal{S}^*} = \widehat{H}_{\mathcal{S}^*}^{-1}(\widehat{g}_{\mathcal{S}^*} - g_{\mathcal{S}^*}) + (\widehat{H}_{\mathcal{S}^*}^{-1} - H_{\mathcal{S}^*}^{-1})g_{\mathcal{S}^*}.$$

The first term is $O_p(N_{\text{all}}^{-1/2})$ since $\|\widehat{H}_{\mathcal{S}^*}^{-1}\|_{\text{op}} = O_p(1)$ and $\|\widehat{g} - g\|_2 = O_p(N_{\text{all}}^{-1/2})$. For the second term, use $\widehat{H}_{\mathcal{S}^*}^{-1} - H_{\mathcal{S}^*}^{-1} = \widehat{H}_{\mathcal{S}^*}^{-1}(H_{\mathcal{S}^*} - \widehat{H}_{\mathcal{S}^*})H_{\mathcal{S}^*}^{-1}$, so

$$\|\widehat{H}_{\mathcal{S}^*}^{-1} - H_{\mathcal{S}^*}^{-1}\|_{\text{op}} \leq \|\widehat{H}_{\mathcal{S}^*}^{-1}\|_{\text{op}} \|\widehat{H} - H\|_{\text{op}} \|H_{\mathcal{S}^*}^{-1}\|_{\text{op}} = O_p(N_{\text{all}}^{-1/2}),$$

and hence the second term is also $O_p(N_{\text{all}}^{-1/2})$. Therefore $\|\tilde{\eta} - \eta^*\|_2 = O_p(N_{\text{all}}^{-1/2})$. Finally, since $\widehat{\eta}$ agrees with $\tilde{\eta}$ w.p.a.1 by Part 2 below, the same rate holds for $\widehat{\eta}$.

Part 2 (Selection of biased sites). By Part 1, $\tilde{\eta} = \eta^* + O_p(N_{\text{all}}^{-1/2})$, so there exists a fixed-radius neighborhood $\mathcal{B} := \{\eta : \|\eta - \eta^*\|_2 \leq r\}$ such that $P(\tilde{\eta} \in \mathcal{B}) \rightarrow 1$. For the full penalized minimizer, the same neighborhood restriction applies w.p.a.1 once we show (below) that all biased coordinates are set to zero, i.e., $\hat{\eta} = \tilde{\eta}$ w.p.a.1. Thus, in the KKT bound we may work on $\mathcal{B}$.

For biased sites $j \notin \mathcal{S}^*$, let $b_{j,N} := |\text{plim}(\hat{\mu}_{t,s_j}^1) - \mu^1|$, so $b_{j,N} \geq b_{0,N}$. Because $\hat{\mu}_{t,s_j}^1 \xrightarrow{p} \text{plim}(\hat{\mu}_{t,s_j}^1)$ and $\hat{\mu}_{ot}^1 \xrightarrow{p} \mu^1$,

$$(\hat{\mu}_{ot}^1 - \hat{\mu}_{t,s_j}^1)^2 \xrightarrow{p} b_{j,N}^2 \geq b_{0,N}^2,$$

and thus $\lambda(\hat{\mu}_{ot}^1 - \hat{\mu}_{t,s_j}^1)^2 \rightarrow \infty$ under [Assumption 4\(b\)](#).

The KKT condition for the penalized objective at coordinate j implies that if $\hat{\eta}_j = 0$ then

$$\left| \frac{\partial \hat{L}}{\partial \eta_j}(\hat{\eta}) \right| \leq \lambda(\hat{\mu}_{ot}^1 - \hat{\mu}_{t,s_j}^1)^2.$$

Since $\hat{L}(\eta) = \eta^T \hat{H} \eta + \hat{g}^T \eta + \hat{c}$ is quadratic,

$$\frac{\partial \hat{L}}{\partial \eta_j}(\eta) = 2(\hat{H}\eta)_j + \hat{g}_j,$$

so for any compact $\mathcal{B} \subset \mathbb{R}^K$,

$$\sup_{\eta \in \mathcal{B}} \left| \frac{\partial \hat{L}}{\partial \eta_j}(\eta) \right| \leq 2 \sup_{\eta \in \mathcal{B}} \|\hat{H}\eta\|_\infty + \|\hat{g}\|_\infty \leq 2\|\hat{H}\|_{\text{op}} \sup_{\eta \in \mathcal{B}} \|\eta\|_2 + \|\hat{g}\|_2.$$

By [Lemma 9](#), $\|\hat{H}\|_{\text{op}} = O_p(1)$ and $\|\hat{g}\|_2 = O_p(1)$, hence $\sup_{\eta \in \mathcal{B}} |\partial \hat{L} / \partial \eta_j(\eta)| = O_p(1)$. Because the right-hand side diverges, the KKT inequality is satisfied for $\eta_j = 0$ w.p.a.1 and any minimizer must set $\hat{\eta}_j = 0$ for all $j \notin \mathcal{S}^*$. Therefore $\hat{\eta} \in \mathbb{R}_{\mathcal{S}^*}$ w.p.a.1, hence $\hat{\eta} = \tilde{\eta}$ w.p.a.1. $\square$

Lemma 12 (Asymptotic Equivalence). *Under [Assumptions 1 to 3](#) and [Assumption 4](#):*

$$\sqrt{N_{\text{all}}} \left(\hat{\mu}_{agg}^1(\hat{\eta}) - \hat{\mu}_{agg}^1(\eta^*) \right) = o_p(1)$$

Proof. The difference is:

$$\sqrt{N_{\text{all}}} \left(\hat{\mu}_{agg}^1(\hat{\eta}) - \hat{\mu}_{agg}^1(\eta^*) \right) = \sum_{j=1}^K (\hat{\eta}_j - \eta_j^*) \cdot \sqrt{N_{\text{all}}} (\hat{\mu}_{t,s_j}^1 - \hat{\mu}_{ot}^1)$$

For $j \in \mathcal{S}^*$: By [Lemma 11\(1\)](#), $|\hat{\eta}_j - \eta_j^*| = O_p(N_{\text{all}}^{-1/2})$. By [Theorem 3](#) with the same rescaling argument as in [Lemma 10](#), $\sqrt{N_{\text{all}}} (\hat{\mu}_{t,s_j}^1 - \hat{\mu}_{ot}^1) = O_p(1)$. Thus each term is $o_p(1)$.

For $j \notin \mathcal{S}^*$: By [Lemma 11](#)(2), $\hat{\eta}_j = 0$ w.h.p., and $\eta_j^* = 0$ by definition (oracle excludes biased sites). Thus the difference is zero w.h.p. Because K is fixed, summing over j yields $o_p(1)$. $\square$

Remark 9 (Growing number of sources K). The arguments above use [Assumption 4](#)(d) (fixed K) in two places: (i) to ensure the plug-in term $\sum_{j=1}^K (\hat{\eta}_j - \eta_j^*) \sqrt{N_{\text{all}}} (\hat{\mu}_{t,s_j}^1 - \hat{\mu}_{ot}^1)$ is $o_p(1)$, and (ii) to treat the variance/covariance estimation problem as finite-dimensional. If $K = K_N \rightarrow \infty$, then additional high-dimensional control is needed, e.g., a sufficient condition for [Lemma 12](#) is

$$\|\hat{\eta} - \eta^*\|_1 \cdot \max_{1 \leq j \leq K} \left| \sqrt{N_{\text{all}}} (\hat{\mu}_{t,s_j}^1 - \hat{\mu}_{ot}^1) \right| = o_p(1),$$

which typically requires both a sparsity assumption on η^* and concentration bounds scaling with $\log K$.

S3.6 Proofs of Main Results for Aggregation

Proof of Theorem 5. Combining [Lemma 10](#), [Lemma 11](#), and [Lemma 12](#):

$$\sqrt{N_{\text{all}}} (\hat{\mu}_{\text{agg}}^1 - \mu^1) = \sqrt{N_{\text{all}}} (\hat{\mu}_{\text{agg}}^1(\eta^*) - \mu^1) + o_p(1) = \frac{1}{\sqrt{N_{\text{all}}}} \sum_{i=1}^{N_{\text{all}}} \varphi_{\text{agg},i}(\eta^*) + o_p(1).$$

As in [Lemma 3](#), the observations are independent with a multi-site structure and site proportions converging to constants. The aggregated influence function contributions $\{\varphi_{\text{agg},i}(\eta^*)\}_{i=1}^{N_{\text{all}}}$ are therefore independent with mean zero and finite variance $L^*(\eta^*)$. Under [Assumption 2](#)(b), these contributions have a uniform $(2 + \delta)$ moment, so the same Lyapunov argument as in [Lemma 3](#) applies and yields:

$$\frac{1}{\sqrt{N_{\text{all}}}} \sum_{i=1}^{N_{\text{all}}} \varphi_{\text{agg},i}(\eta^*) \xrightarrow{d} \mathcal{N}(0, L^*(\eta^*)).$$

For variance estimation consistency: By [Lemma 11](#)(1), $\|\hat{\eta} - \eta^*\|_2 = O_p(N_{\text{all}}^{-1/2})$. By the uniform convergence of $\hat{L}$ to L^* and local Lipschitz continuity of the quadratic map $L^*(\eta) = \eta^T H \eta + g^T \eta + c$ in a neighborhood of η^* (e.g., on $\{\eta : \|\eta - \eta^*\|_2 \leq R\}$, $|L^*(\eta_1) - L^*(\eta_2)| \leq (2\|H\|_{\text{op}}R + \|g\|_2)\|\eta_1 - \eta_2\|_2$):

$$|\hat{L}(\hat{\eta}) - L^*(\eta^*)| \leq |\hat{L}(\hat{\eta}) - L^*(\hat{\eta})| + |L^*(\hat{\eta}) - L^*(\eta^*)| = O_p(N_{\text{all}}^{-1/2}) + O_p(N_{\text{all}}^{-1/2}) = o_p(1).$$

Finally, write

$$\sqrt{\frac{N_{\text{all}}}{\widehat{V}}} (\widehat{\mu}_{\text{agg}}^1 - \mu^1) = \left(\sqrt{\frac{L^*(\eta^*)}{\widehat{V}}} \right) \cdot \left(\frac{1}{\sqrt{L^*(\eta^*)}} \sqrt{N_{\text{all}}} (\widehat{\mu}_{\text{agg}}^1 - \mu^1) \right).$$

We already proved $\sqrt{N_{\text{all}}}(\widehat{\mu}_{\text{agg}}^1 - \mu^1) \xrightarrow{d} \mathcal{N}(0, L^*(\eta^*))$, hence the second factor converges in distribution to $\mathcal{N}(0, 1)$. By variance consistency, $\widehat{V} \xrightarrow{p} L^*(\eta^*)$, so the prefactor $\sqrt{L^*(\eta^*)/\widehat{V}} \xrightarrow{p} 1$. Therefore the product converges in distribution to $\mathcal{N}(0, 1)$ by Slutsky's theorem. $\square$

Proof of Proposition 1. Since η^* minimizes $L^*(\eta)$ and $\eta = 0$ is feasible, we have $L^*(\eta^*) \leq L^*(0)$.

For strict improvement, consider one informative source $j \in \mathcal{S}^*$. The optimal projection is:

$$\eta_j^* = -\frac{\text{Cov}(\varphi_{ot}, \varphi_{t,s_j} - \varphi_{ot})}{\text{Var}(\varphi_{t,s_j} - \varphi_{ot})}.$$

If the covariance is non-zero and the denominator is positive, the minimizing weight is non-zero and strictly decreases variance relative to $\eta = 0$, so $L^*(\eta^*) < L^*(0)$. $\square$